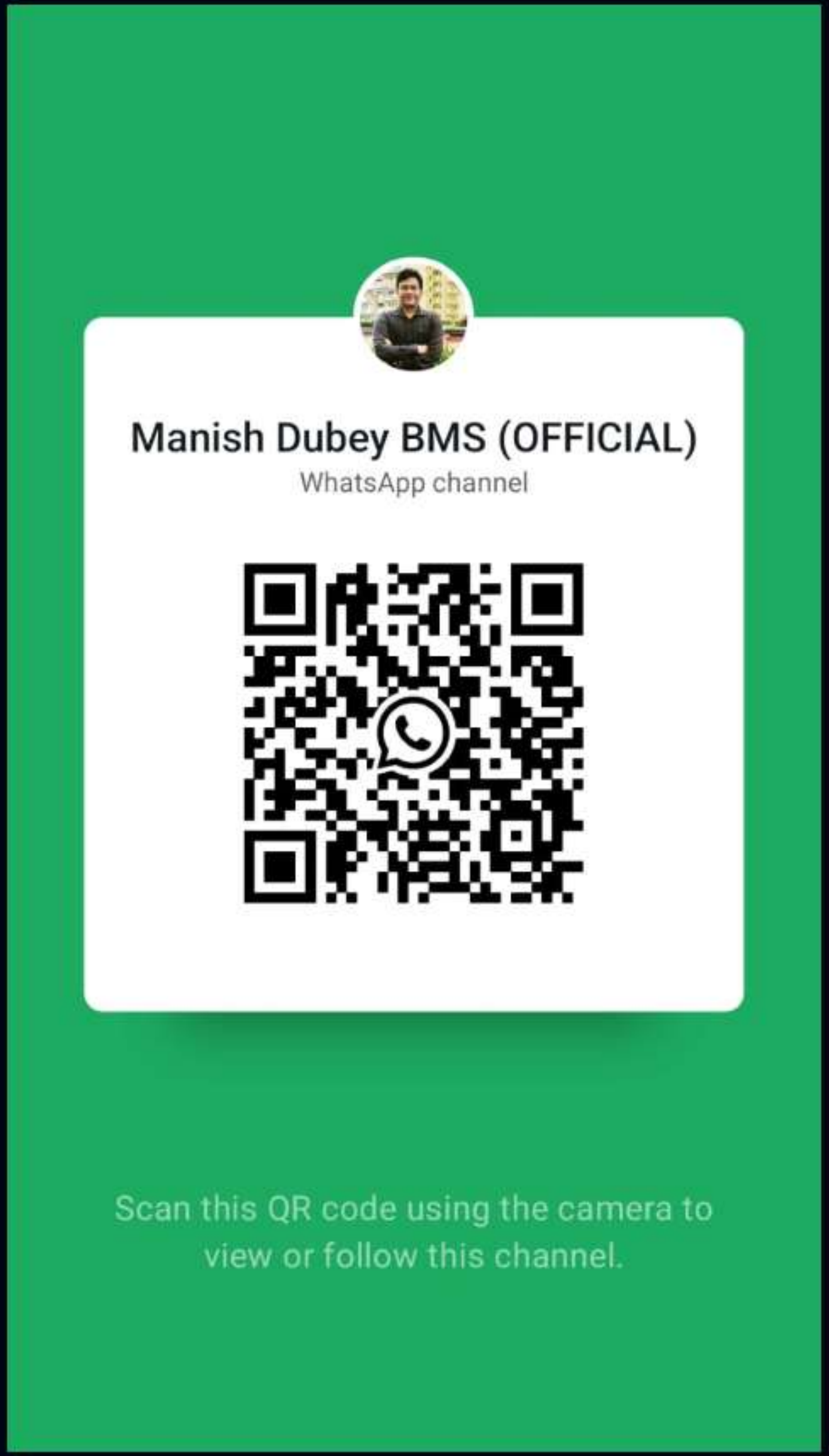

RENEET 2026
MAHA
MARATHON
Complete
ZOOLOGY
Oneshot
2
PART
MD Sir
SANKALP NEET
VEDANTU
Vedantu
ZOOLOGY
ANIMAL KINGDOM
HUMAN PHYSIOLOGY
EVOLUTION

1) Evolution
२) Biotech Both
3) Human Reproduction
4) Reproductive Health
5) Human Health & Diseases



Origin of Universe = $\boxed{13.8 \text{ B.Y.A}}$ (According to Big Bang theory).

Origin of Earth = $\boxed{4.5 \text{ B.Y.A}}$

Origin of Life = $\boxed{4.0 \text{ B.Y.A}}$ [$\boxed{500 \text{ million}}$ after formation of Earth]

- Non cellular
  (3.0 B.Y.A)
  - Giant Molecules
    - R.NA → DNA
    - Protein
    - Polysaccharides.
      etc
- Cellular
  [2.0 B.Y.A]
  - Unicellular
    ↓
  - Multicellular

Theories to Explain Origin of Life

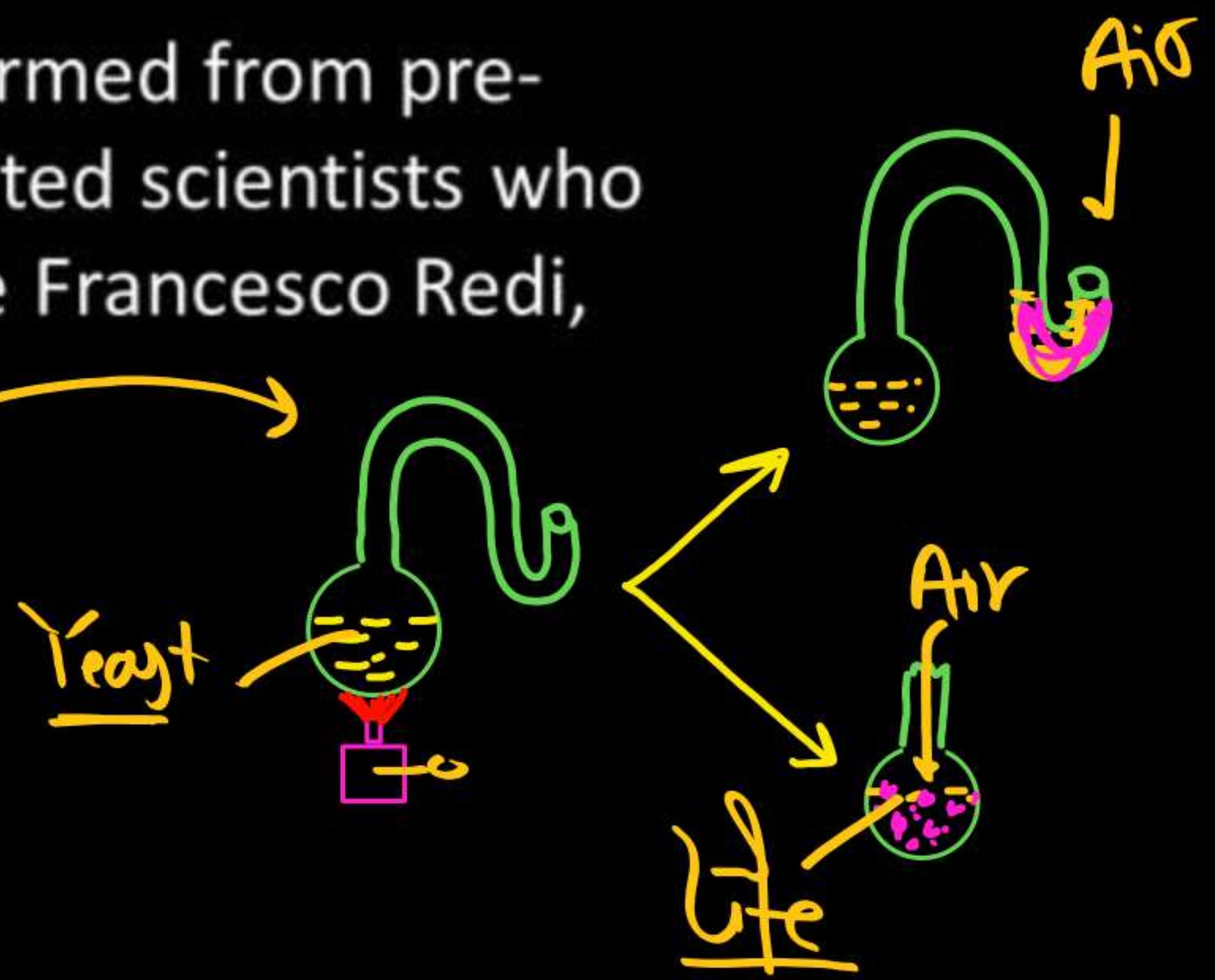
. Theory of Biogenesis.

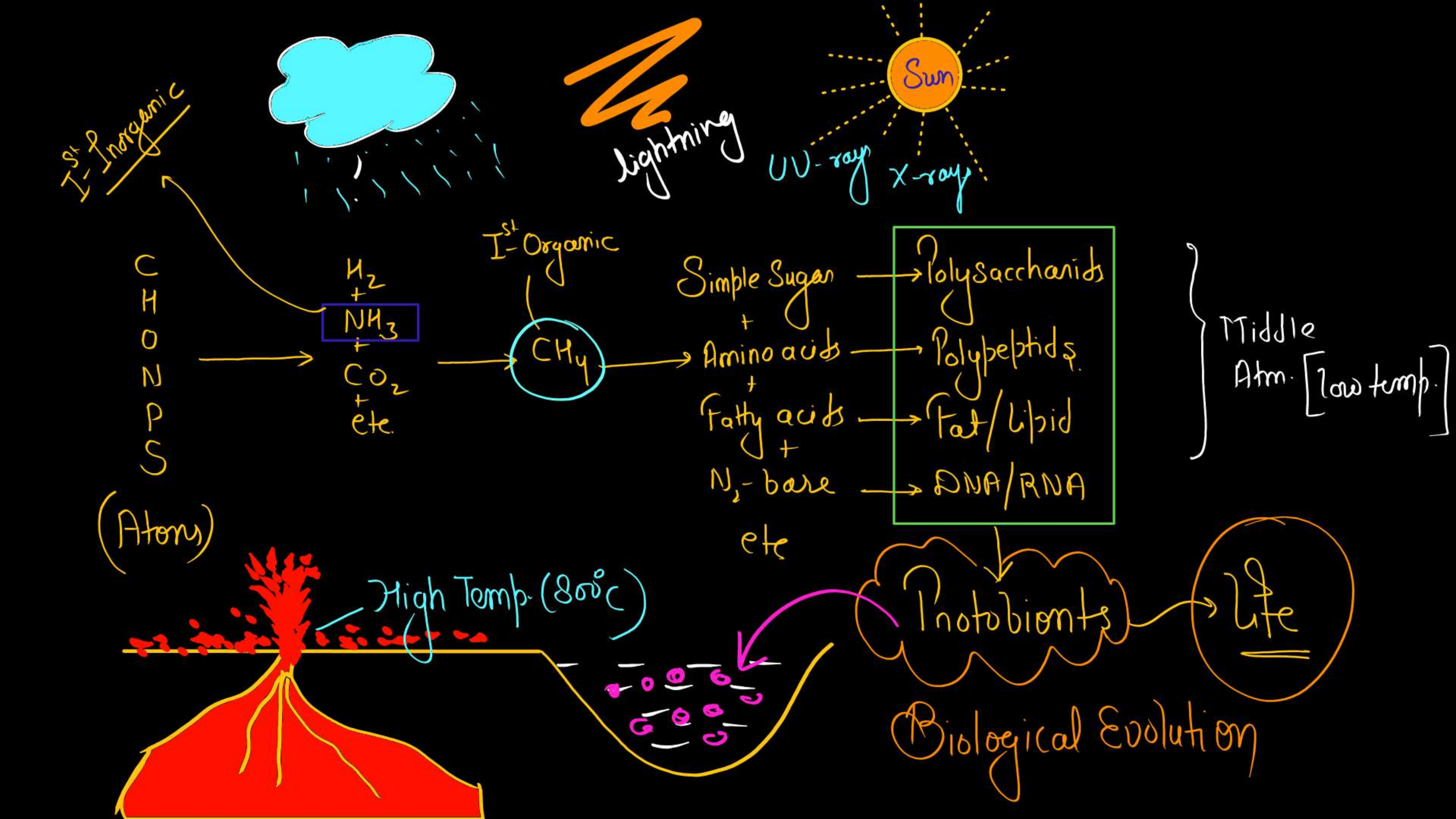
The theory of spontaneous generation was disproved by many scientists.

They proved that new organisms can be formed from pre-existing ones, i.e., omnis vivum ex vivo. Noted scientists who experimentally challenged the theory were Francesco Redi, Spallanzani and Louis Pasteur.

. Oparin-Haldane Theory. Imp

Pri. abiogenasis
or
Chemical Evolution





I^{st}- Inorganic
lightning
Sun
UV- ray
X-ray
C H O N P S
(Atoms)
H_2 + NH_3 + CO_2 + etc.
I^{st}- Organic
CH_4
Simple Sugar + Amino acids + Fatty acids + N_2- base
etc
Polysaccharids
Polypeptids
Fat/Lipid
DNA/RNA
Middle Atm. [low temp.]
High Temp. (800°C)
Protobionts
Life
Biological Evolution

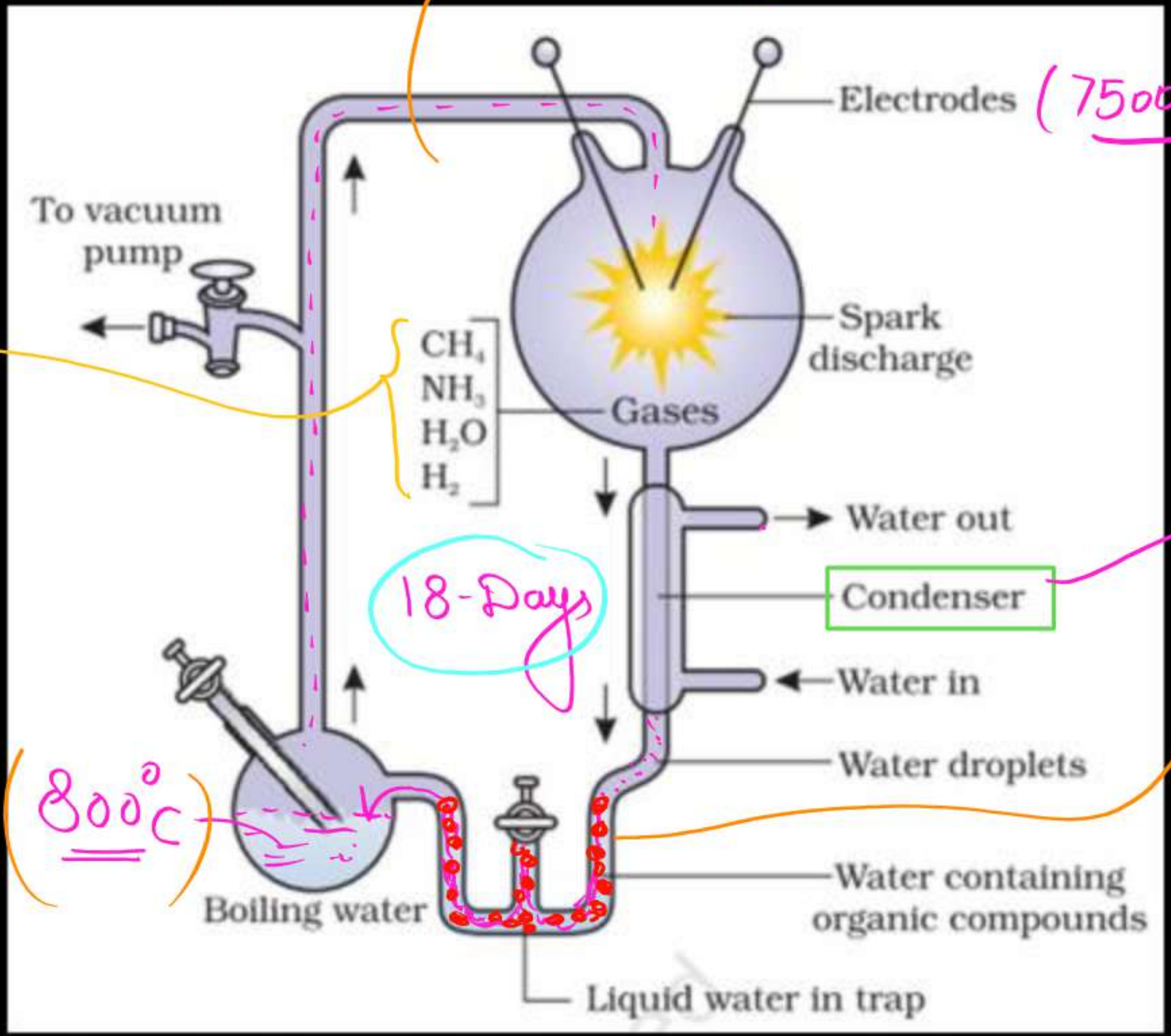
Miller's Experiment- (1953)

Devoid of Radiation (UV/X-Rays)

Gases

$CH_4 : NH_3 : H_2$

$2 : 1 : 2$



Low Temp. (Middle Atm)

[Amino Acids]

Glycine
+
Alanine
+
Aspartic acid

(1) Palaeontological Evidences (Evidences from Fossil Record)

Fossils - Fossils can be defined as the remains or traces of the non-degraded parts of ancient living things found within rock life preserved in the Earth's crust.

→ Sedimentory rocks

Age of Fossils

Radioactive dating Tech. →

R.A. 1000 gm → R.A. 500 → RA 250, NRA 250

R.A. 1000 gm → NRA 500

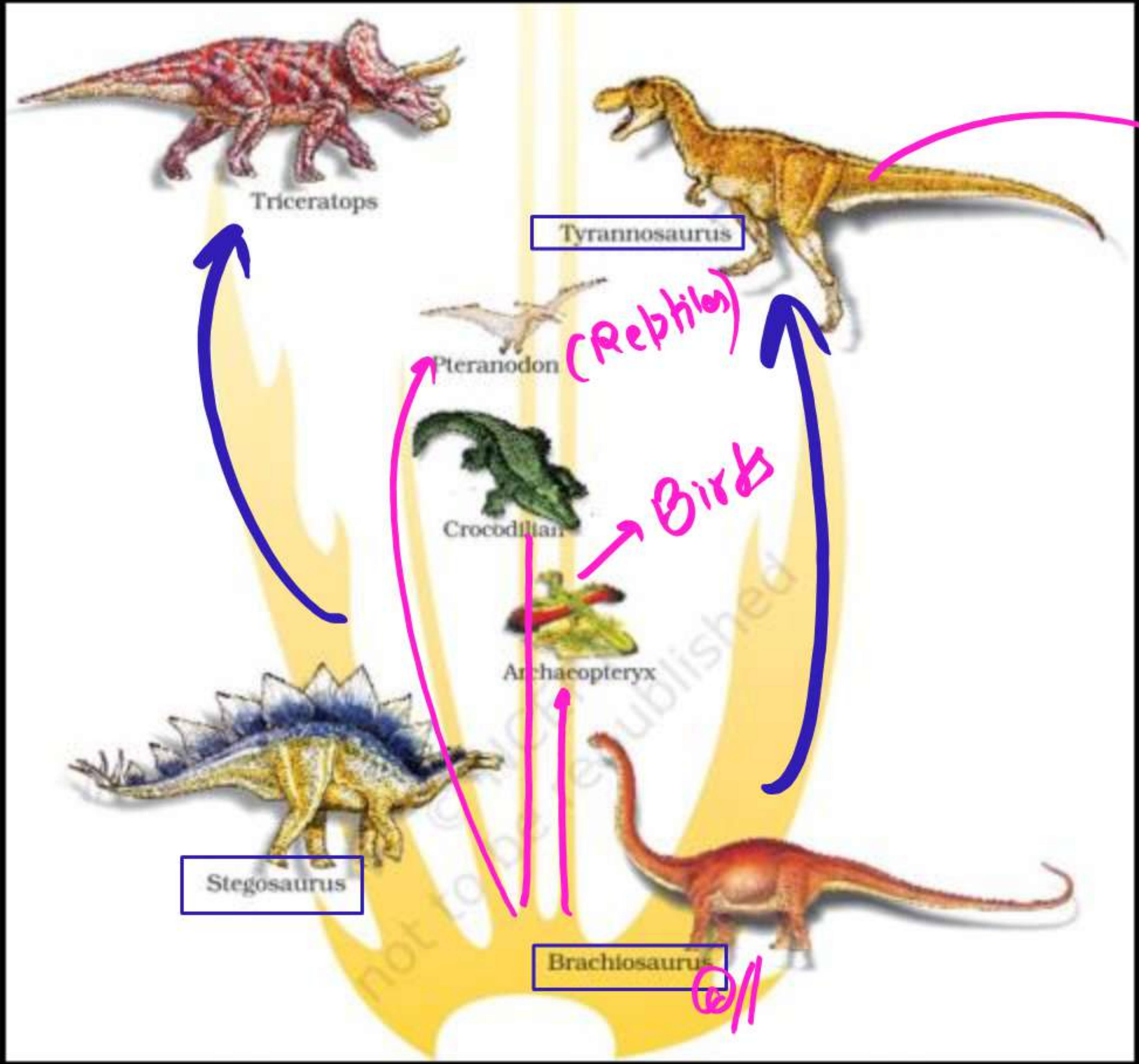
1 ½ life

2nd ½ life

Ratio of RA : NRA in —

1 ½ life = 1 : 1

2 ½ life = 1 : 3


Triceratops
Tyrannosaurus
70 feet
Pteranodon
(Reptiles)
Crocodilian
Birds
Archaeopteryx
Stegosaurus
Brachiosaurus
not to be republished

(2) Evidences from Comparative Anatomy and Morphology

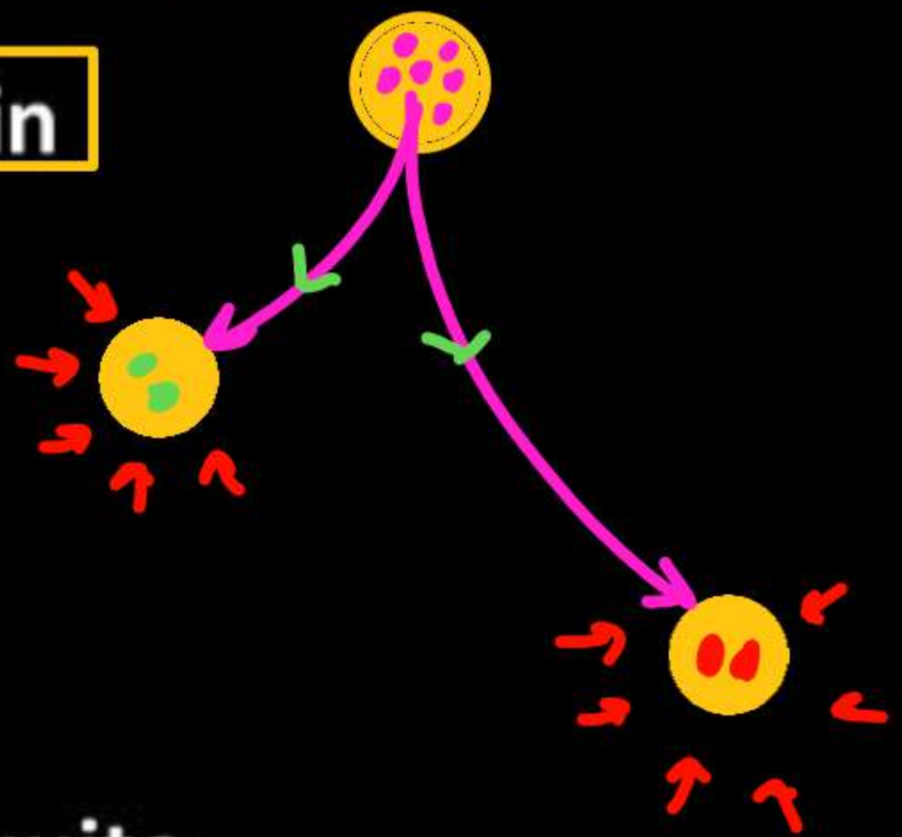
1 Q-100%

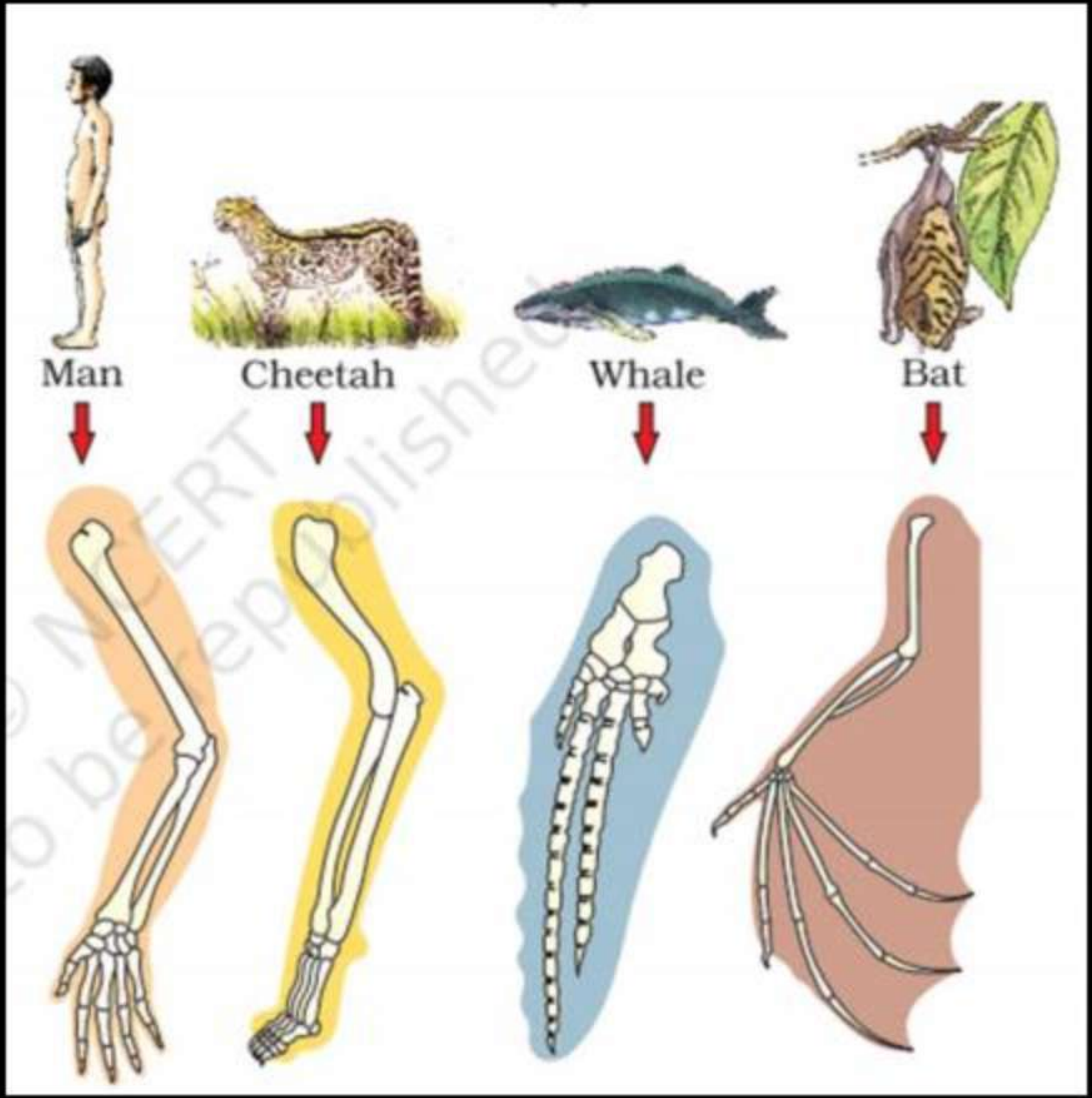
ReNEET 2026

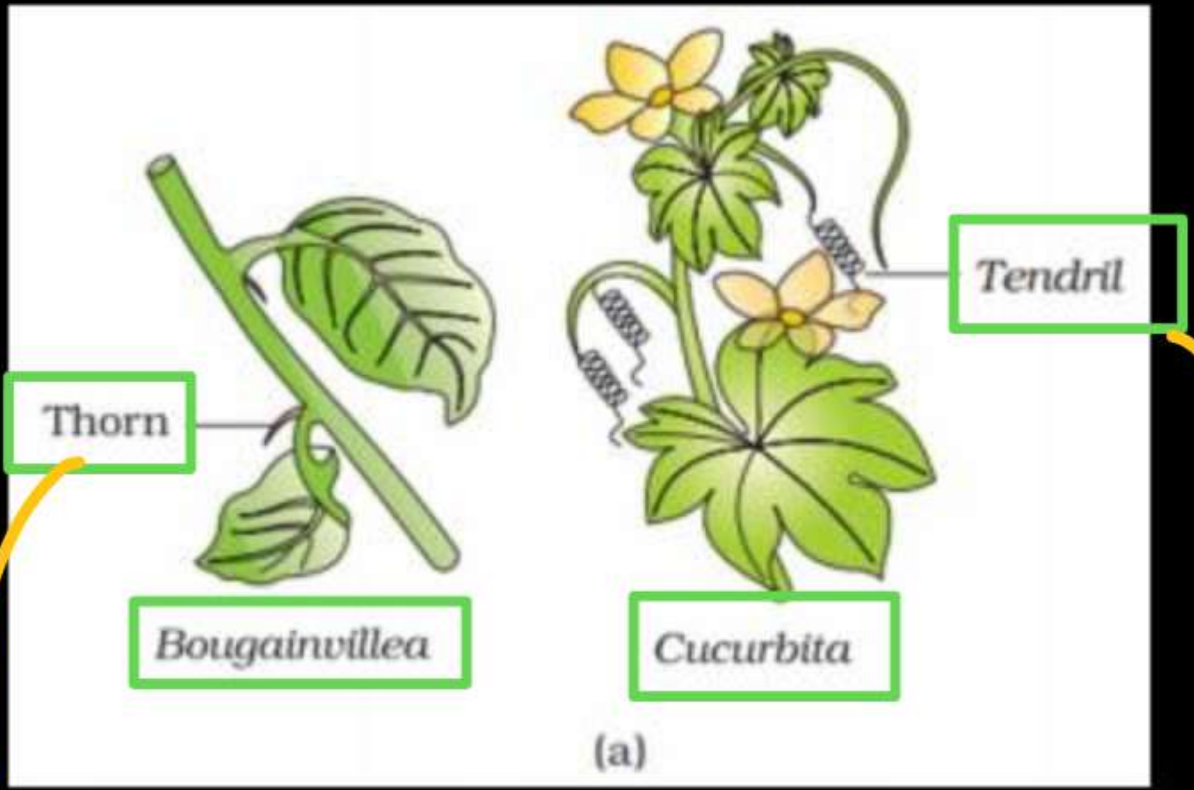
Homologous organs-

- Similar fundamental structure but are different in functions
- Divergent evolution
- Common ancestry
- Eg.-

1. Forelimbs of man, cheetah, whale and bat.
2. The mouth parts of cockroach, honey bee, mosquito and butterfly.
3. Thorn and tendrils of *Bougainvillea* and *Cucurbita* represent homology







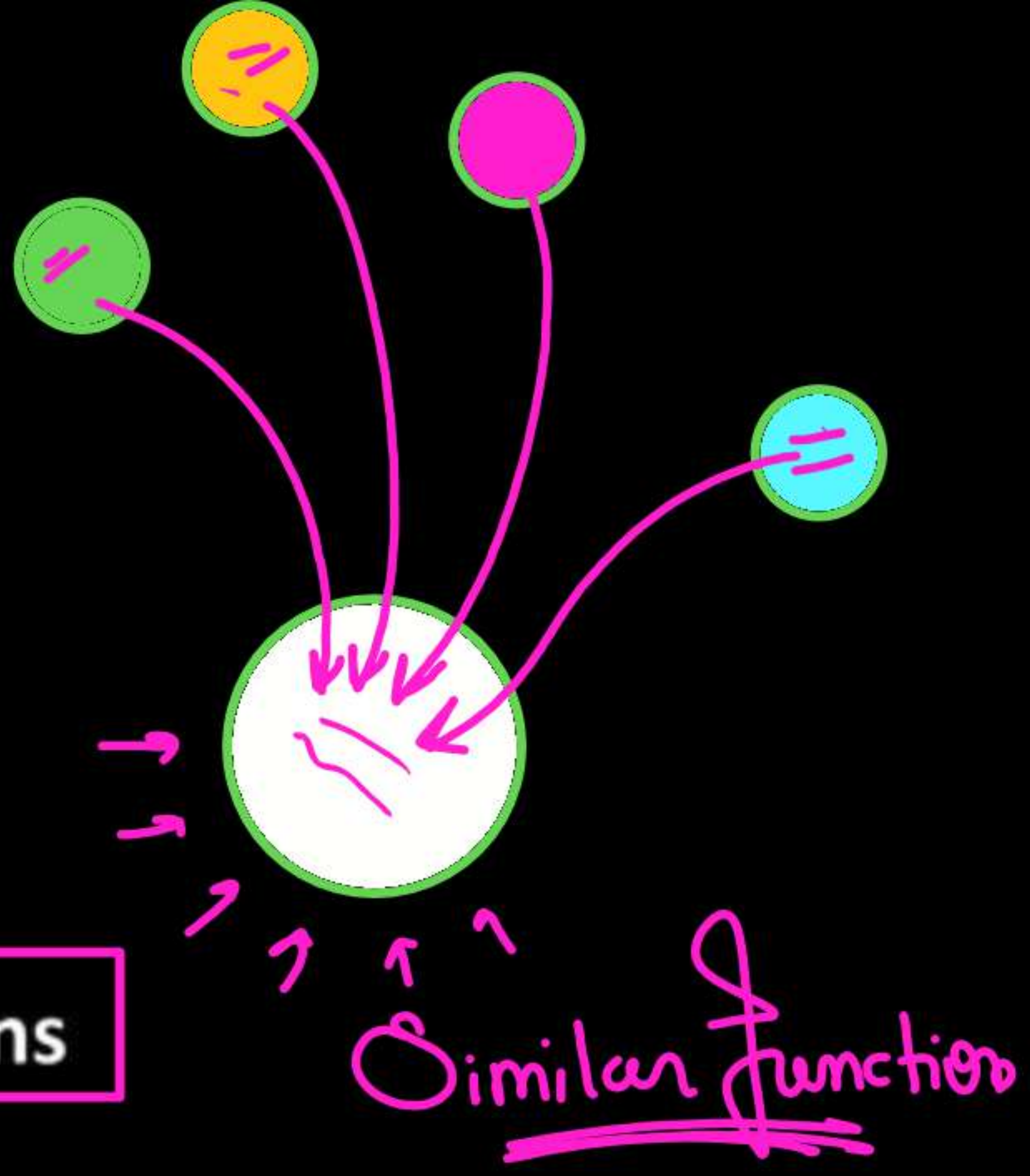
Protection

Support

Analogous Organs

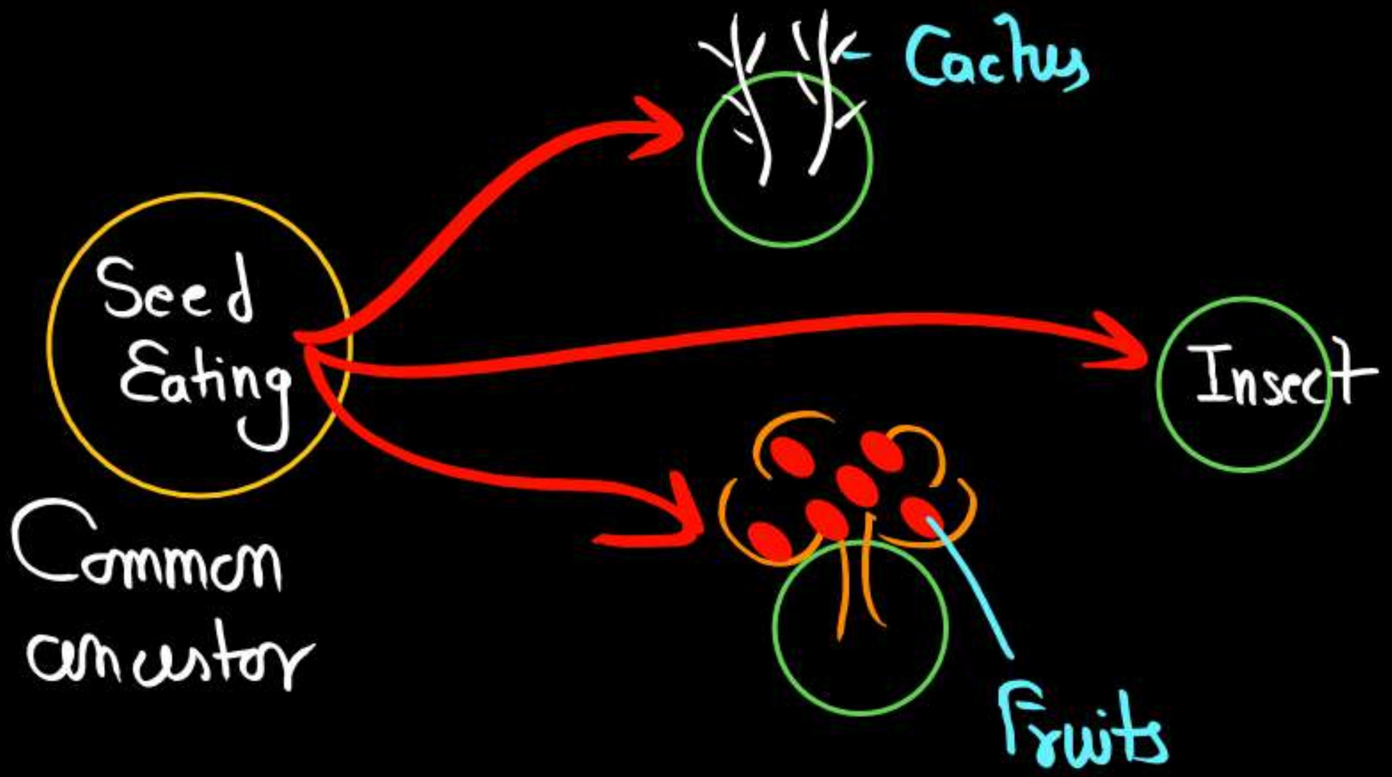
- Different their anatomical structural but
- Similar in functions due to same needs.
- Result of convergent evolution
- Eg.-

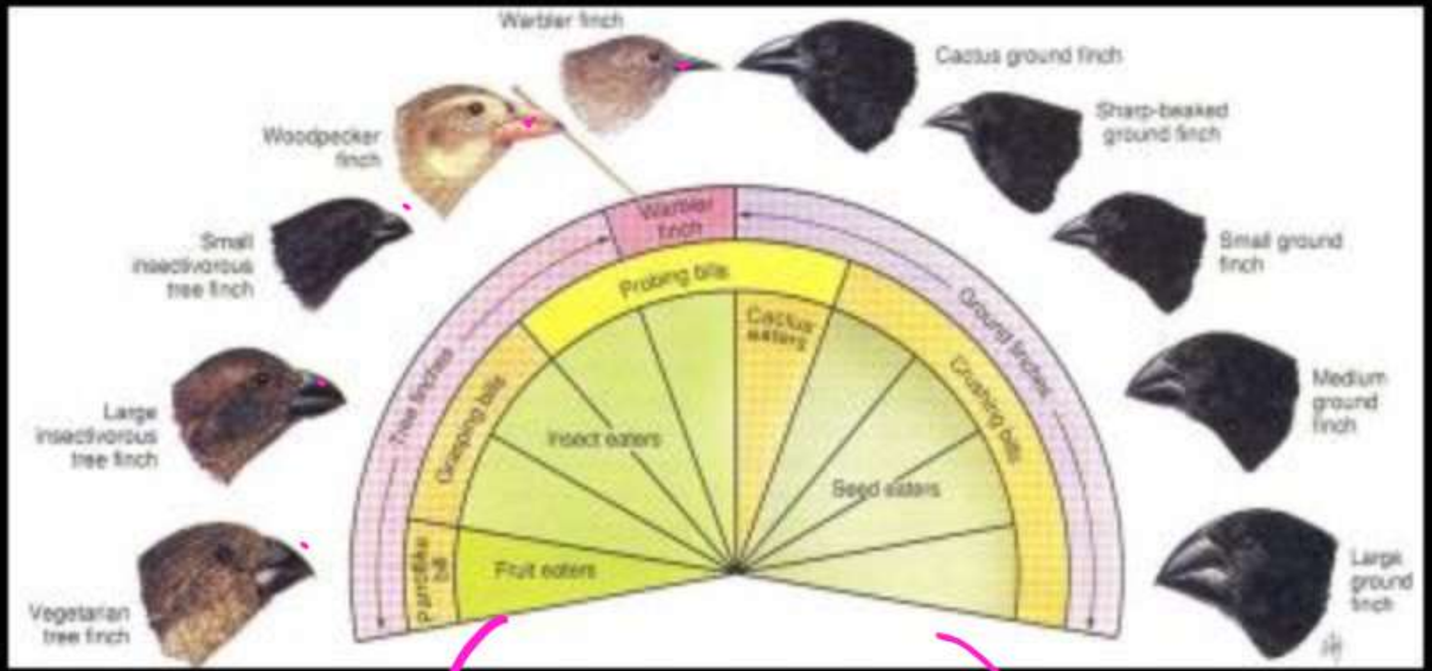
1. Wings of butterfly and of birds
2. Pectoral Fins of sharks and flippers of Dolphins
3. Stings of honey bee and scorpion
4. Eye of octopus and eye of mammals
5. Sweet potato (root modification) and potato (stem modification)



ADAPTIVE RADIATION –

- During his journey (1831) Darwin went to Galapagos Islands - 22.
- Ship -"H.M.S. Beagle
- Duration of journey 5 years.





13- Types

Australian marsupials -

This process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called adaptive radiation.

Eg. Darwin's finches.
Placental mammals
marsupials of Australia

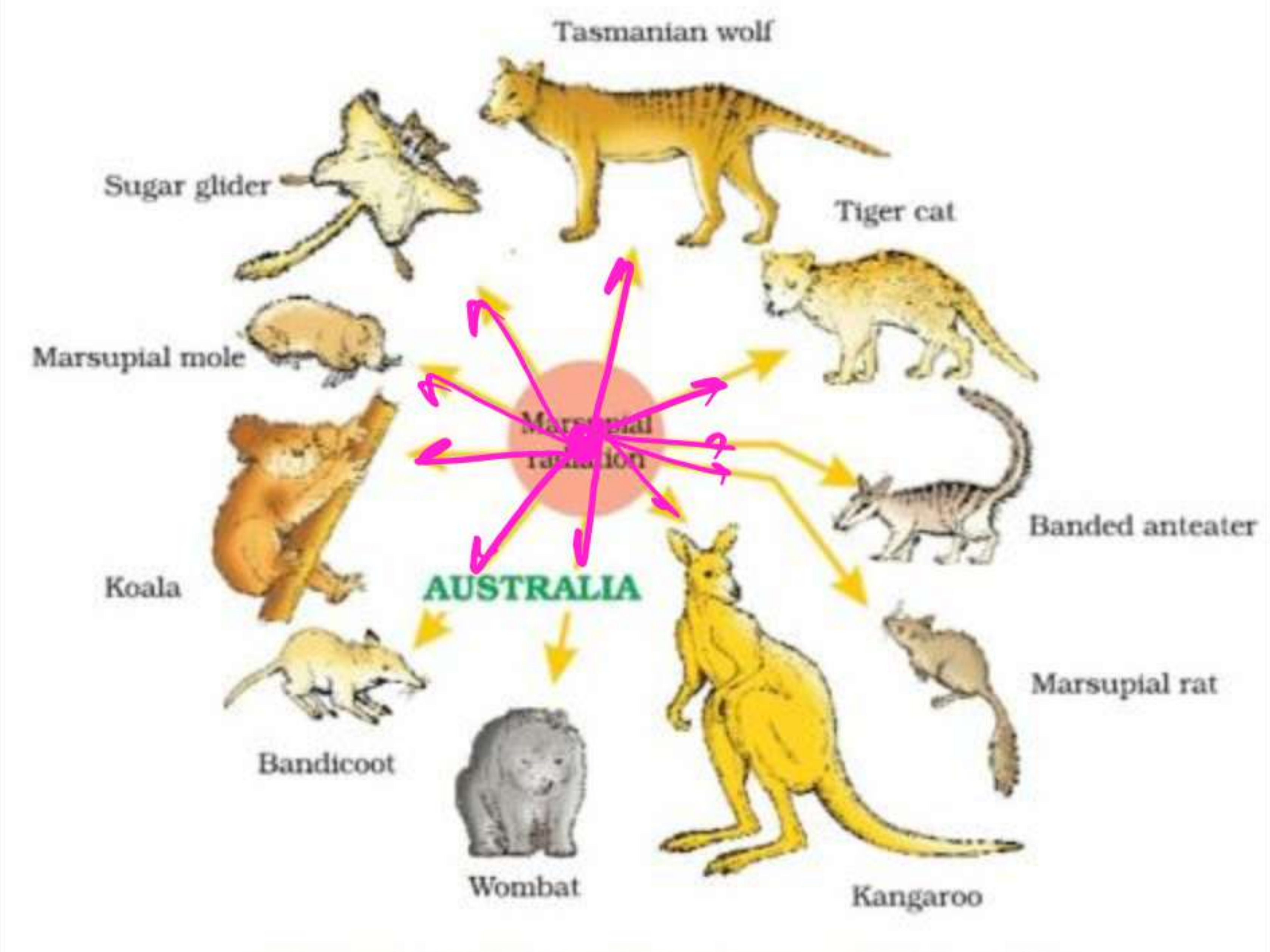
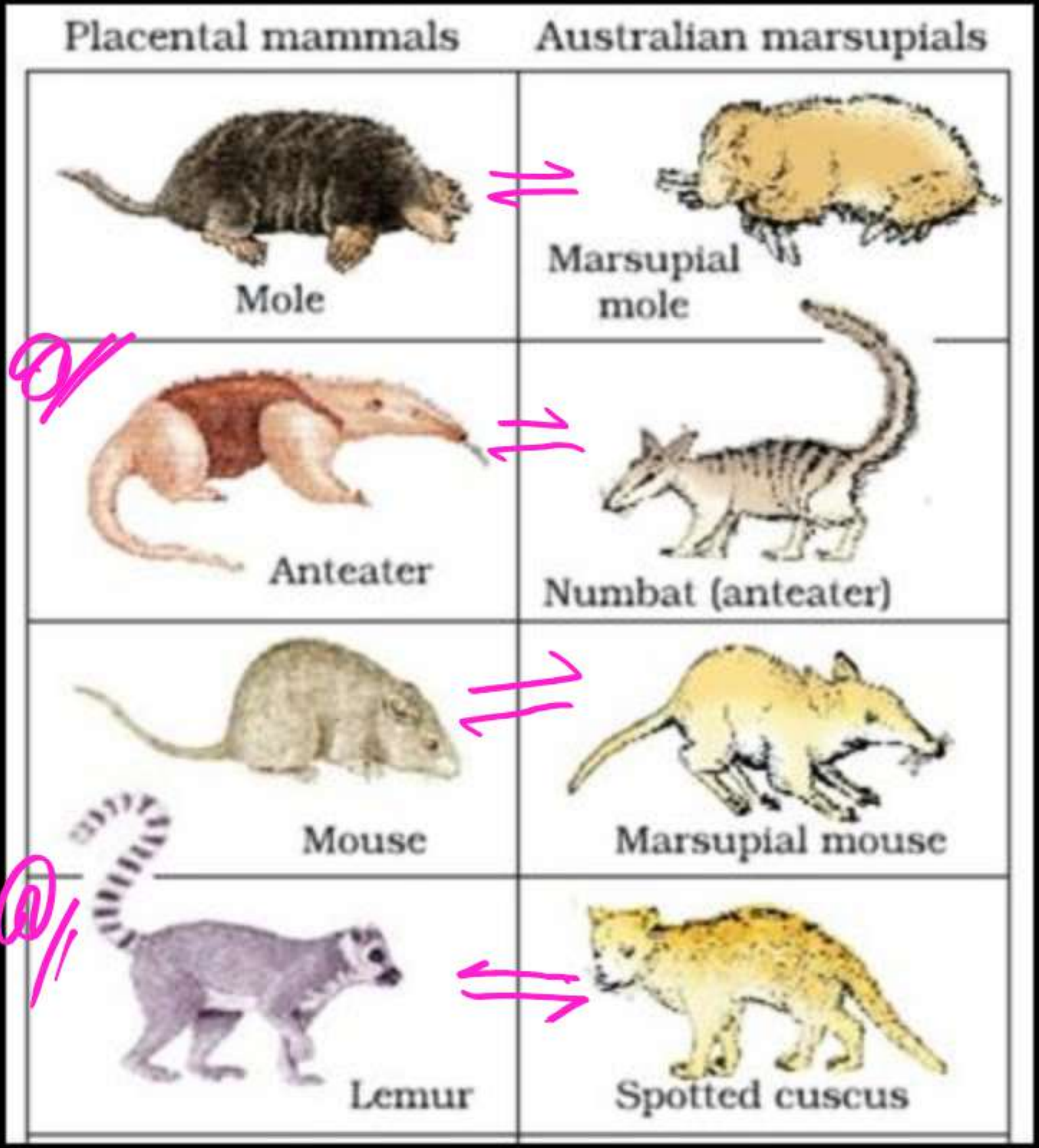


Figure 7.6 Adaptive radiation of marsupials of Australia

- When more than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution.



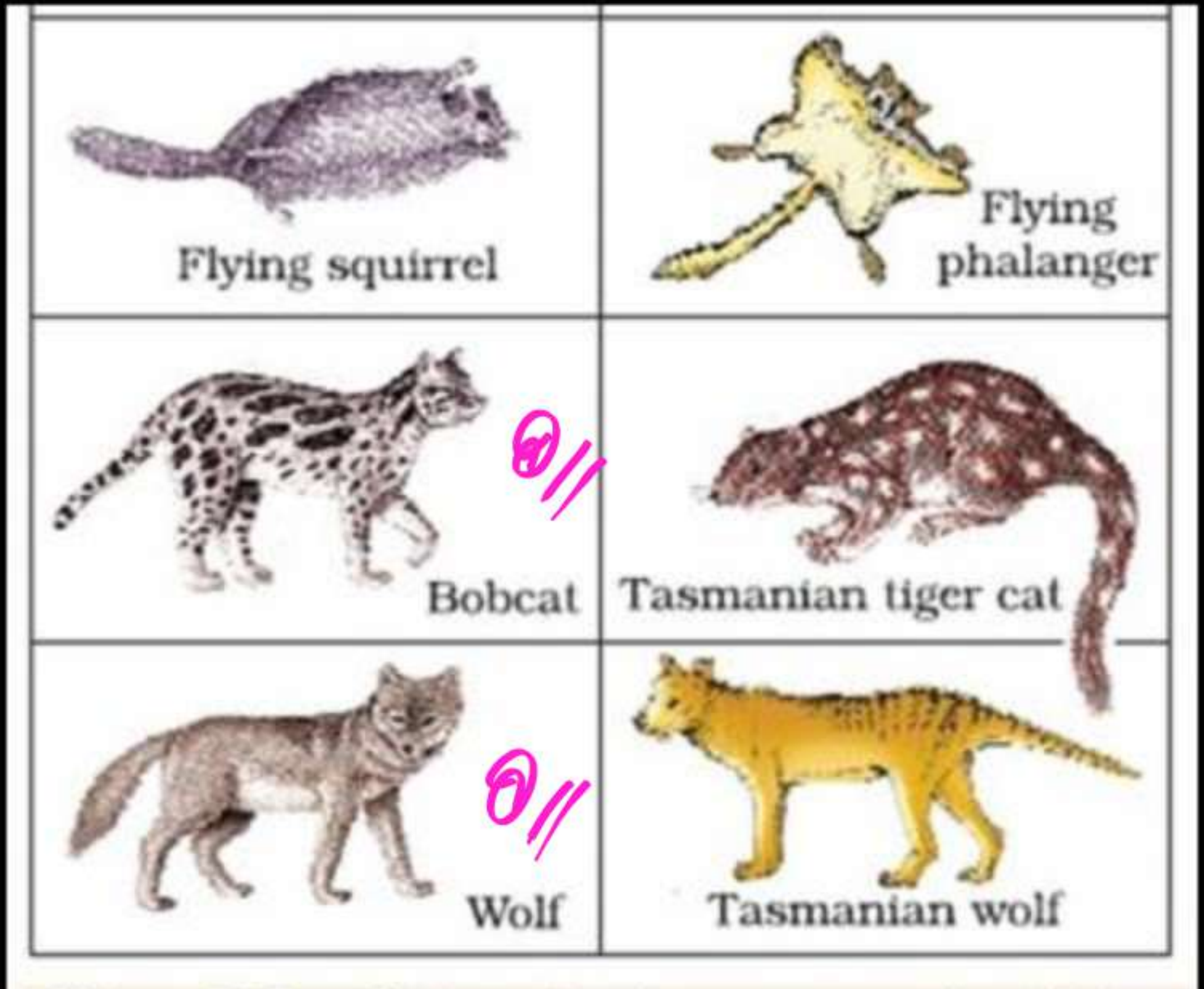
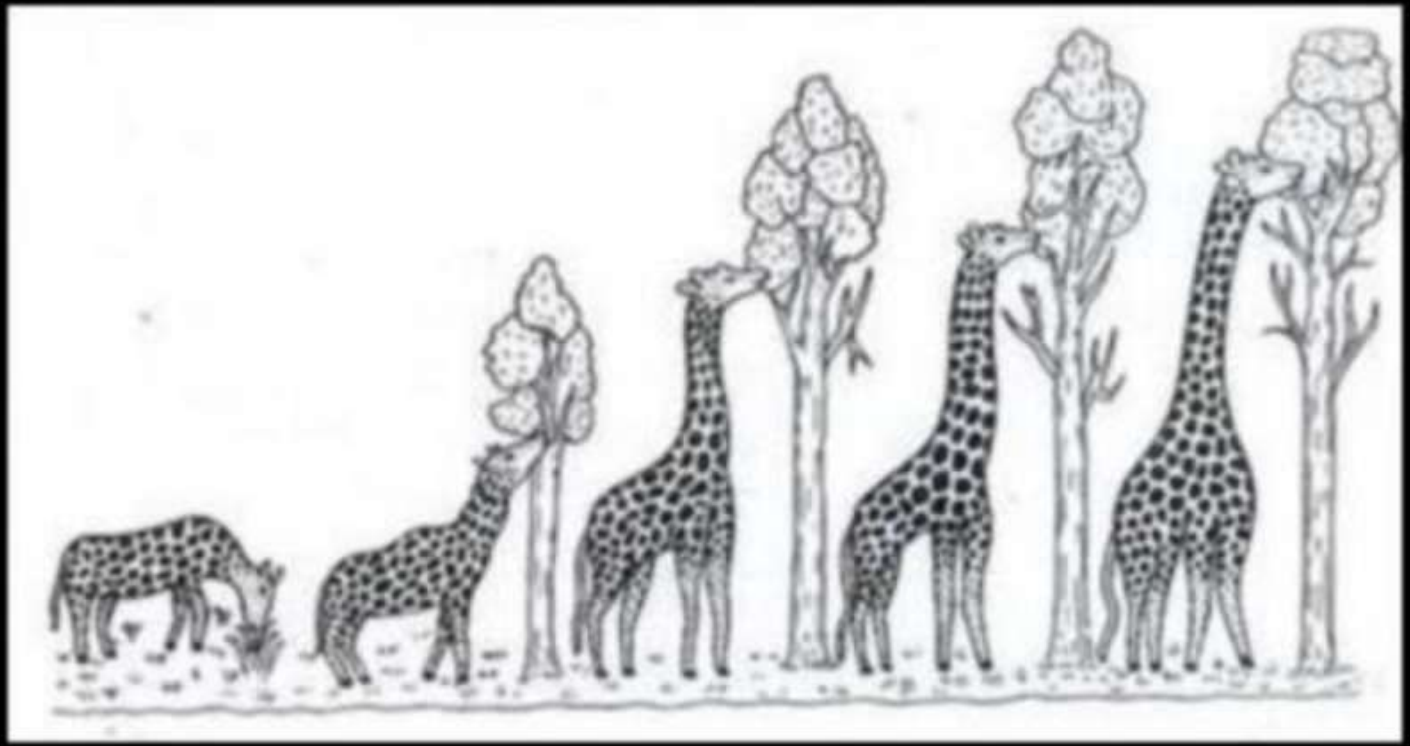


Figure 7.7 Picture showing convergent evolution of Australian Marsupials and placental mammals

Biological Evolution –

(1) Lamarck's Theory of Evolution -



* Theory of use & disuse of Organs

or

Inheritance of Acquired character

(2) Darwinism -

Theory of Natural Selection
or
Origin of Species by N.S. } 1859

Darwin was influenced by =
An Essay on principle of Population
T.R. Malthus

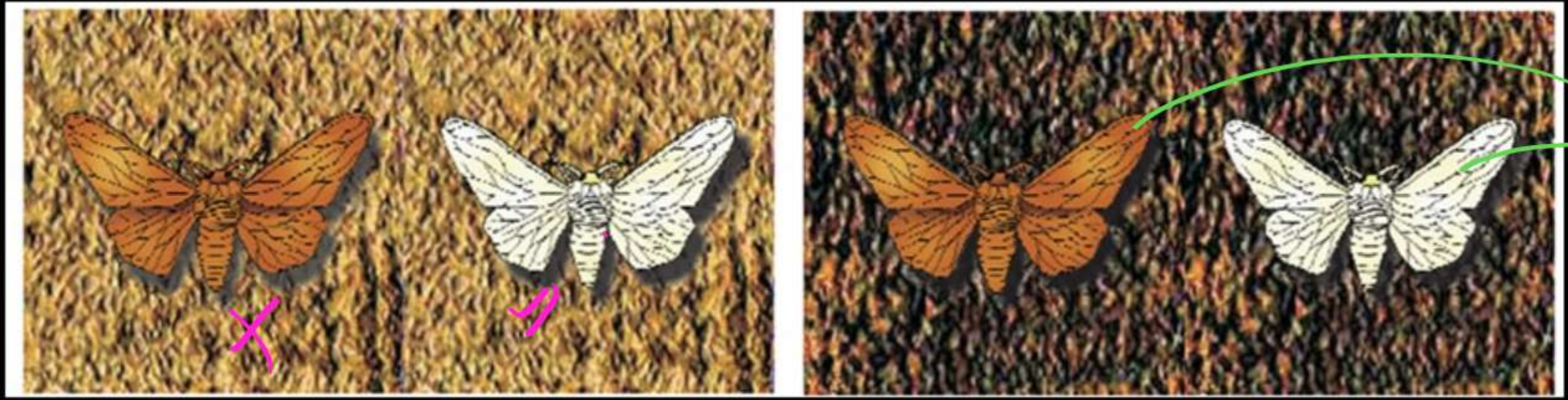
Alfred Wallace (Naturalist) } Malay Archipelago
Similar Conclusion around same time

Main Points of Theory of Natural Selection

1) - High Rate of Reproduction
2) - Total No. almost Cons.
3) - Struggle for existenc (Competition)
4) - Variation (Continuous Variation) → Adaptive ✓ / Non adaptive ✗
5) - Natural Selection = Max^m useful Variation → Selected by Nature
6) - Survival of fittest = End Result of Selection

Two key Concepts of D. Theory
- Branching Descent [Pattern of Evolu.]
- Natural Selec. [Mechanism of Evolution]

Industrial Melanism



Peppered Moth
or
Biston butularia

[England]

[Before Industrilisation]
[1850]
↓
- ↑ No. of white Wings Moth
- ↓ No. " Black " "

[After Industrilization]
[1920]
↓
- ↑ No. of Black Wings Moth
- ↓ No. of white " "

(3) Mutation Theory → Hugo de Vries

└ work on - Evening primrose (Oenothera lamarckiana)

According to this theory →

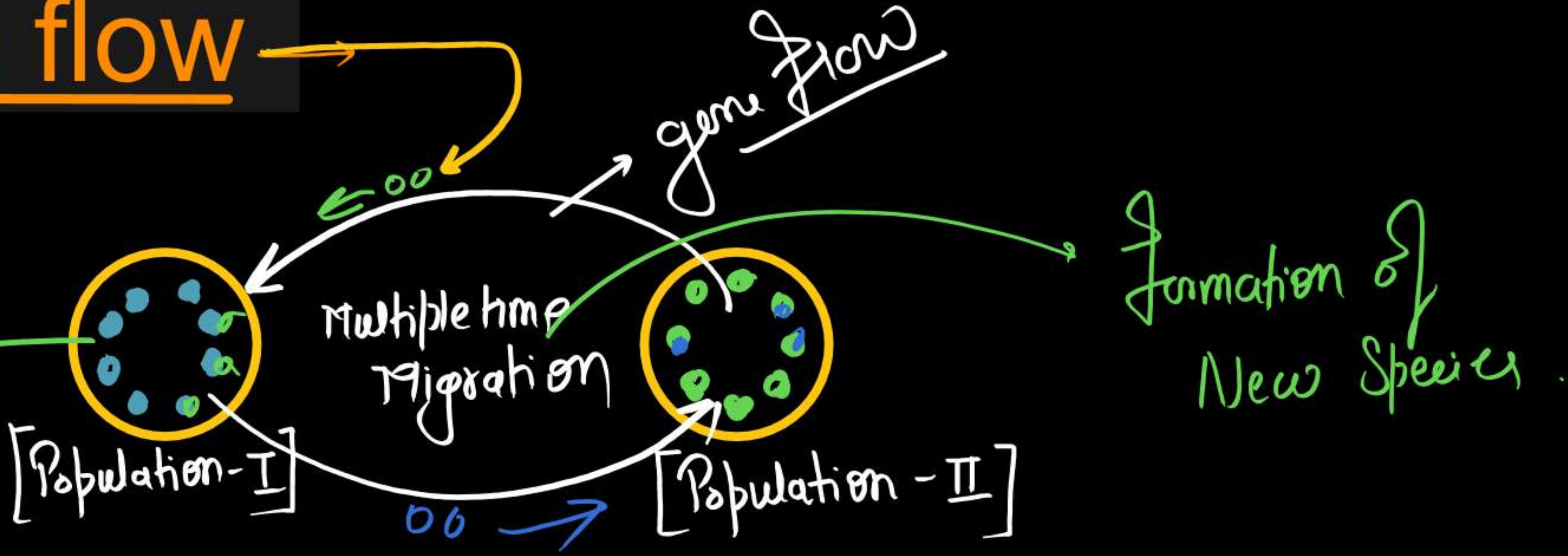
* Mutation are responsible for Evolution
* Mutation provide raw material for evolution
* Variation are large, Discontinuous & Nondirectional, Randum
* Evolution is Sudden & jerky Process. → Called - Saltation (Single Step Large Mutation)

MODERN/SYNTHETIC THEORY

- Many factors are collectively responsible for formation of new species :⟶

1) - Gene migration/gene flow
2) - Genetic Drift
3) - Mutation ✓
4) - Genetic Recombination ✓
5) - Natural Selection ✓

(1) Gene flow



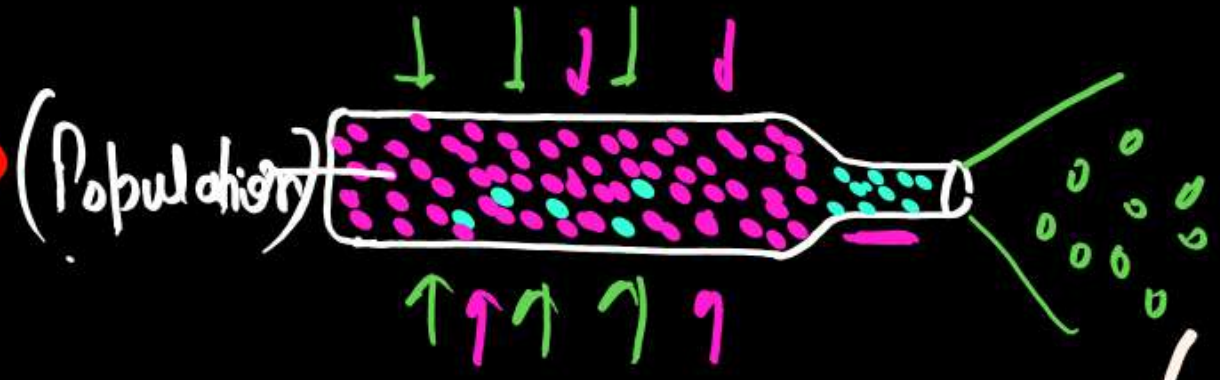
Gene pool

[Collection of all gene and its alleles in a population]

(2) Genetic Drift → Bottle Neck Effect

v. imp

Small Population



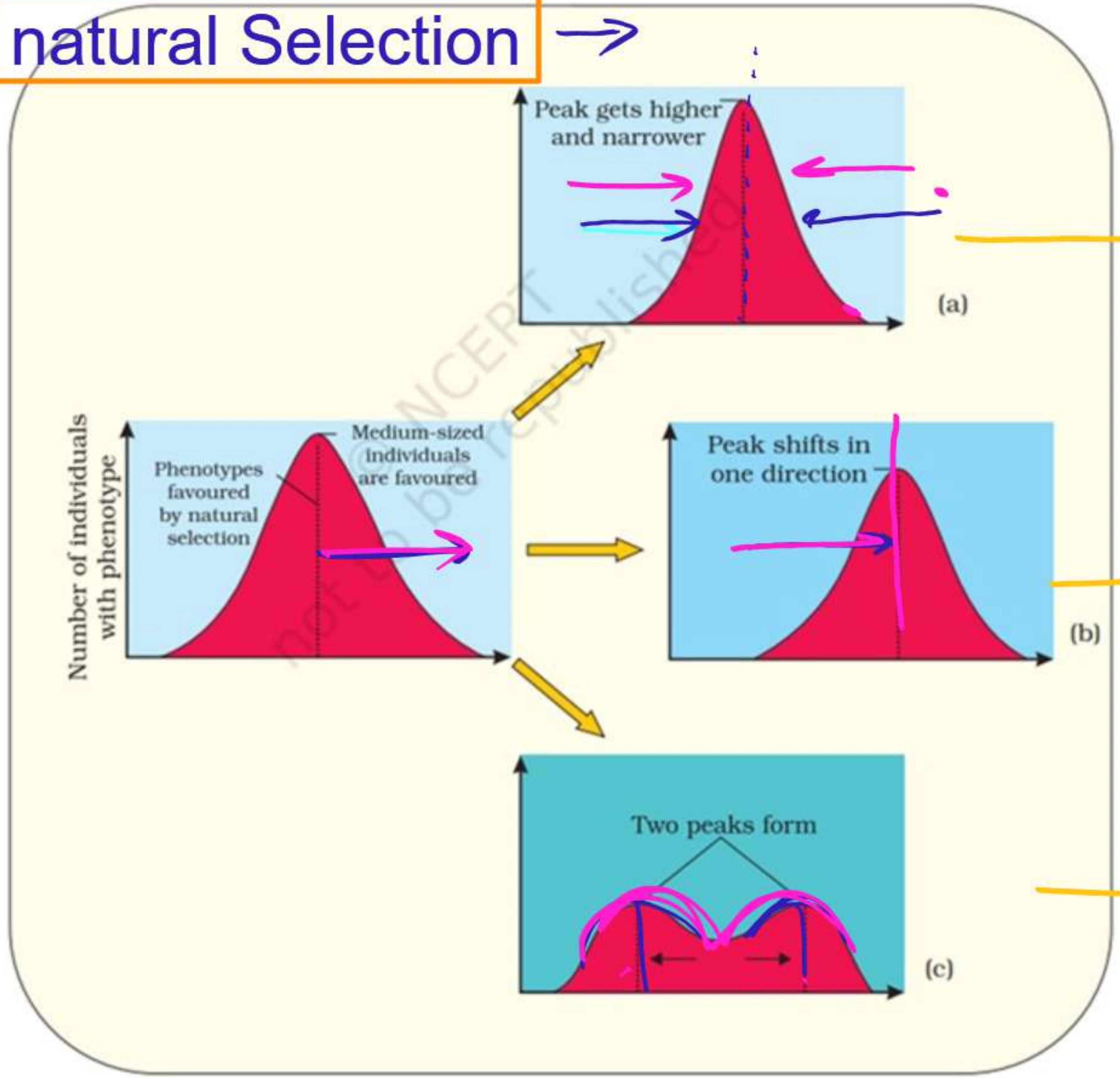
- Randum / Nondirectional changes of gene frequncies in small Population by chance.
- May - ↑ or ↓ Due to (Size of Population)
  - → Geographical isolation
  - → Migration by chance
  - → Natural Calamity.

Founder effect

New Species.

natural Selection →

Types of Natural Selection



7.8 Diagrammatic representation of the operation of natural selection on different traits : (a) Stabilising (b) Directional and (c) Disruptive

Stabilising Selection

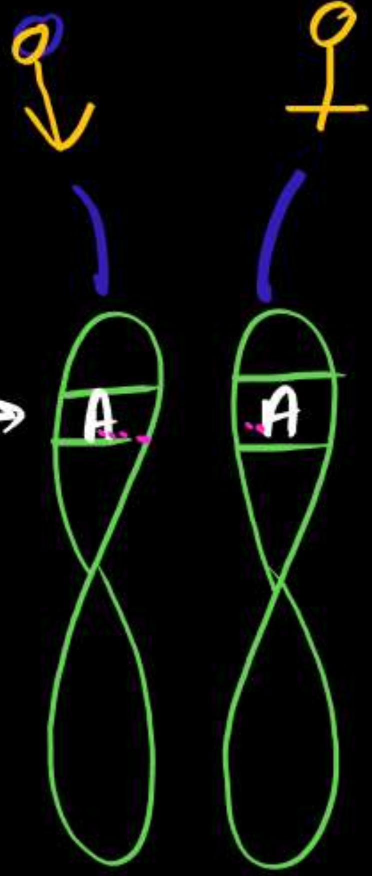
Directional Selection

Disruptive Selection

HARDY-WEINBERG PRINCIPLE -

Population (Large)

Frequency of occurrence of allels of gene or Locus or Individual



→ AA
→ aa
→ Aa

A = P
a = q

Gene and allilic frequency in a large population remains constant if —
- Randum Mating
- Absence of 5-Factors

1)- Mutation 2) Gene migration, 3) Recombination
4)- Genetic Drift 5) Natural Selection

- $p^2 + q^2 + 2pq = 1$

or

$(p+q)^2 = 1$

or

$(p+q=1)$

A Brief account of Evolution →

Ist acellular life = 3 BYA

Ist cellular life = 2 BYA

Invertebrate become active = 500 MYA

Ist Vertebrate (jaw less fish) = 350 M.Y.A

Ist jawed Vertebrate = 350 M.Y.A.

eg = $\boxed{\text{Latemaria / Coelacanth / Lobe fin fish}}$

→ Caugh in South africa (1938)

→ Living fossil

→ Though to be extinct

Ist Amphibia — Extinct

— Invade land after Plant

→ Frog & Salamander

Early Reptile — Extinct

- Dinosours — Domenant Land Reptiles in jurassic Period ⇒ 150 mya
  - Bigest = Tyrannosaurus rex [20-feet]
  - Some back in water ⇒ 200 mya ⇒ Icthyosaurs
  - about 65-mya = Dinosours Disappered from Earth
- Turtle
- Lizard
- Snakes
- Tuatara
- Crocodile

Survived

Birds → Mordern Birds

Mammals — After jurassic
- Placental
- Marsupial

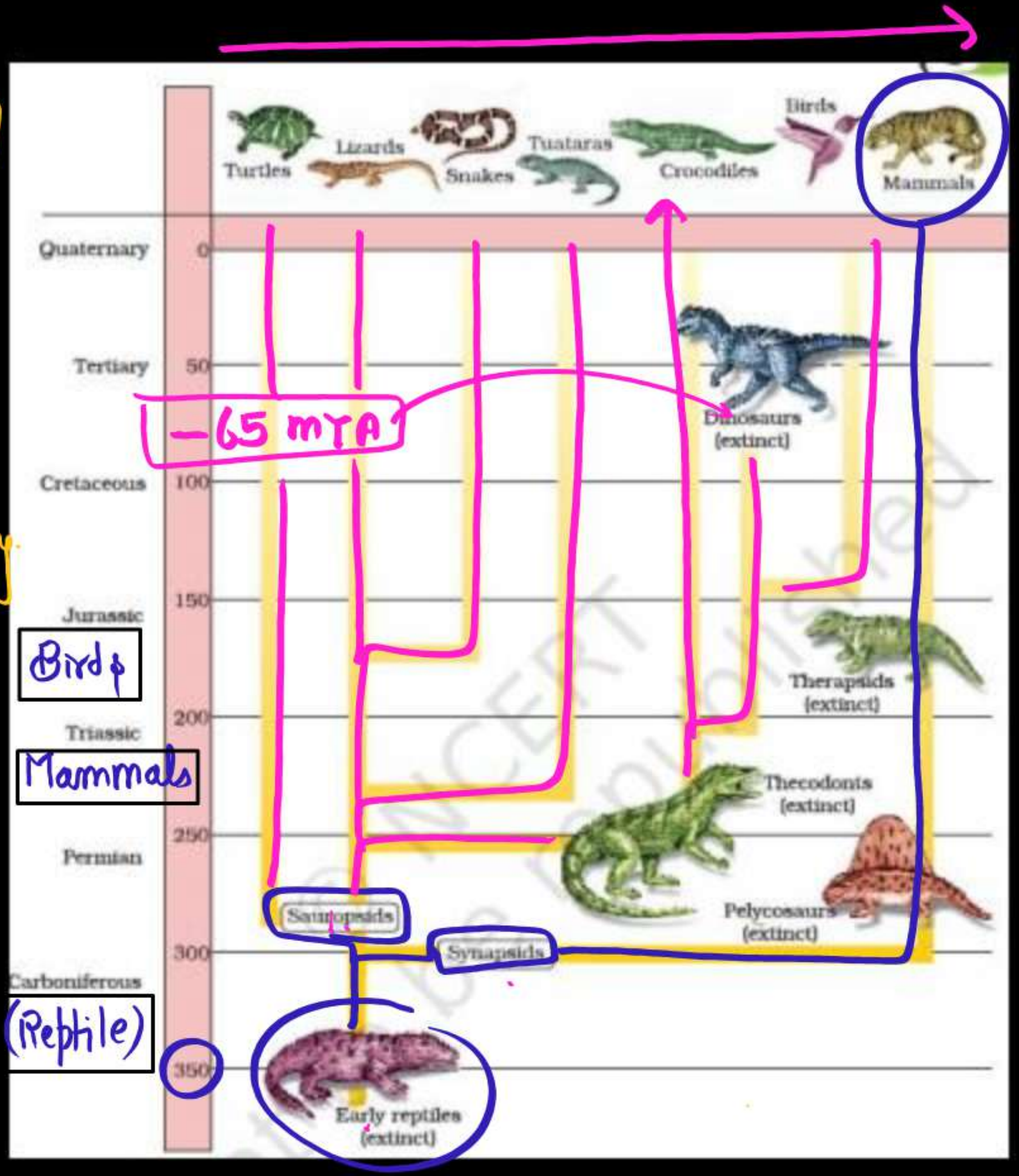
Flowering Plant } Angiosperm

} In Cretaceous Period

ERA		Period	
Cenozoic	Cenozoic	Quaternary Tertiary	Age of Angiosperm
		Bry	
Mesozoic	Mesozoic	Cretaceous Jurassic Triassic	Origin of - Angiosperm Age of - Bry/Pteri/Gy.
Paleozoic	Paleozoic	Permian Carboniferou Devonian Silurian	Origin of - Bryophyte Pteridophyte Gymnosperm

Devonian → Amphibia

Silurian → Fish



Primates → Monkeys

Primates → Apes → [Gibbon → Orangutan → Gorilla → Chimpanzee]

Chimpanzee — Closed to Human

Primates ↓ **Human**

Dryopithecus = Apelike }

Rama. Shiva pithecus = Manlike } 15 mya = 3-4 mya ⇒ Ethiopia & Tanzania

4 feet - Hight

Australopithecus $\boxed{400-550\ cc}$ 2.5 mya (Ape-Man)

Pre historic Man → Homohabilis $\boxed{650-800\ cc}$ 2 mya → Skillful, Not eat meat

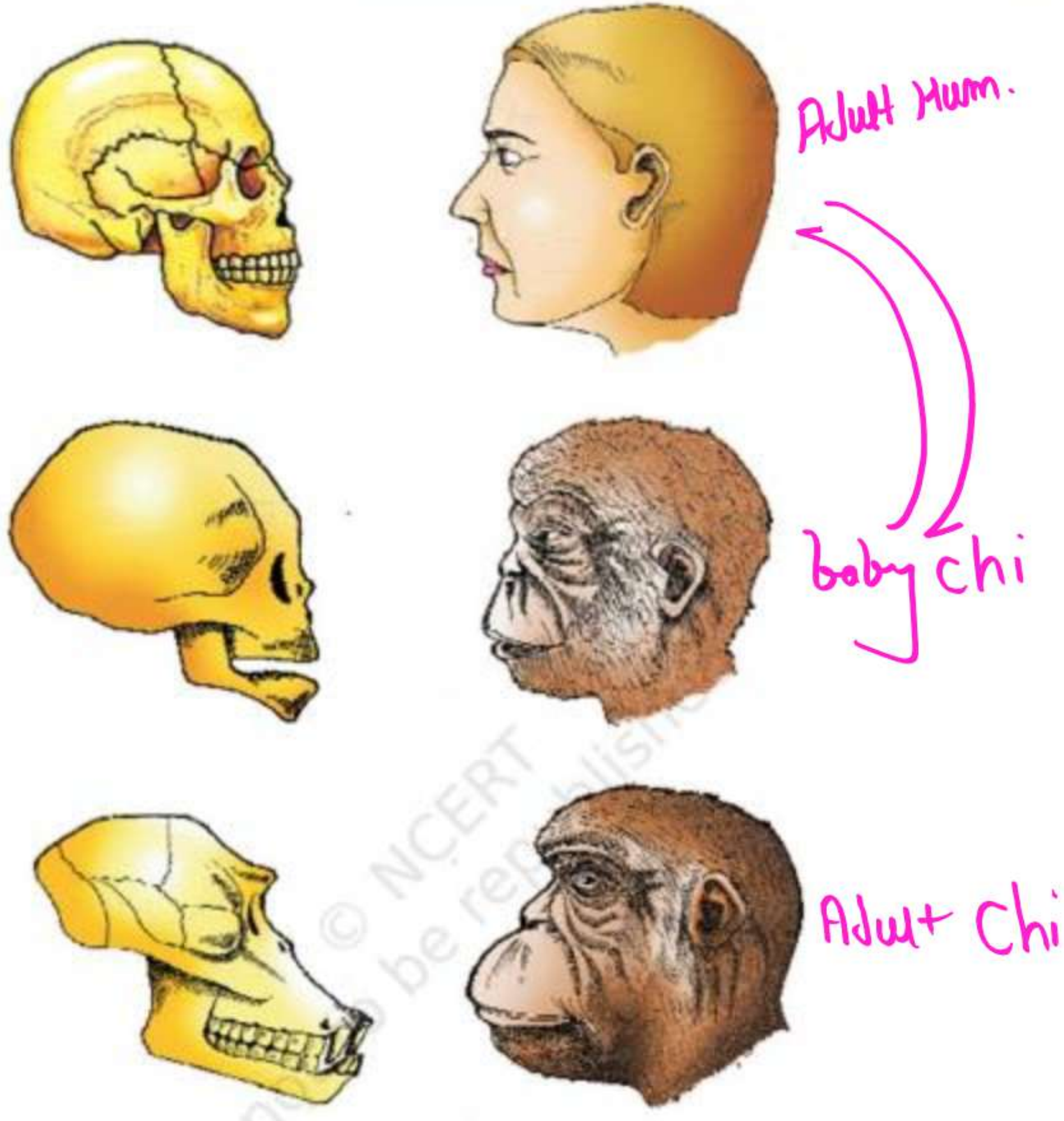
→ Homoerectus $\boxed{900\ cc}$ = 1.5 mya → Jawa (1891), Fire, Eat Meat

Modern Man → Neanderthal $\boxed{1400\ cc}$ = 1,00000 - 40000 YB = Hides, buried their dead, Speach center

→ Cromagnon $\boxed{1650\ cc}$ = 75000 - 10000 YB [ice age] = 18000 YB, Cave Panting [tnt in M.P]

→ Homosapiens - Sapiens $\boxed{1350\ cc}$ = 10000 - Y.B = Agricul.

Modern Man ⇓ Parallel Evolution of H. Brain & Language



A comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee. The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull

Manish Dubey BMS (OFFICIAL)

WhatsApp channel

Scan this QR code using the camera to view or follow this channel.

Old Biotechnology –

techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.

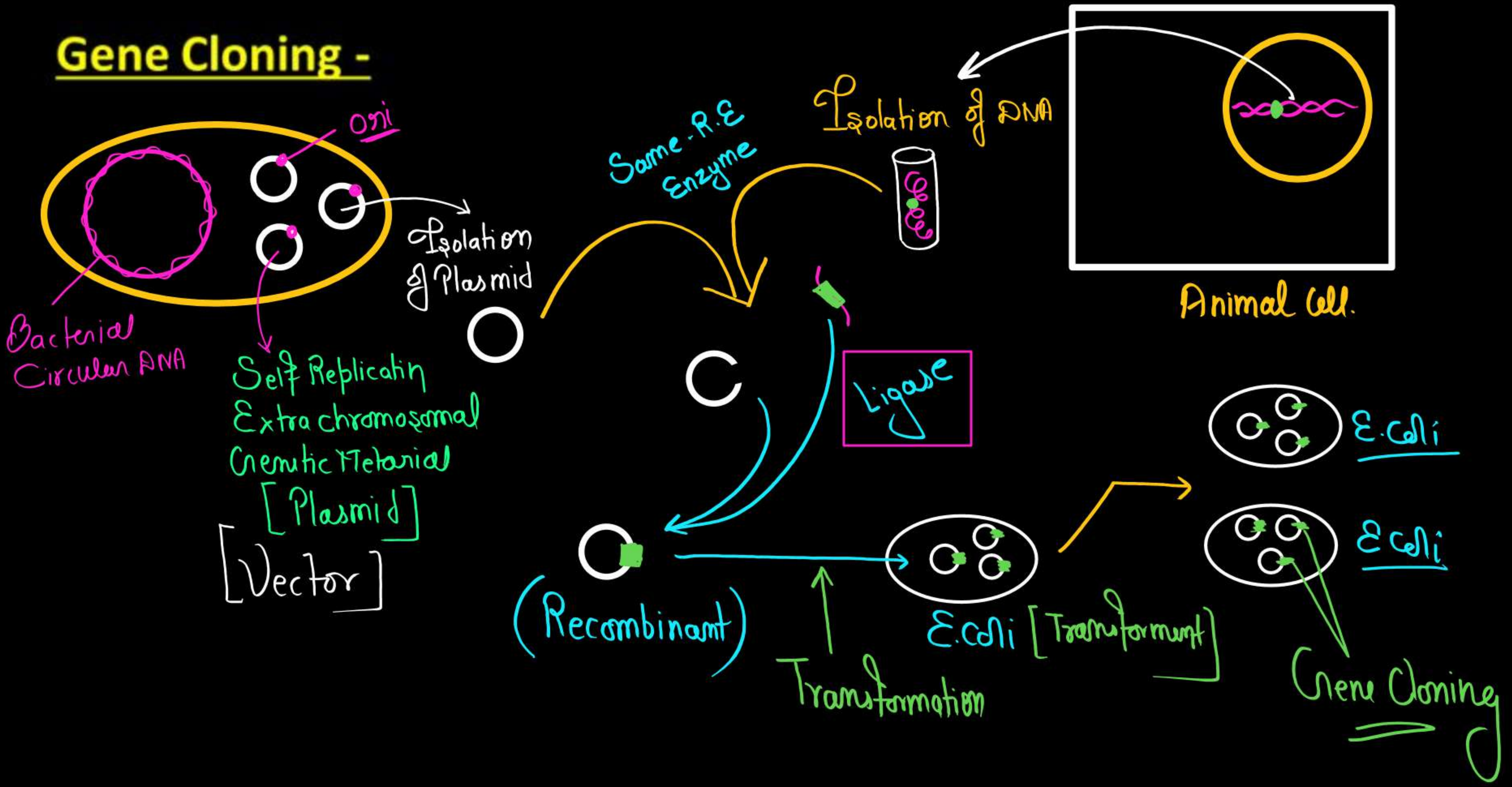
Today Biotechnology -

in vitro fertilization leading to a ‘test-tube’ baby, synthesizing a gene and using it, developing a DNA vaccine or correcting a defective gene, are all part of biotechnology.

According to The European Federation of Biotechnology (EFB)

‘The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services’.

Gene Cloning -



Herbert Boyer vimp

↓

Studies on a Couple of Restriction Enzymes

↓

Cut the DNA in a perticular fashion

↓

Sticky ends → Pasting togeter

Stanley Cohen vimp

↓

Work on Plasmid

↓

Developed a Mathod of Removing & Reinserting the plasmid in Bacteria

Constriction of Ist - Artificial Recombinant DNA in 1972

1972 → Native Plasmid isolate from Salmoneolla typhimurium + Antibiotic Resistant gene

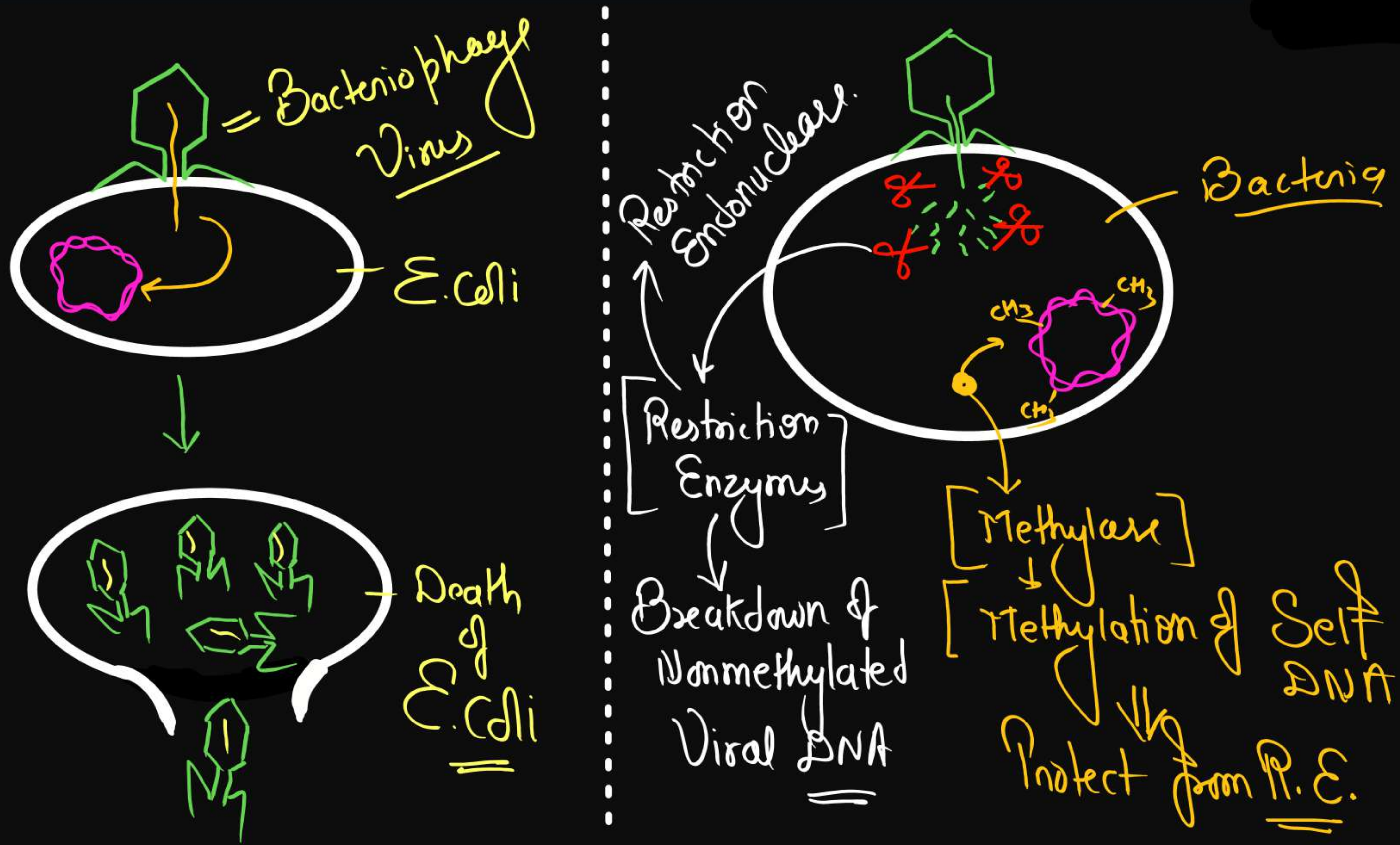
Ist Recombinant

1963 → Isolation of Two Enzyme
- Endonuclease
- Methylase

Restricting the growth of Bacteriophage in E. Coli

Methylase → - Methylation of DNA/RNA -

Endonuclease → - Cut Non Methylated DNA | RNA


= Bacteriophage Virus
E. Coli
Death of E. Coli
Restriction Endonuclease.
Restriction Enzymes
Breakdown of Nonmethylated Viral DNA
Bacteria
CH3
CH3
CH3
Methylase
Methylation of Self DNA
Protect from R.E.

[redacted]

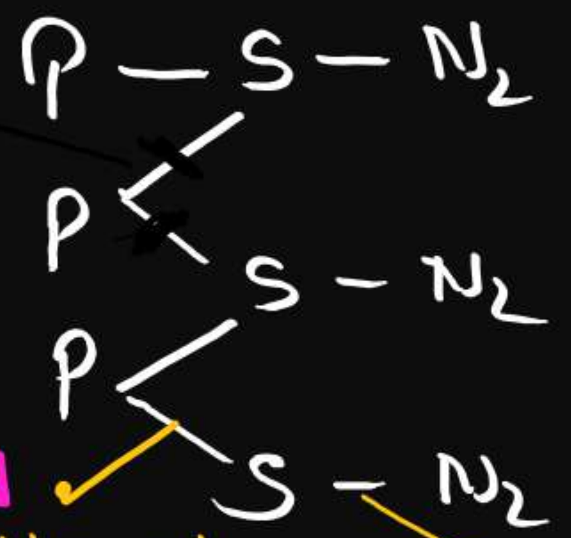
TOOLS OF RECOMBINANT DNA TECHNOLOGY

(1) DNA manipulating enzymes -

- Restriction enzymes (molecular scissors) = R.E.
- Polymerase enzymes ⟶ DNA - Poly - I
- Ligases (molecular glue) ⟶ T4 - Ligase [Isolate from - T4 - Bacteriophage] Virus

(2) Vectors (molecular Vehicle)

(3) Host organism



Phospho-ester bond or (P - Sugar bond)

N-Glycocidic bond.

Restriction Endonuclease

2025, 2024, 2022, 2021, 2020, 2019, 2017

900 → 230 BS

R.E.

Hind-II (Ist Restriction Endo.)

5' – GTT | AAC – 3'
3' – CAA | TTG – 5'

↓

GTT   AAC
CAA   TTG

(Blunt End)

Eco.R-I

- E → Genus
- co → Species
- R → Strain
- I → Order of isolation

5' – G | AATTC – 3'
3' – CTTAA | G – 5'

Palindromic Sequence – or Recognising Site (4-8)

↓

– G   AATTC –
– CTTAA   G –

(Sticky End)

Essential features of a vector are –

(1) Origin of replication (ori):

(2) Selectable marker : In addition to 'ori', the vector requires a selectable marker, which helps in identifying and eliminating non transformants and selectively permitting the growth of the transformants.

Eg.

- Antibiotics resistance genes.
- Lac Z gene for B –galactosidase enzyme (produce colour in the presence of a chromogenic substrate)

(3) Cloning sites : Palindromic sequences (restriction site 4-8 nucleotides) present with in selectable marker

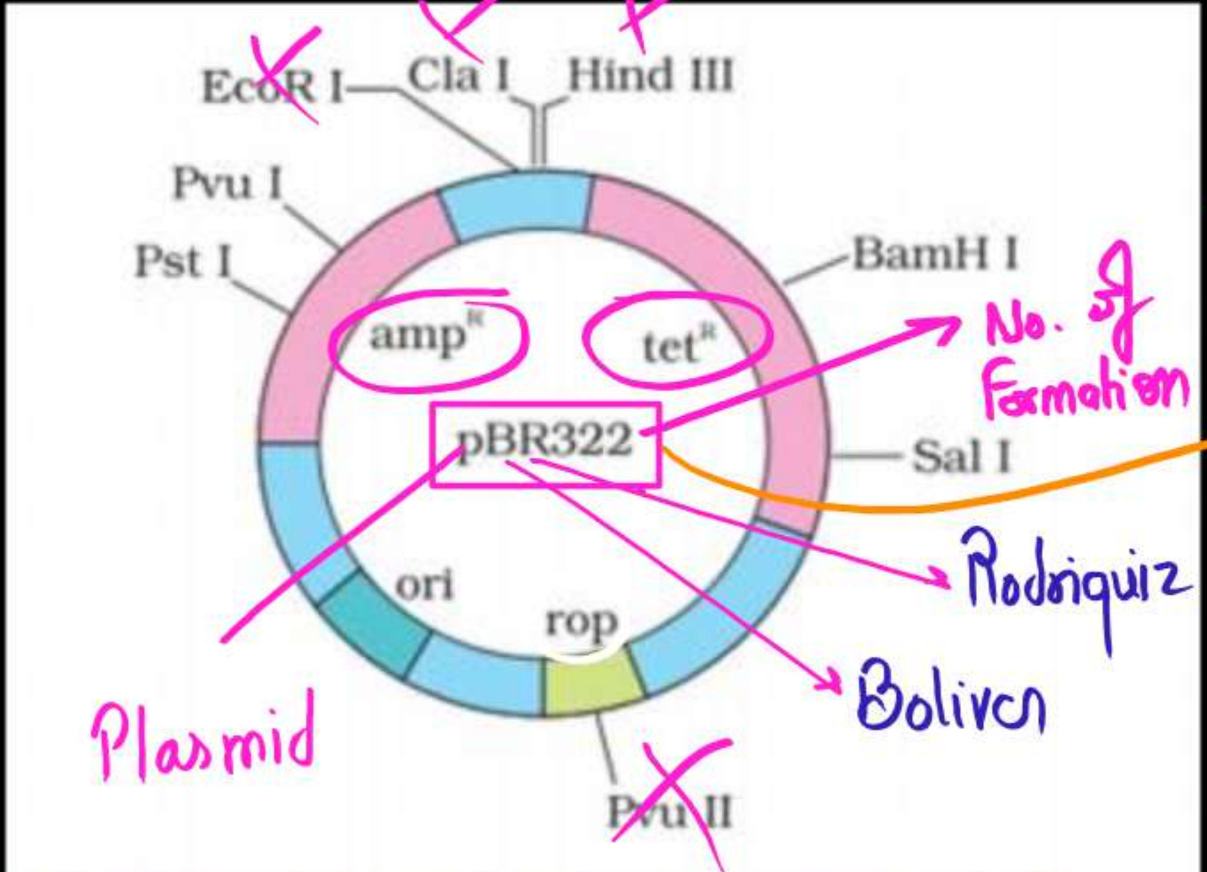


Figure 11.4 *E. coli* cloning vector pBR322 showing restriction sites (*Hind* III, *Eco*R I, *Bam*H I, *Sal* I, *Pvu* II, *Pst* I, *Cla* I), ori and antibiotic resistance genes (amp^R and tet^R). rop codes for the proteins involved in the replication of the plasmid.

vvvImp

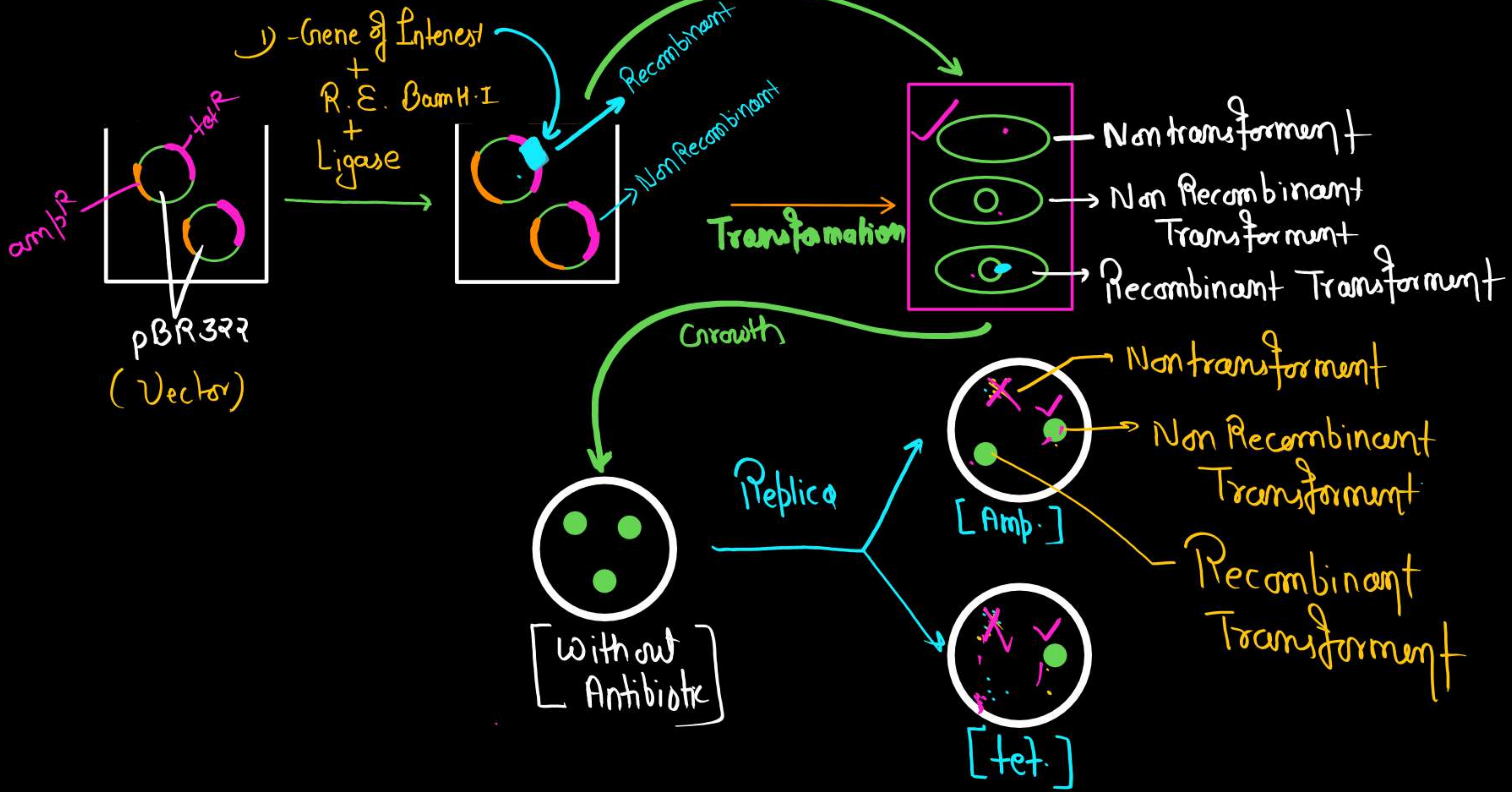
Salactable Marker [Antibiotic - Resis. Gene]
→ amp^R — [Puu-I, Pst-I]
→ tet^R — [BamH-I, Sal-I]
} → Restriction Site

Total Restriction Site = 8

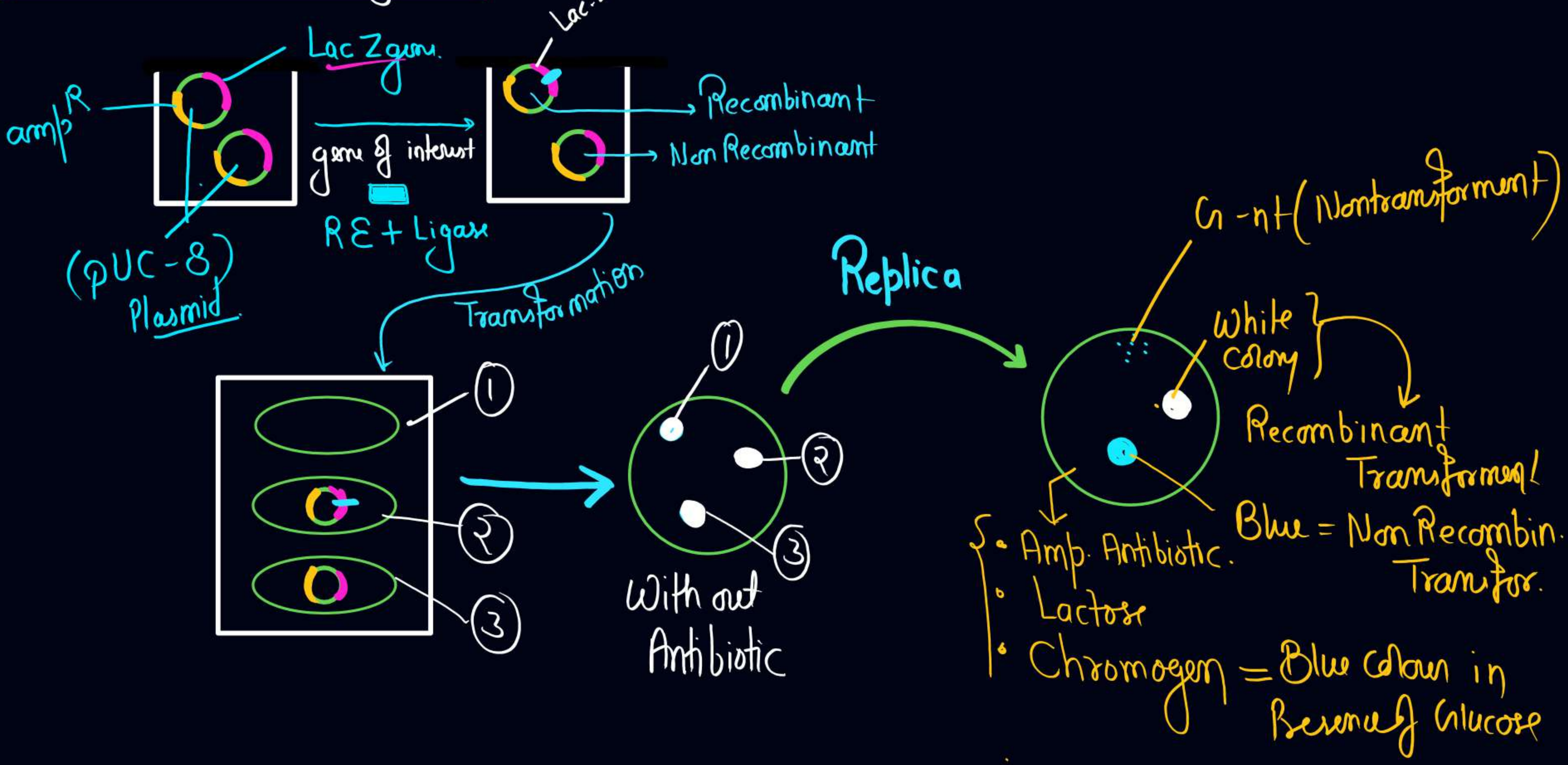
Useful (within S.Marker) = 4

Gene - rop ⟹ Pvu-II

Selectable Marker = Antibiotic Resistant Gene



S. Marker (Lac - Z gene)



Vectors for cloning genes in plants and animals :

- **Plants -** *Agrobacterium tumifaciens*, a pathogen of several dicot plants is able to deliver a piece of DNA known as **'T-DNA'** to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen.

- The tumor inducing (Ti) plasmid of *Agrobacterium tumifaciens*

Plant = Ti-plasmid - Vector
Animal = Retro Virus - Vector

Competent Host

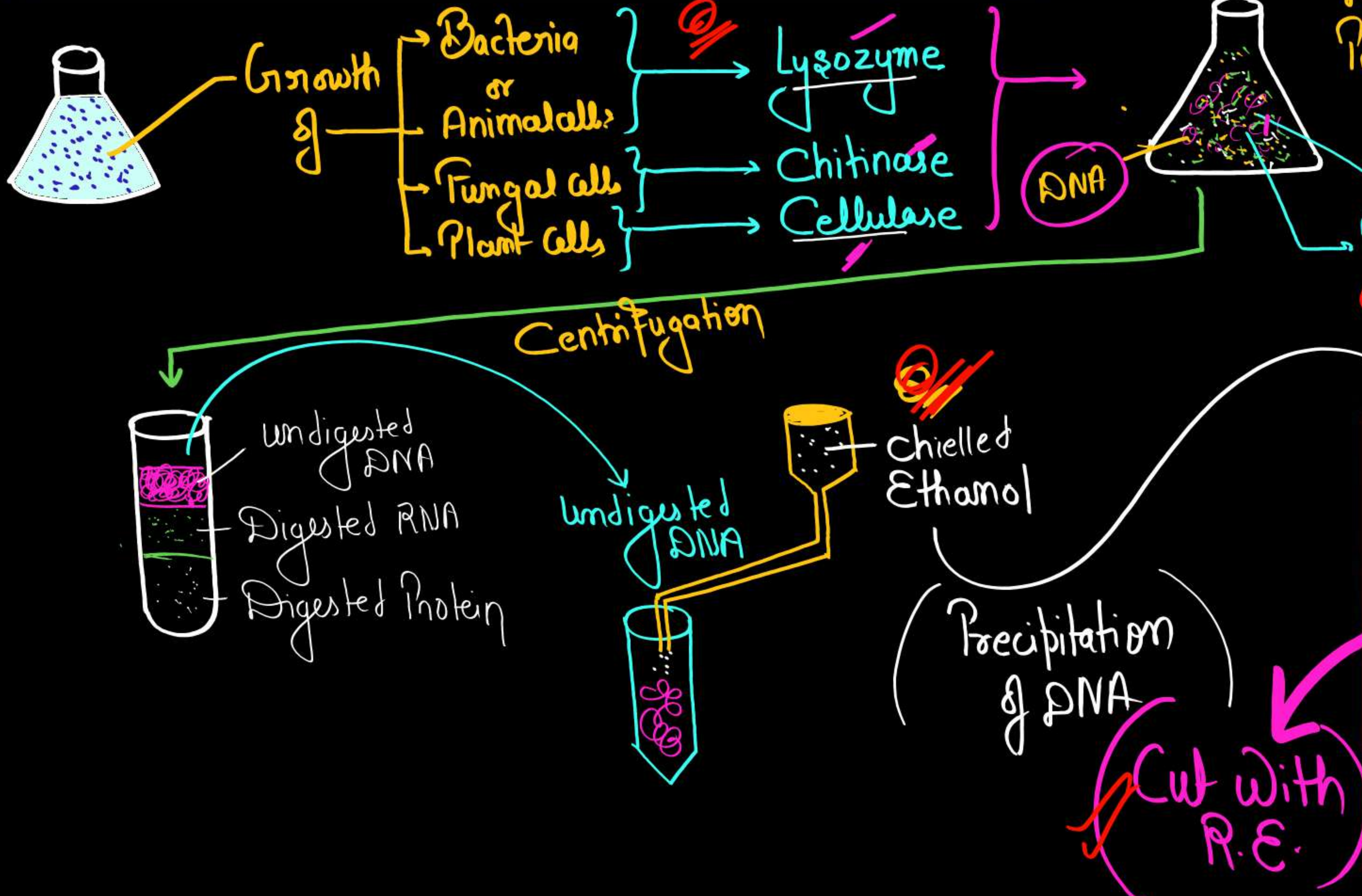
- Heat Shock (Bacteria)
  - $\rightarrow Ca^{++} + DNA^{-ve}$
  - $\rightarrow 42^{\circ}C \rightleftharpoons ICE$
- Animal Cell = Micro injuction
- Plant Cell = Biolistic / Gene gun } VImp
  - $\downarrow$ Gold or Tungsten^{+} + DNA^{-}

PROCESSES OF RECOMBINANT DNA TECHNOLOGY

Recombinant DNA technology involves several steps –

a) Isolation of DNA
b) Fragmentation of DNA by restriction endonucleases
c) Isolation of a desired DNA fragment by gel electrophoresis
d) Amplification of DNA fragment by PCR
e) Ligation of the DNA fragment into a vector
f) Transferring the recombinant DNA into the host
g) Culturing the host cells in a medium at large scale and extraction of the desired product.

(1) Isolation of the Genetic Material (DNA)



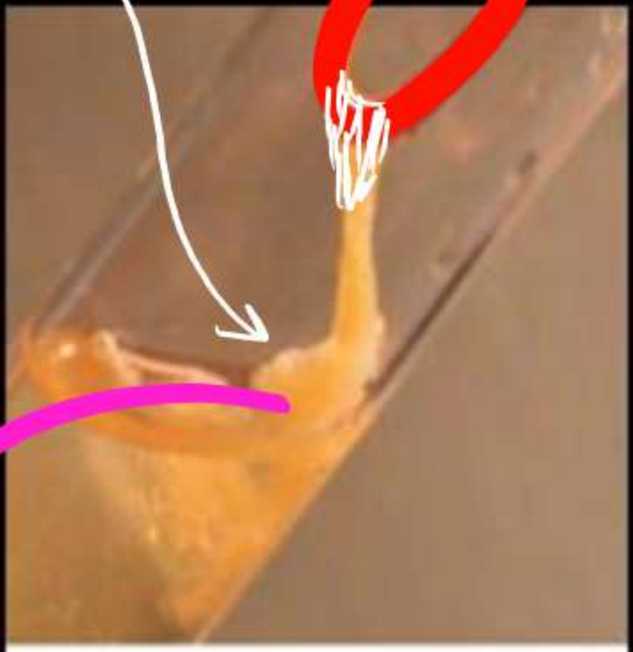
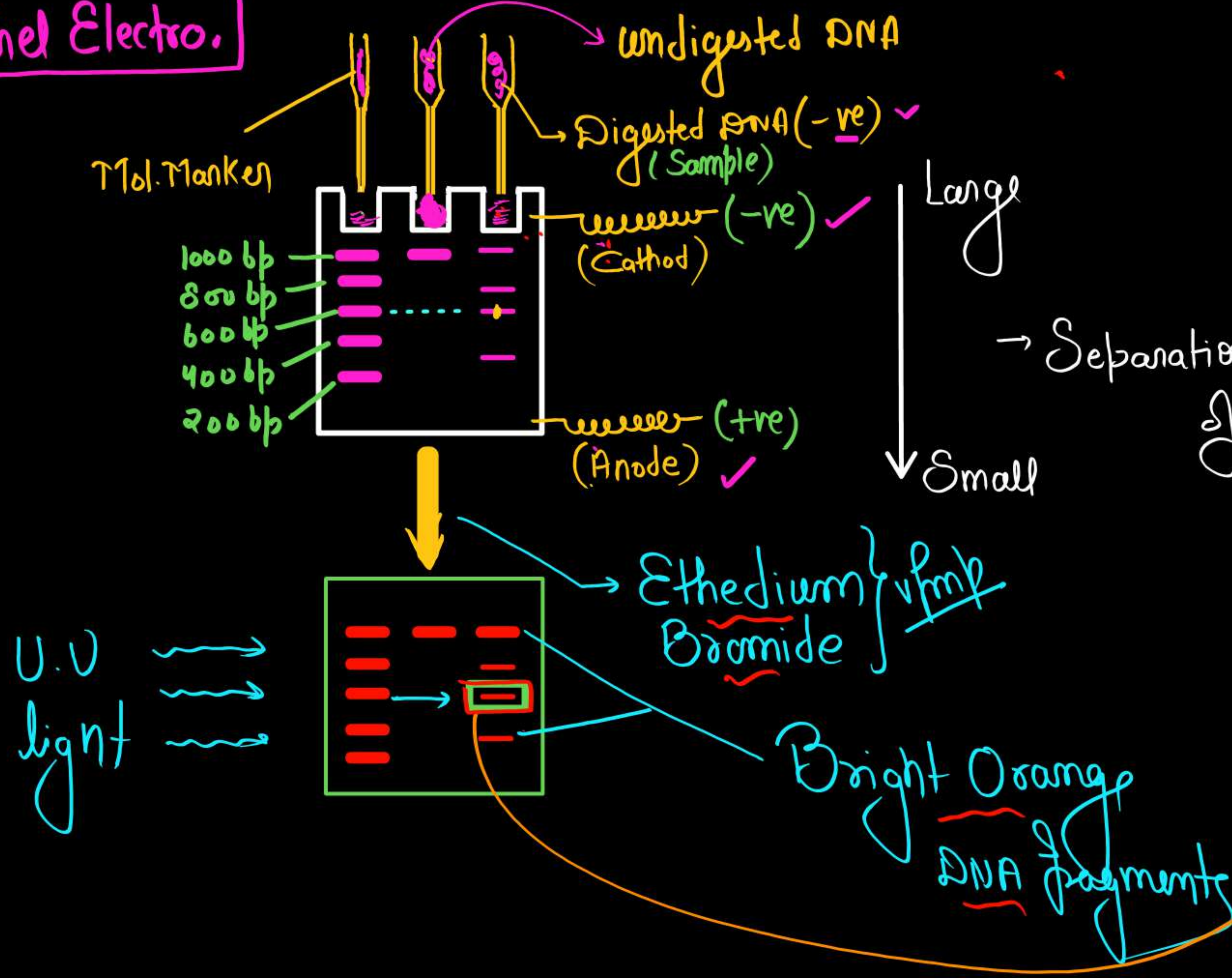


Figure 11.5 DNA that separates out can be removed by spooling

Gel Electro.



→ Separation of fragment on the basis of Size of fragment.

Elution

Sample

undigested DNA

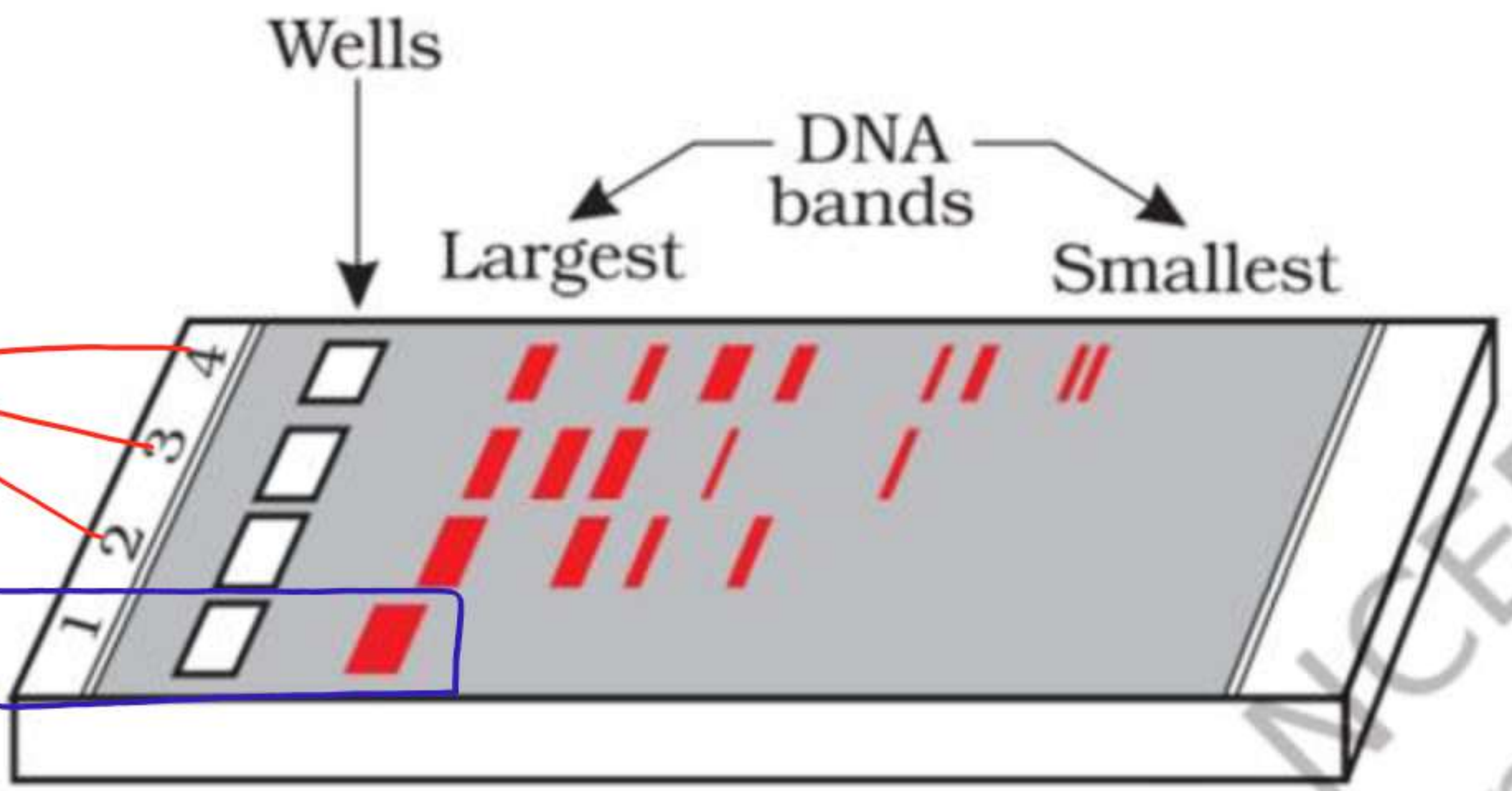
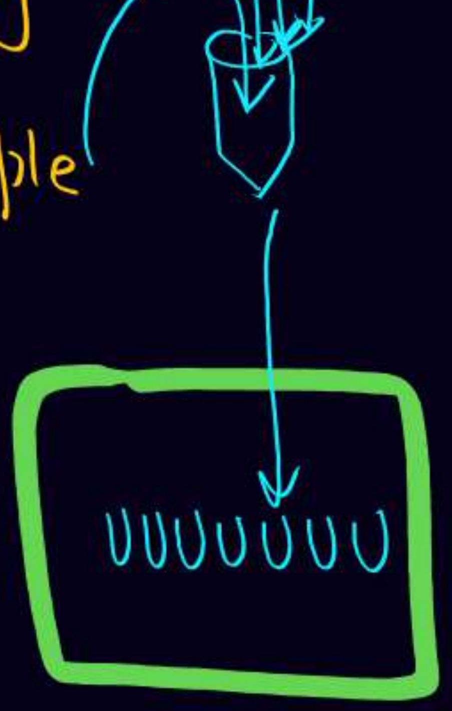


Figure 11.3 A typical agarose gel electrophoresis showing migration of undigested (lane 1) and digested set of DNA fragments (lane 2 to 4)

2025, 2022, 2021,

P.C.R → (Thermo Cycler)

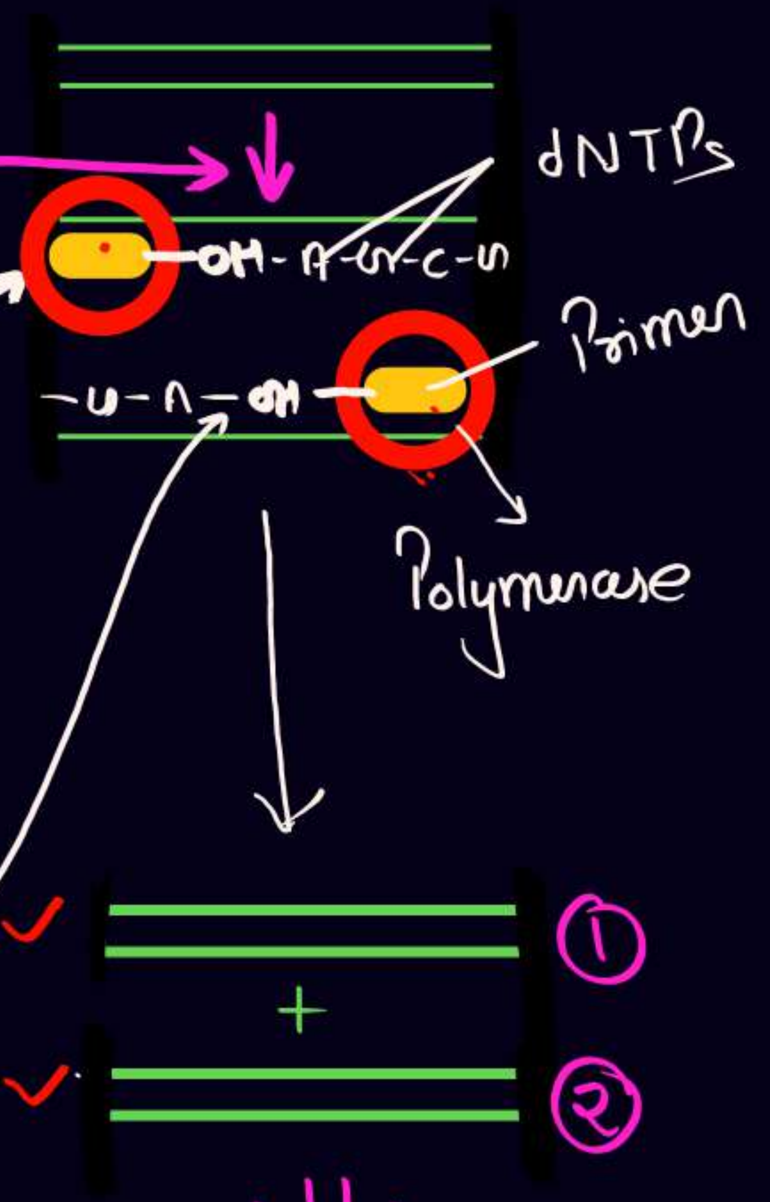
Primers
+
Taq Polymerase
+
dNTPs + Mg^{++}
+
DNA Sample



→ 95°C → = Denaturation

→ 50-60°C = Annealing

→ 72°C = Extension



~~vvvImp~~ { 30-Cycle = 1 billion DNA

$[2^n]$ - 2025

Bio-Rector

= Volume = 100 - 1000 L (2025, 2024, 2019)

→ Control Growth by Maintain → [Temp., pH, Sub., Salts, Vit., O_2]

- Simple Stirred Tank Bio.
  - Culture, Impeller, Foam braker
  - acid / base Balance
- Sparged Stirred Tank Bio.
  - ↑ - Surface area for ariation (O_2)
  - Bubbles

Downstream Processing

= Separation + Purification } Imp

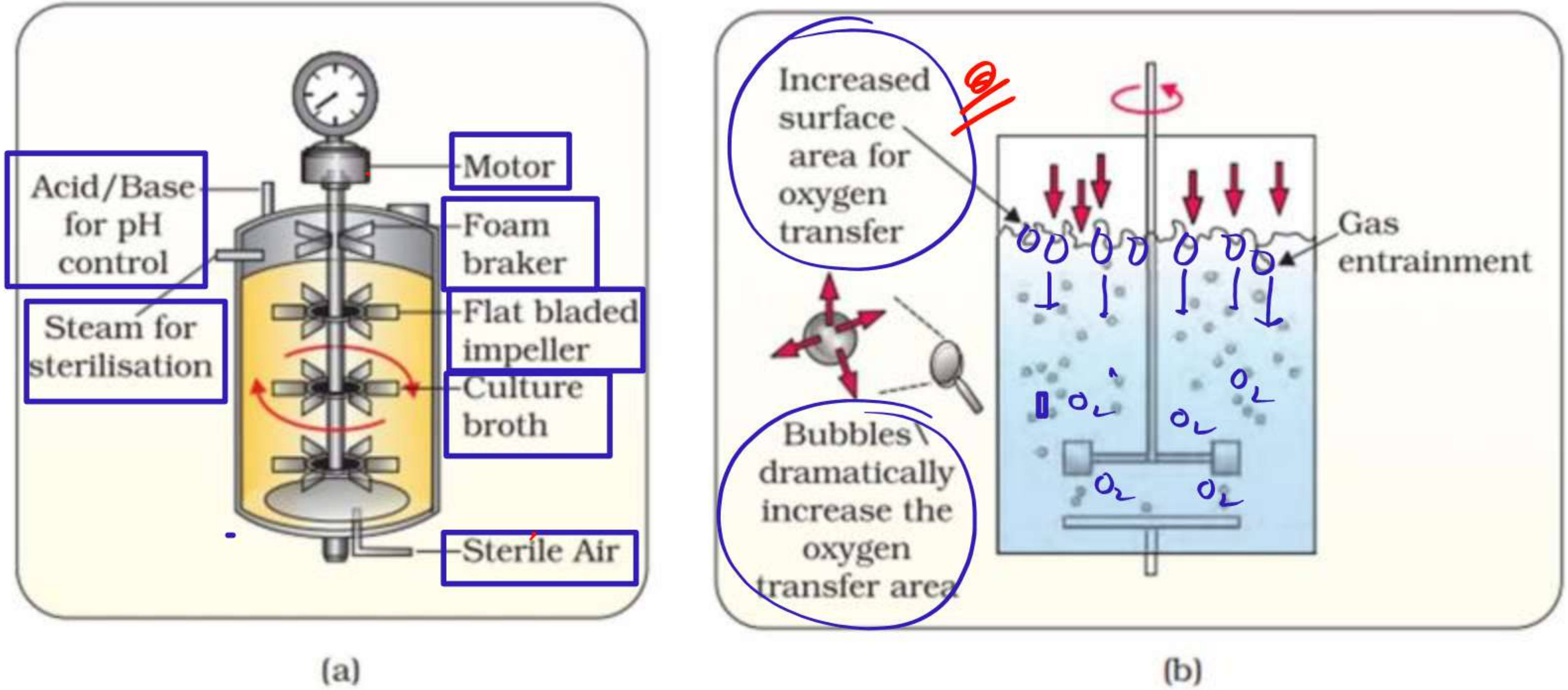


Figure 11.7 (a) Simple stirred-tank bioreactor; (b) Sparged stirred-tank bioreactor through which sterile air bubbles are sparged

CHAPTER 12

BIOTECHNOLOGY AND ITS APPLICATIONS

vvvImp

Biotechnology, as you would have learnt from the previous chapter, essentially deals with industrial scale production of biopharmaceuticals and biologicals using genetically modified microbes, fungi, plants and animals. The applications of biotechnology include therapeutics, diagnostics, genetically modified crops for agriculture, processed food, bioremediation, waste treatment, and energy production. Three critical research areas of biotechnology are:

(i) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.

(ii) Creating optimal conditions through engineering for a catalyst to act, and

(iii) Downstream processing technologies to purify the protein/organic compound.

Let us now learn how human beings have used biotechnology to improve the quality of human life, especially in the field of food production and health.

Biotechnology application in agriculture

① - Golden Rice (Enhanced Nutritional Value) → Vit - A
(Oryza Sativa)
→ Isolation of β-Carotene gene → Daffodil
→ Plasmid → Ti - plasmid
→ Normal Rice → Golden Rice { Golden Colour ↓ β-Carotene → Vit-A }

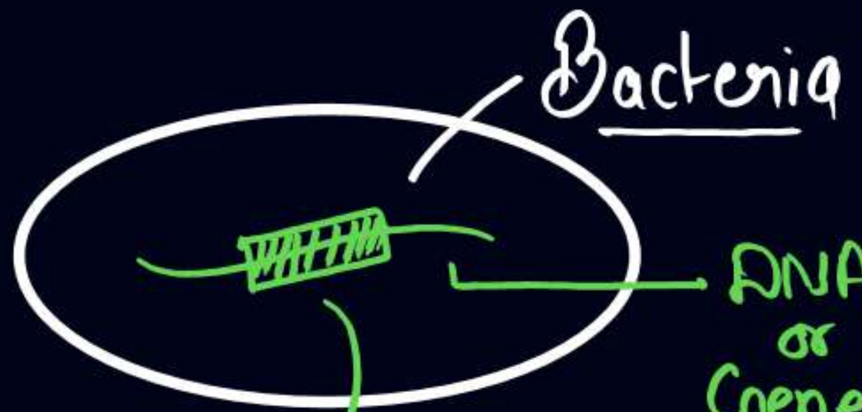
② · Flaver Savr (Transgenic Tomato) →
→ Reduced post Harvest losses
→ RNAi - used for Silencing of Polygalacturonase

3) - Bt-Toxin (Bt Cotton) = {Pest Resistant Crop} {2024, 2020, 2019}

- Bacteria = Bacillus thuringiensis
- Gene = Cry gene
- Inactive Protein = Cry Protein (Endotoxin) → Active in only $\boxed{\text{Alkaline Medium of Mid Gut of Insect}}$ → Active Toxin

Attach with Wall of Midgut → Solubilised → Create Pores → Swelling & Lysis

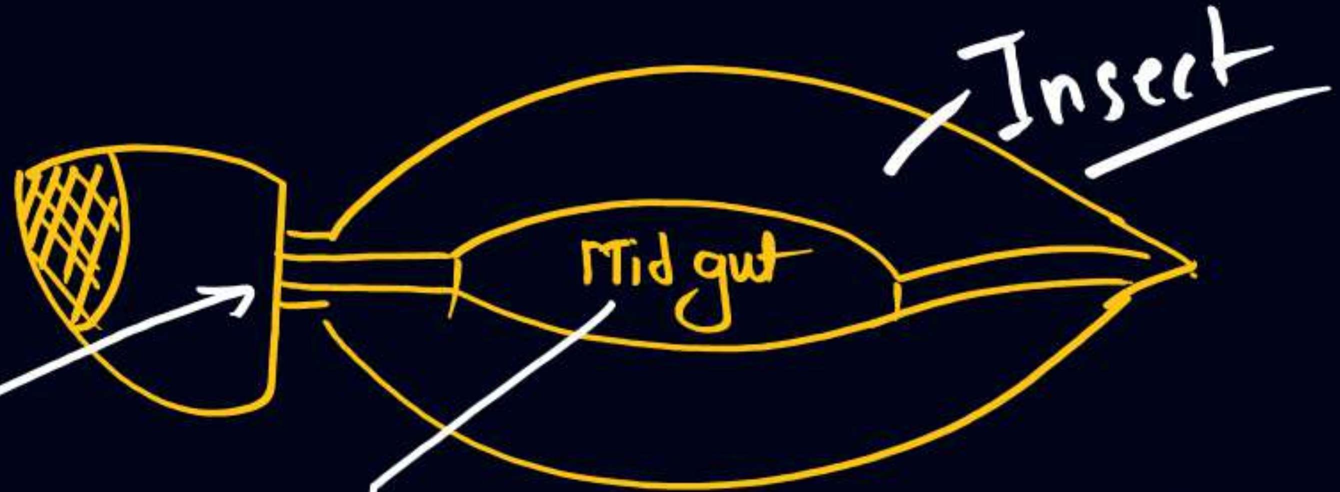
- Insect Resistant Crops → Cotton, Corn, rice, Tomato, Potato, Soyabean
- Insect → Lapidopterans → Tobacco budworm, armyworm, Corn borer, Cotton boll worm
  - Coleopterans → Beetles
  - Dipterens → Flies & Mosquito



DNA or Gene } Cry gene

{ Cry protein [Inactive Protein] or Inactive endotoxins. }

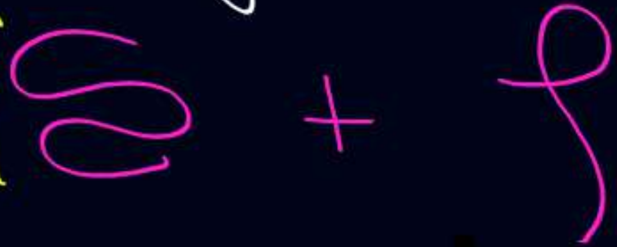
} → Enter in Mid gut of Insect



(Alkaline Medium in Midgut)

↓

In highly alkaline medium, in presence of Some enzyme →



Active Endotoxins → attach with gut Epithelium → Create Pore In. Wall of gut

Death of Insect

imp Cry gene → Cry-I Ab → Corn borer ✓

→ Cry-II Ab + Cry-I Ac → Cotton boll worm ✓

NCERT

Note → Choice of gene depend upon → Crop & Targeted Pest

imp

@// Most Bt toxins are insect-group specific.

4) - RNA interference (RNAi) → [Pest / Nematode Resistant Plant]

{2025, 2020, 2019}

→ Nematode = Meloidegyne incognitia (Round worm) @//

→ Plant = Root of Tobacco @//

→ RNA interference - Take place in all Eukaryotic Organism as a method of Cellular Defense

- Method ⇒ Silancing of a Specific m-RNA due to Complementory ds RNA → Prevent translation of m-RNA (Silancing)

- Source of Complementory R.NA ⇒ Virus (RNA) + Transposons

vvvImp

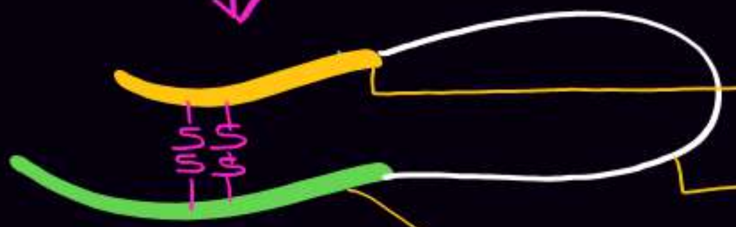
2026

Biotechnological Application in Medicine

- $\boxed{30}$ Recombinant therapeutics = World %
- $\boxed{12}$ of these = India

D-Insulin

- Insulin used in diabetes Extracted from
  - Cattle
  - Pig
- Insulin from animal Source = Allergy
- Pro insulin (Inactive) Synthesised by — β-Cells of Pancrease



A - peptide = 21 a.a.
C - peptide = 33 a.a.
B - peptide = 30 a.a

$\boxed{84}$ amino acid.

Maturation → Removal of C-Peptide [33 aa]

} 51 amino acid [Mature Insulin] Q

Disulphide bridge

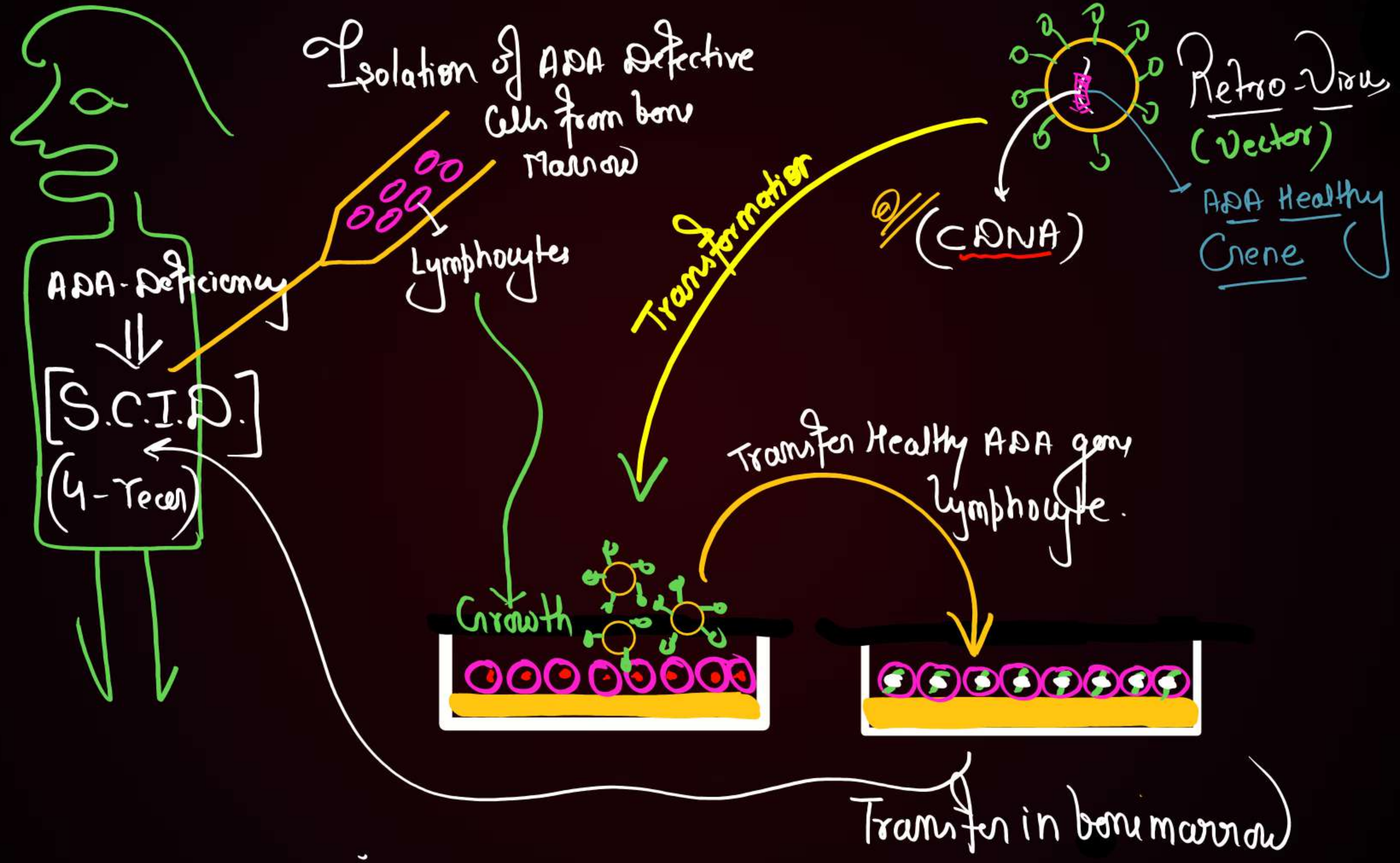
Genetically Engineered Insulin (Humulin) {2025, 2022, 2021, 2020, 2016}

- 1983 = Eli Lilly ⇒ American Company
- Host = E. Coli
- Vector = pBR 322
- A & B chain produced Separately → Extracted → Combined by Disulphide bond

2) Gene Therapy {2024 (Multi.), 2022, 2021, 2020, 2018, 2016} → 1990 → Ist Clinical gene therapy

4-Year old girl → Adenosine Deaminase Deficiency } ADA

(SCIDS) → {ADA Enzyme Stimulate} → Humoral Immu., C.M.I



Isolation of ADA Defective
Cells from bone
Marrow
ADA-Deficiency
S.C.I.D.
(4-Year)
Lymphocytes
Transformation
Retro-Virus
(Vector)
(CDNA)
ADA Healthy
Gene
Transfer Healthy ADA gene
Lymphocyte.
Growth
Transfer in bone marrow

In children ADA Deficiency Can be Cured by

1) - Bone marrow transplantation
2) - Enzyme Replacement theraphy
3) - Gene theraphy → Isolation of ADA Defective lymphocytes

↓

Functional ADA cDNA of ADA + Retro virus → Introduced in lymphocytes → Subsequently Returned to patient

RNA
↓ Reverse trans
DNA $\boxed{\text{cDNA}}$

Note - Gene therapy at Early Embryonic stage, it Could be a permanent Cure

Molecular Diagnosis

vimp

2023, 2021 → 3

Early Diagnosis

- → P.C.R
- → Autoradiography (R.D.T.)
- → E.L.I.S.A

Autoradiography

Molecular Probe [P^{32}]
[SS RNA/ssDNA]

↓

Hybridise with to its Complimentary DNA for Detection of Mutated gene on Photographic film.

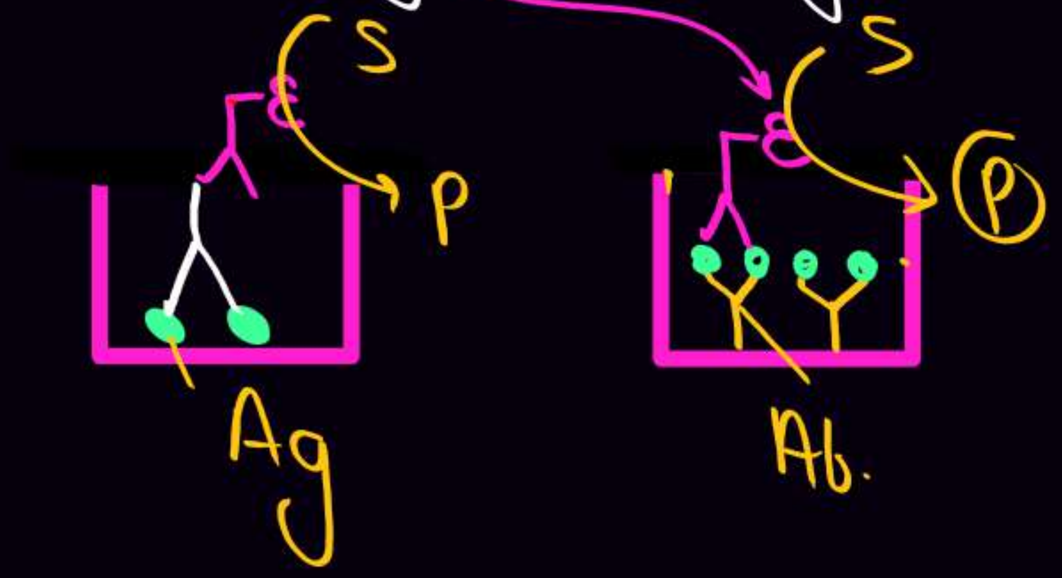
→ Mutation -nt

ELISA

Based on antigen-antibody interaction

Detect Pathogen
- Ag
- Ab

by - Peroxydase Enzyme linked Ab.



Transgenic Animals

Transgenic Animals → Rat, Rabbit, Pig, Sheep, Cow & Fish

$\boxed{95\% = \text{Mice}}$ Q//

Use of Transgenic Animals

1) - Normal physiology & Development of gene Expration

2) - Study of Diseases → Cancer, Cystic fibrosis, Rheumetoid arthritis & Alzheimer's

vvvImp 3) - Biological Products → Emphysema — α·1 antitripsin
Phenylketonurea (PKU), Cystic fibrosis

Q//

$\boxed{1997}$ - Ist Transgenic Cow ⇒ $\boxed{\text{Rosie}}$ — Protein enriched Milk

Q (2.4 gm/L)

↓

Q Human α - Lactalbumin

4) - Vaccine Safty → Polio-Vaccine

5) Chemical ,,

Ethical Issues

2019 → **GEAC** = Genetic Engineering Approval Committe

Total Varities of Rice in India = 200,000

Varieties of Basmati rice = 27

Q// **1997** = American Company got patent right on Basmati

Low Calorie Sweetener Protein = Brazzin (2000 times Sweet as Sugar)

Indian Patents Bill = Q// $\boxed{\text{Second amendment}}$

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Q: cry I Ab and cry II Ab produce toxins that control

Corn

Cotton

A Cotton bollworms and corn borer respectively

B Corn borer and cotton bollworm respectively

C Tobacco budworms and nematodes respectively

D Corn borer and Tobacco budworms respectively

B

Vedantu
Learn LIVE Online

Q: Bt toxin kills insects by

C

A inhibiting protein synthesis

B generating excessive heat

C creating pores in the midgut epithelial cells, leading to cell swelling and lysis

D obstructing a biosynthetic pathway

Q: Some strains of *Bacillus thuringiensis* produce proteins that kill insects like

Cry

A lepidopterans

B coleopterans

C dipterans

D All of these

D

Vedantu
Learn LIVE Online

Q: A novel strategy was adopted to prevent Meloidogyne incognita infection in tobacco plants that was based on the process of

A DNA interference

B RNA interference

C RNA initiation

D DNA initiation

B

Q: Which of the following nematodes infects the root of the tobacco plants which reduces the production of tobacco?

A *Wuchereria*

B *Ascaris*

C *Meloidegyne incognita*

D *Enterobius*

C

NCERT

Q: Transposons are also known as

A silenced genes

B mobile genetic elements

C pleiotropic genes

D Both (a) and (b)

D

Q: What is the demerit of using bovine insulin (from cow) and porcine insulin (from pig) in diabetic patients?

A It leads to hypercalcemia

B It is expensive

C It may cause allergic reactions

D It may lead to mutations in human genome

C

Q: In 1983, which of the following companies prepared human insulin?

A Genetech

B Eli Lilly

C GEAC

D None of these

Q: Treatment of a genetic disorder by manipulating genes is called

A gene therapy

B rDNA technology

C bone marrow transplantation

D enzyme replacement therapy

Q: For the first time, gene therapy was tried on a 4 year old girl in 1990 to treat which of the following enzyme deficiency?

A Cytosine Deaminase (CDA)

B Adenosine Deaminase (ADA)

C Tyrosine oxidase

D Glutamate trihydrogenase

B

Q: PCR is used to

A detect HIV in suspected AIDS patients

B detect mutations in the genes of suspected cancer patients

C diagnose many genetic disorders

D All of the above

D

Q: A single strand of nucleic acid tagged with a radioactive molecule is called

A plasmid

B vector

C probe

D selectable marker

Q: Technique used to detect mutated genes is called

A gel electrophoresis

B polymerase chain reaction

C gene therapy

D autoradiography

Q: Which of the following techniques is based on the principle of antigen-antibody interaction?

A PCR

B ELISA

C Recombinant DNA technology

D Gene therapy

B

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Human Reprotuc tion

Vulva/Pudendum/ External genitalia

vvfmp

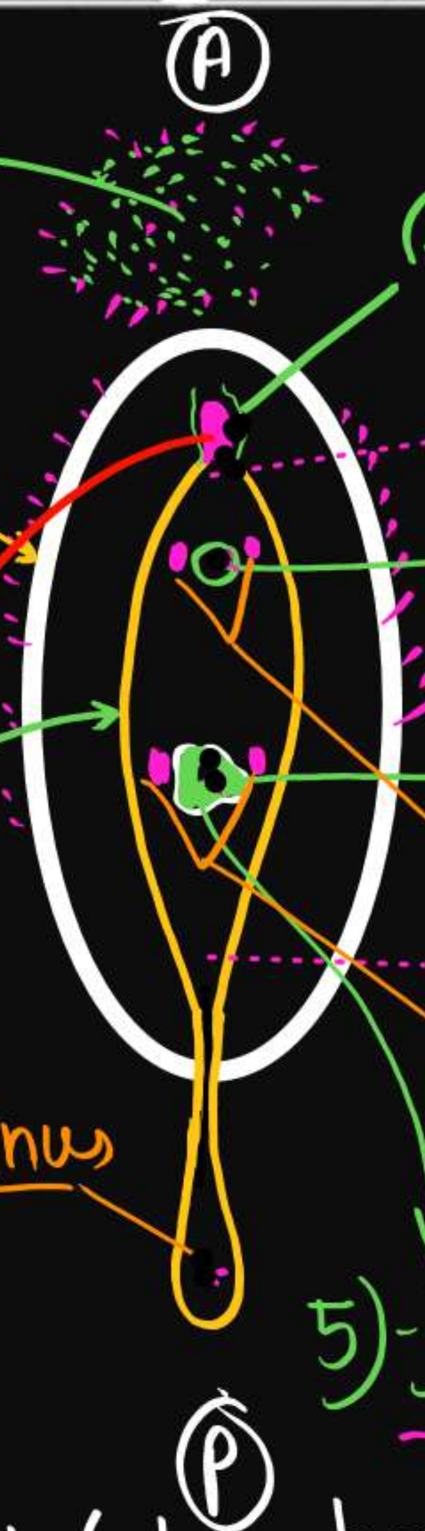
1Q

1)- Mons Pubis
- Fatty Tissue
- Hairs

2)- Labia Majora
- Flashy fold
- Fatty Tissue
- Hairs.

3)- Labia Minora
- Thin Skin.

4)- Clitoris → Erectile Tissue
- Anterior Side of L. Minora.
- Homology with Glans penis.



Fore Skin

Urethral Orifice ✓

Vaginal Orifice ✓

Vestibulum

Paraurethral Glands (Sken's glands)

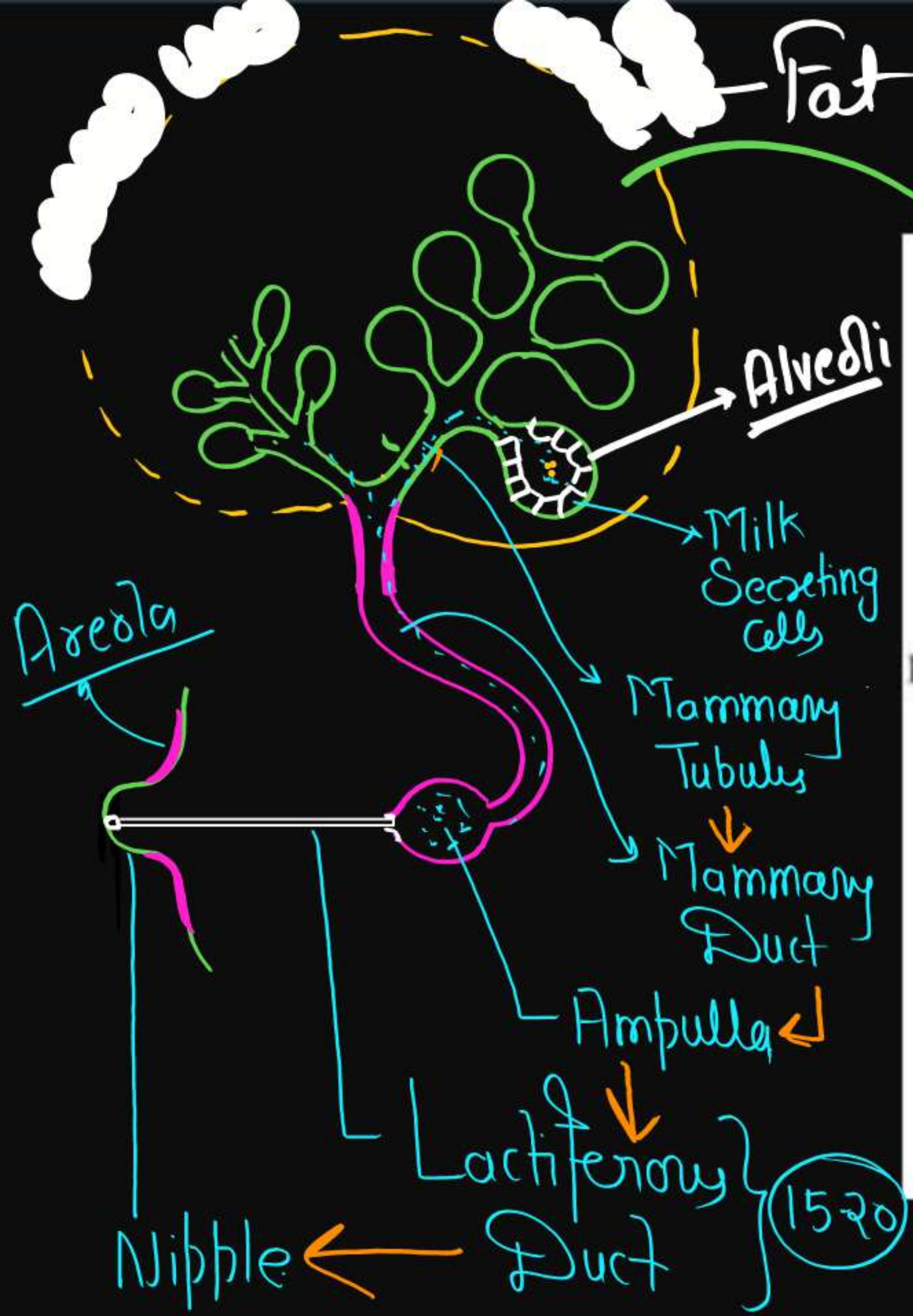
Q10

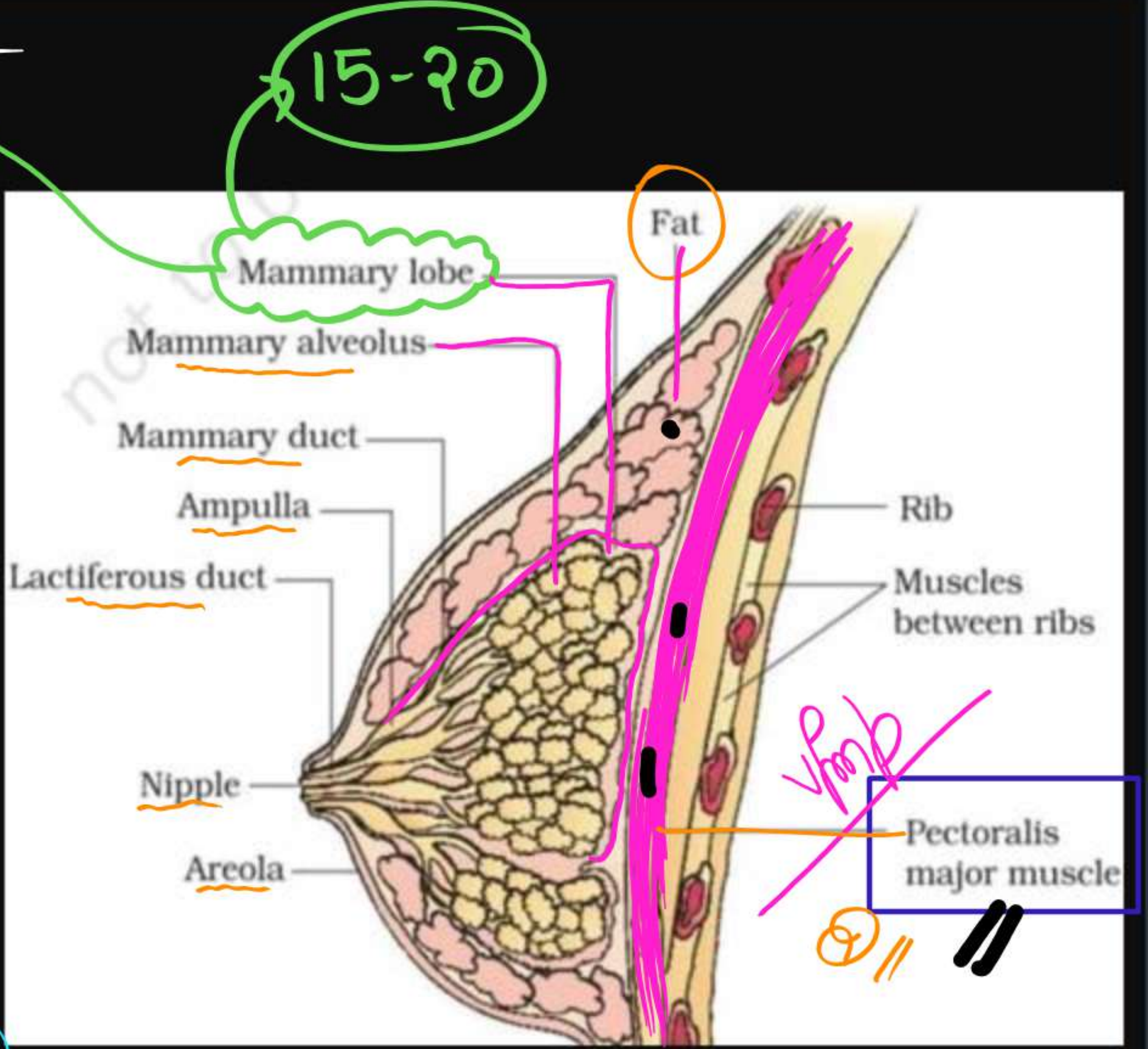
Birtholine glands

Q11

5)- Hymen

Mammary Glands





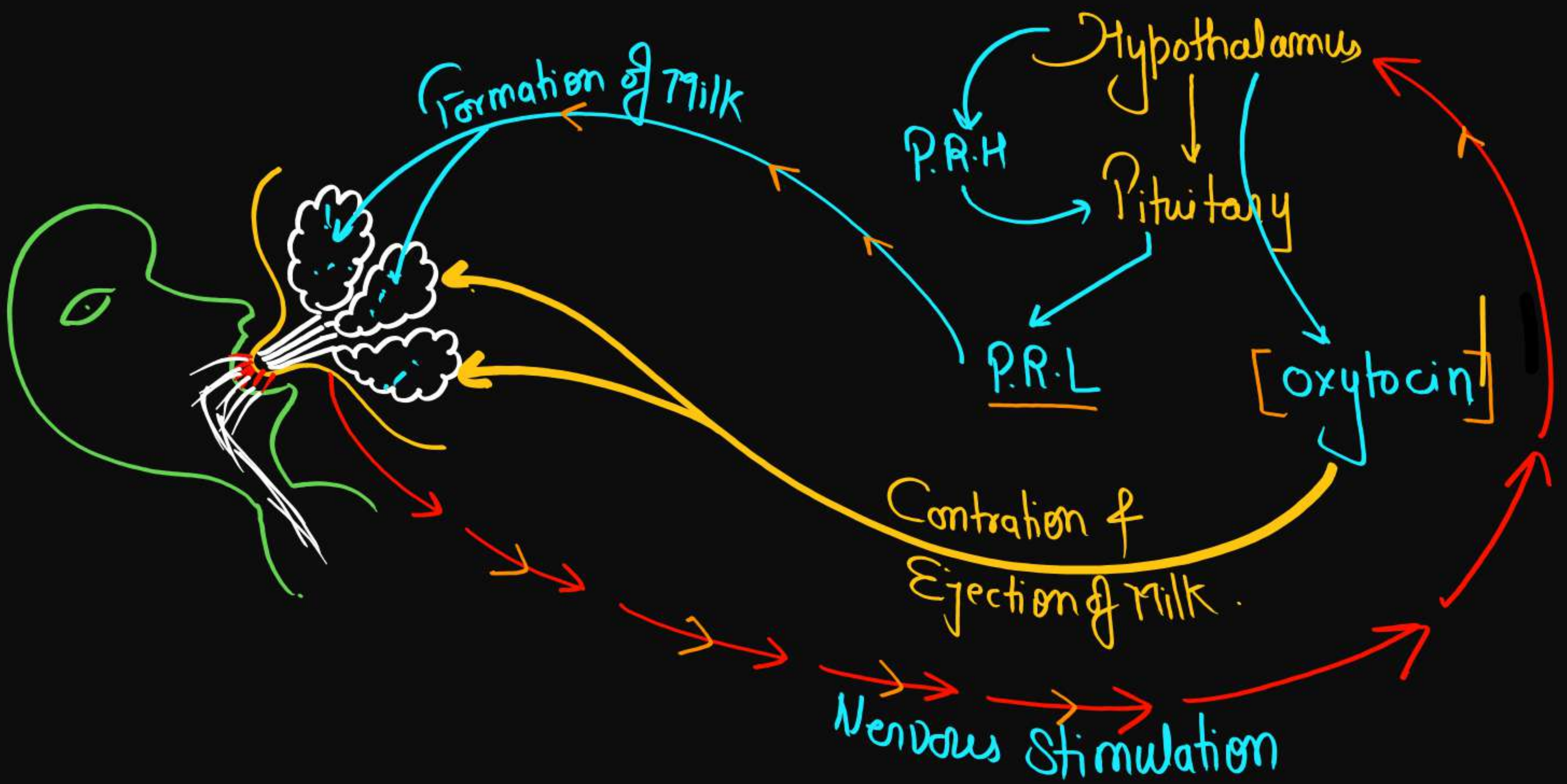
Lactation-

- Growth of mammary glands during puberty is called thelarche, which regulate by- Estrogen and during pregnancy regulate by Progesterone and Lactogenic hormone (PRL & hPL).

IgA

The mammary glands of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called **lactation**. This helps the mother in feeding the new-born. The milk produced during the initial few days of lactation is called **colostrum** which contains several antibodies absolutely essential to develop resistance for the new-born babies. Breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.

The principal stimulus in maintaining prolactin secretion during lactation is the sucking action of the infant. = Neuro - Endocrine Mechanism.



Spermatogenesis –

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It begins during puberty and continues till death. In humans, this process takes $\boxed{65\text{-}75}$ days.

Spermatogenesis

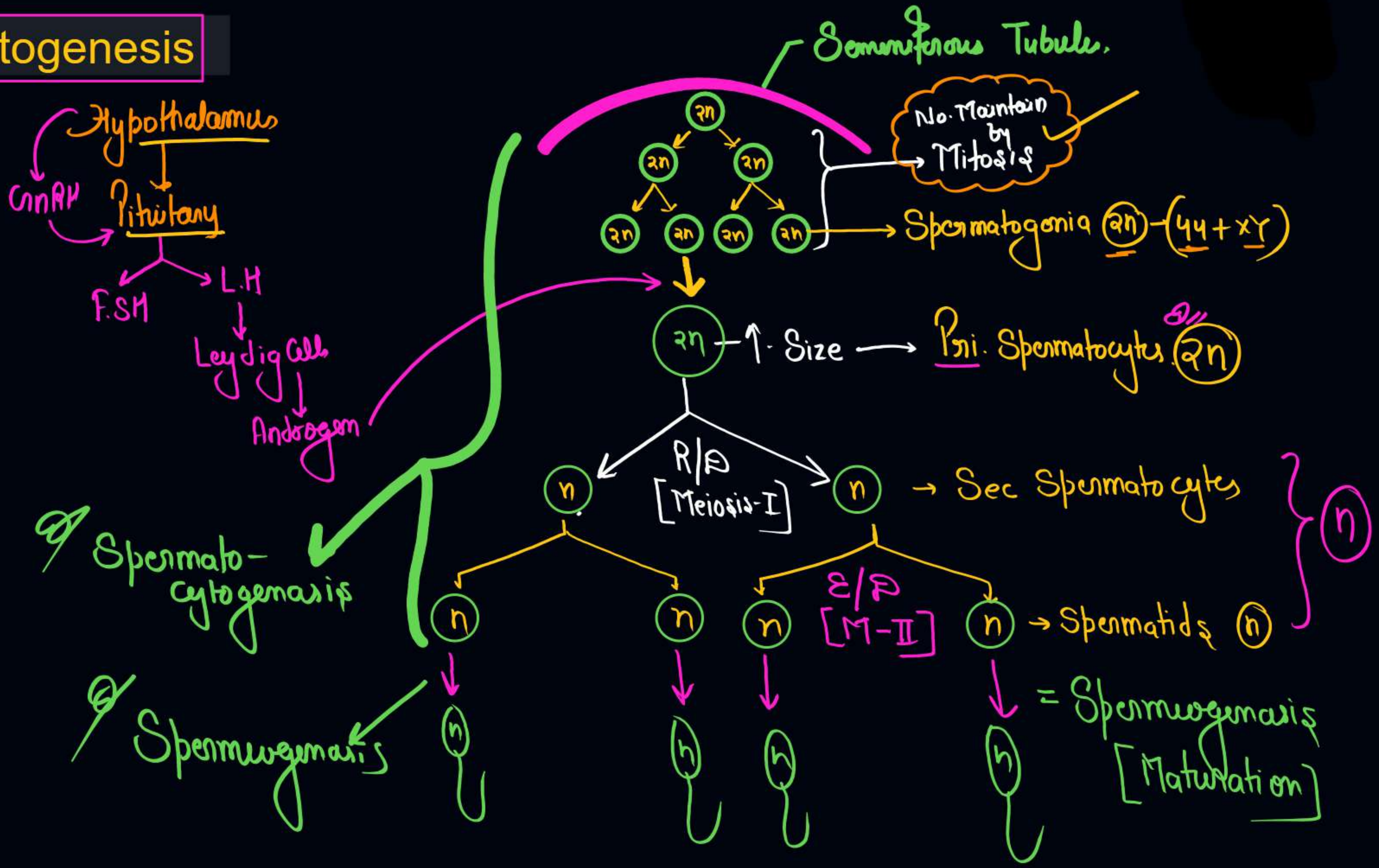
(1) Spermatocytogenesis → Division of Cytoplasm < Mitosis / Meiosis

It is formation of spermatids from spermatogonia.

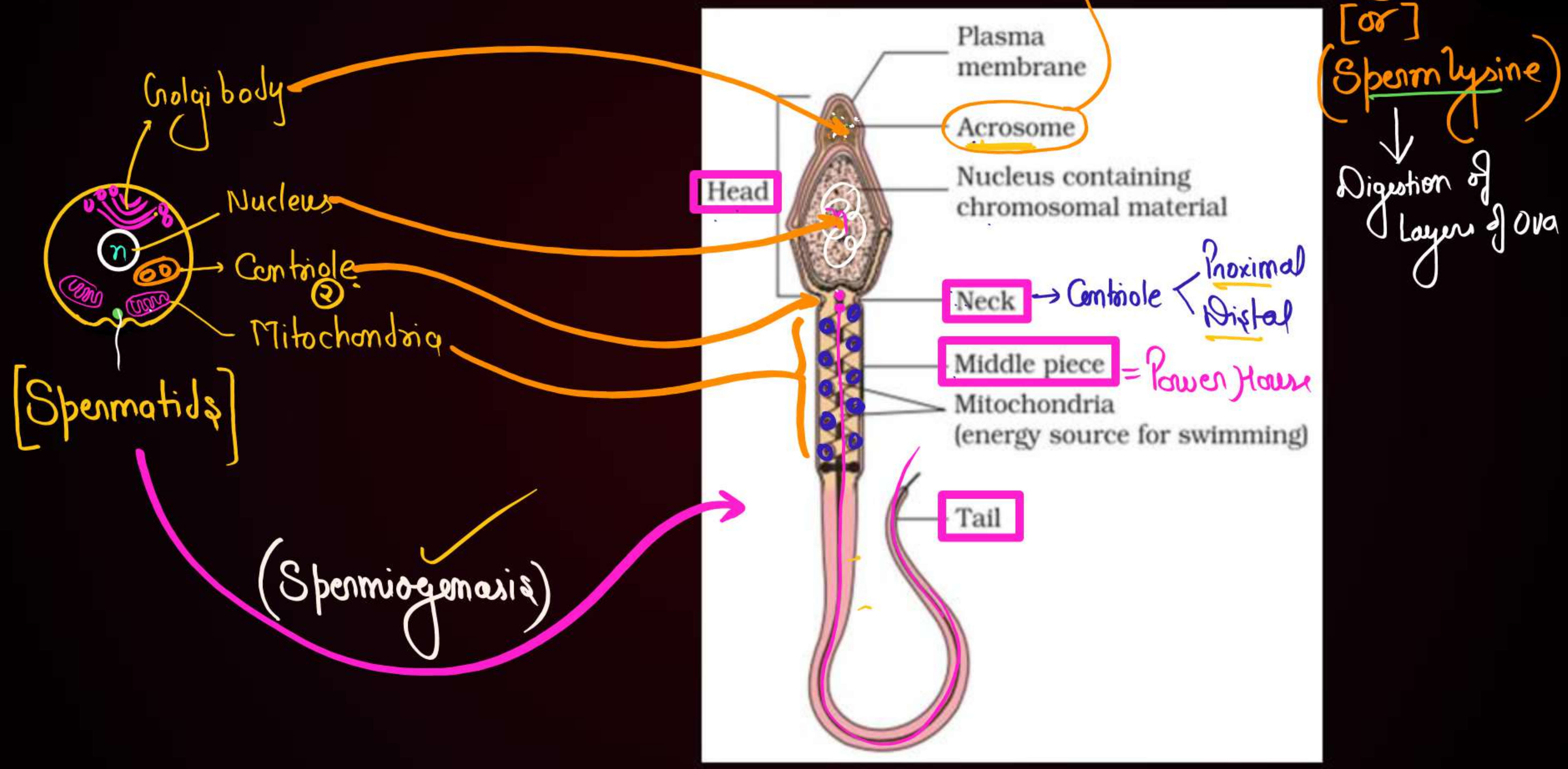
(2) Spermiogenesis/Spermateliosis

(Longest phase) -It is differentiation of spermatids (metamorphosis) into sperms.

Spermatogenesis



(2) Spermiogenesis/Spermateliosis



- After spermiogenesis, sperm heads become embedded in the Sertoli cells, and are finally released from the seminiferous tubules by the process called spermiation.

Enter in Epididymis

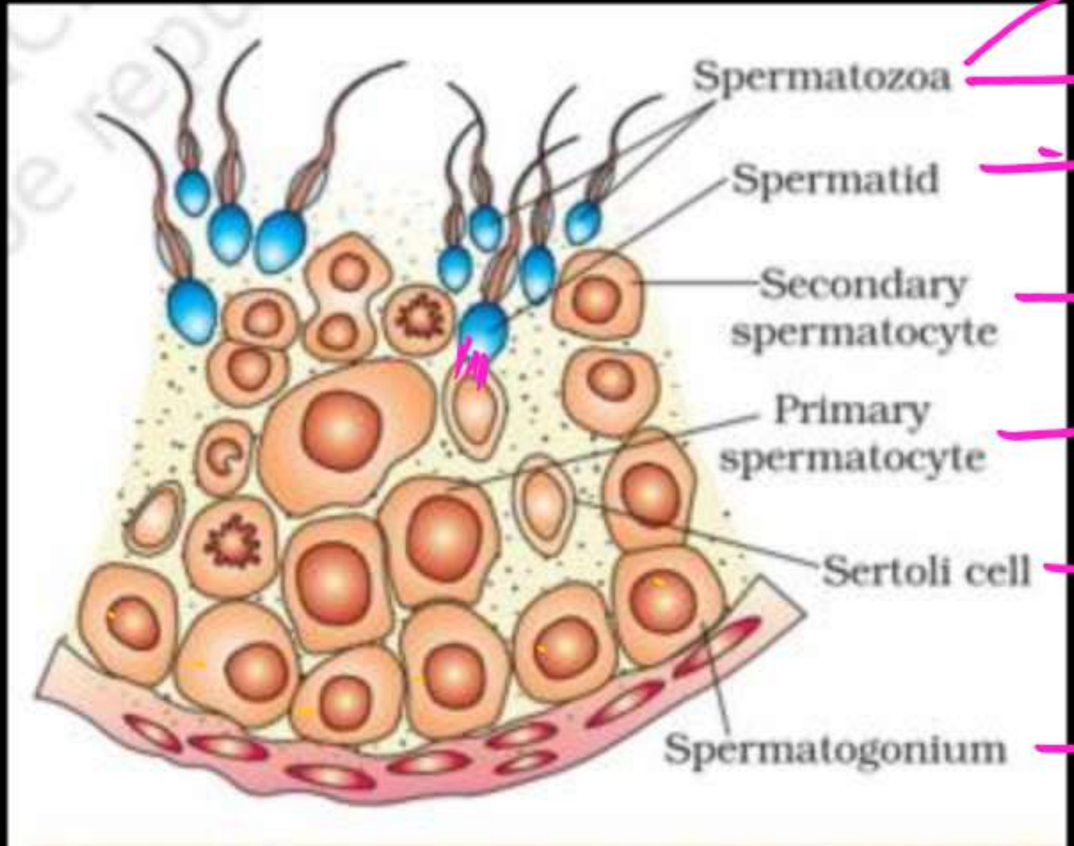


Figure 3.5 Diagrammatic sectional view of a seminiferous tubule (enlarged)

n
n
n
2n
2n
2n

- For formation of $\boxed{\text{100 sperms}}$
- How many meiotic division are required = 25
- How many Pri. spermatocytes are required = 25
- How many Sec. spermatocytes are required= 50
- How many spermatids are required = 100

Oogenesis

vvImp

2-Million Follicles — at the time of Birth

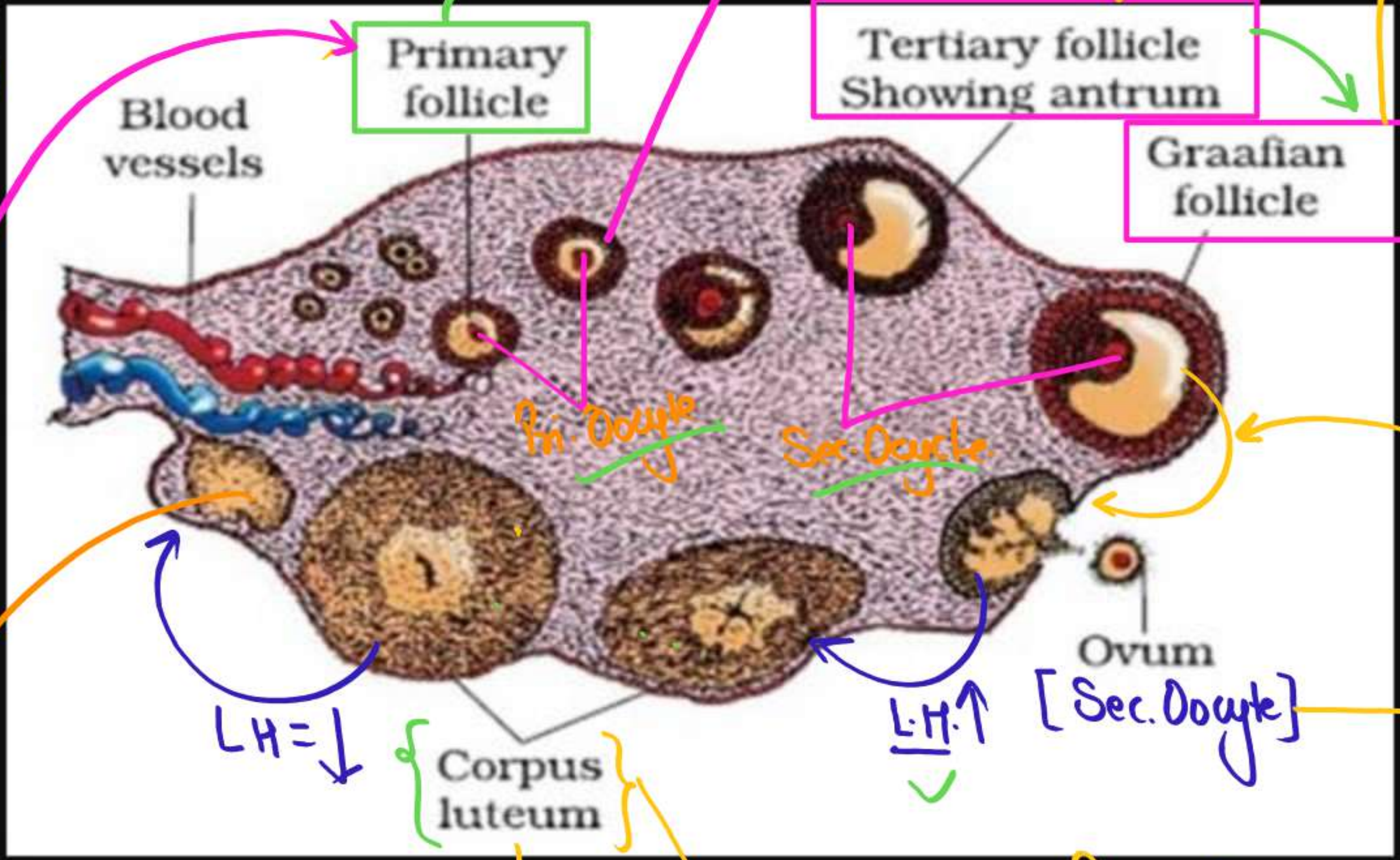
→ Atrasia (Degeneration)

60,000 - 80,000 Pri. Follicles in Each Ovary (at the time of Puberty)

F.S.H

Sec. Follicle

Estrogen



Pri. Oocyte

Sec. Oocyte

Q// Ovulation (L.H) (14 day)

L.H. ↑

[Sec. Oocyte] → Enter in Fallopian Tube

LH = ↓

(Corpus Albicans)

Q// (Temp. Endo. Gland)

Release — [Progesteron, Estrogen etc. etc]

Pri. Follicle $\xrightarrow{F.S.H}$ Sec. Follicle — Ter. Follicle — Graafian Folli.

Pri. Follicle

Pri. Oocyte (2n)

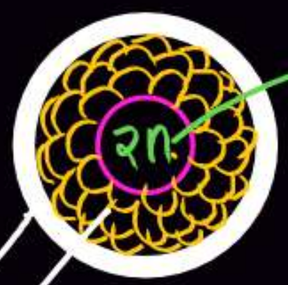
Single layer of granulosa cells

Oocyte = Pri. Oocyte

Meiosis arrest in Prophase - I

Sec. Follicle

Pri. Oocyte (2n)



→ Multilayer of granulosa cells

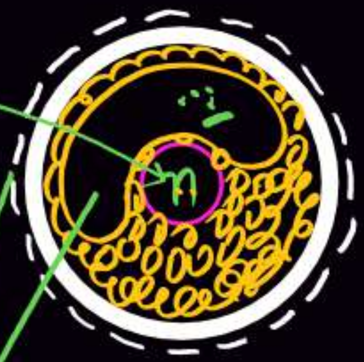
→ Formation of Theca layer

Oocyte = Pri. Oocyte

Arrested Meiosis } → Start

Ter. Follicle

Sec. Oocyte (n)



→ Fluid fild Cavity Antrum tnt

→ Theca < Externa, Interna

(2n) Pri. Oocyte → (n) Sec. Oocyte + (n)

↓ Meiosis-I Complete

(n) → Formation of Ist Polar body

Graafian Folli.

Sec. Oocyte (n)

Q// Zona pellucida

Q// [Non cellular]

Q// Sec Oocyte

= [Meiosis arrest in Meta-II] ✓

Reproductive Cycle – (Menarche to Menopause)

12-Y 50-Y

The reproductive cycle in the female primates (e.g. monkeys, apes and human beings) is called menstrual cycle. The first menstruation begins at puberty and is called **menarche**. In human females, menstruation is repeated at an average interval of about 28/29 days, and the cycle of events starting from one menstruation till the next one is called the **menstrual cycle**. One ovum is released (ovulation) during the middle

(1) On the basis of Ovarian cycle (series of events in the ovaries)

- Follicular Phase
- Ovulation
- Luteal Phase

(2) On the basis of Uterine (menstrual) cycle (series of changes in the endometrium)

- Menstrual phase
- Proliferative phase
- Secretory phase

Menstrual phase (1-5 days)

(3-5)

Breakdown of Endometrium

↓

↓- Level of Progesteron

Maintain Endometrium

↓

Degeneration of Corpus luteum

↓

[↓- Level of L.H] - 14 days

Maintain Corpus luteum

* Blood loss
* ↑ - Prostaglandin

Follicular phase } 6-13 days (7-8 Days)

or

[Proliferative Phase]

Development of Follicles

~~F.S.H~~ →

Inhibit

Dominant Follicle } Release Inhibin

Release

[Estrogen]

↓

Proliferation of Endometrium
(Division)

Ovulation (14- Day)

At the end of Follicular phase.
Size of Follicle = ↑

Estrogen Peak (13th)

Hypothalamus
✓ GnRH
Pituitary

✓ L.H
LH
F.SH

L.H Surge
Peak

(Ovulation)
14th

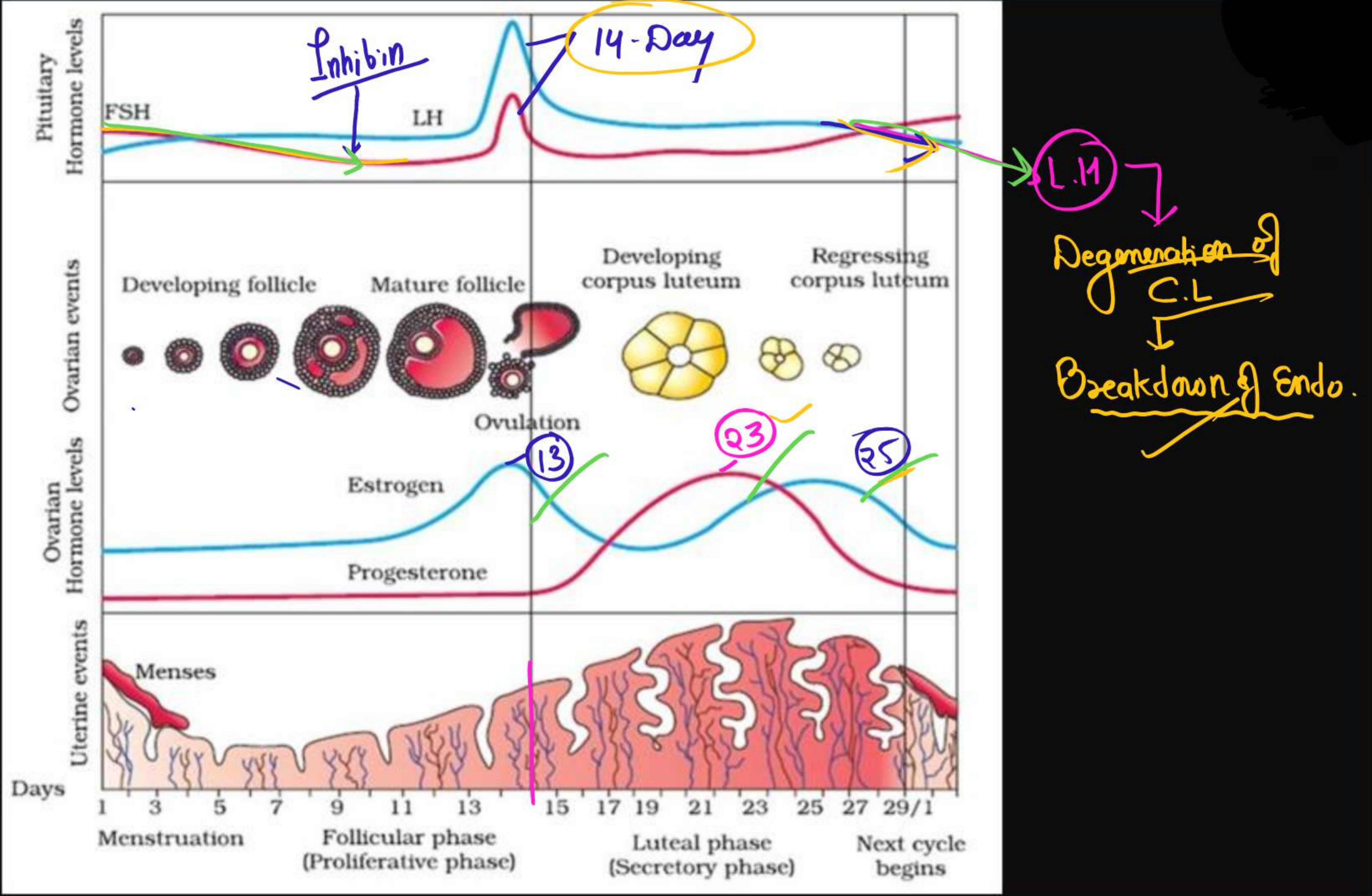
Luteal Phase (Secretory Phase)
(15-28)

Formation of Corpus luteum

Release

① Progesteron
↓
↑ Maintain Endometrium
↓
Endometrium become -
Highly - Glandular &
Secretory
Highly Vascular

② Estrogen
↓
Again Proliferation
or
Thickning
of Endometrium



Pituitary Hormone levels
FSH
LH
Inhibin
14-Day
L.H
Degeneration of C.L
Breakdown of Endo.
Ovarian events
Developing follicle
Mature follicle
Developing corpus luteum
Regressing corpus luteum
Ovulation
Ovarian Hormone levels
Estrogen
Progesterone
13
23
25
Uterine events
Menses
Days
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29/1
Menstruation
Follicular phase (Proliferative phase)
Luteal phase (Secretory phase)
Next cycle begins

3.5 Fertilisation and Implantation

During copulation (coitus) semen is released by the penis into the vagina (insemination). The motile sperms swim rapidly, pass through the cervix, enter into the uterus and finally reach the ampullary region of the fallopian tube (Figure 3.11b). The ovum released by the ovary is also transported to the ampullary region where fertilisation takes place. Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region. This is the reason why not all copulations lead to fertilisation and pregnancy.

[redacted]

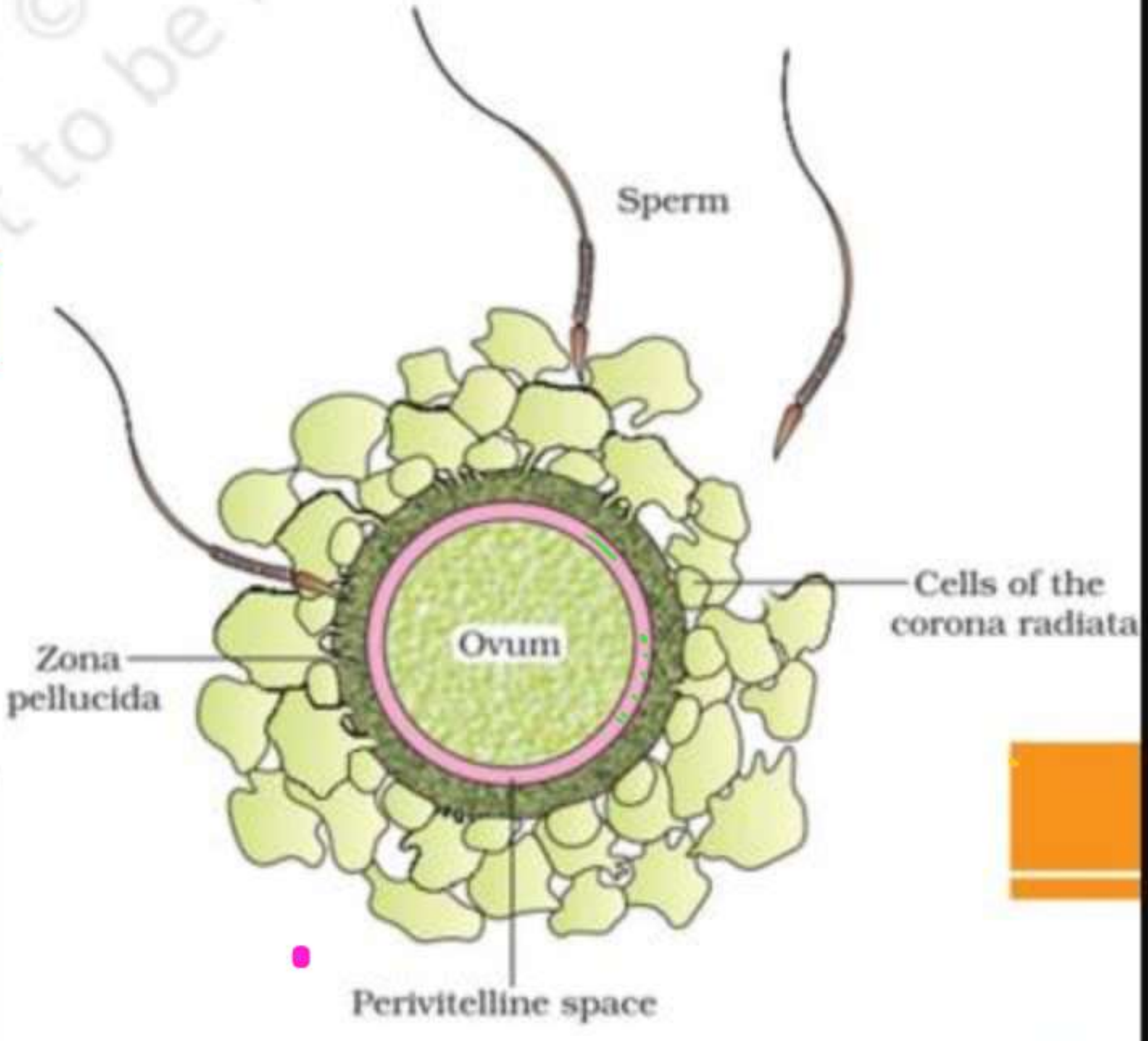
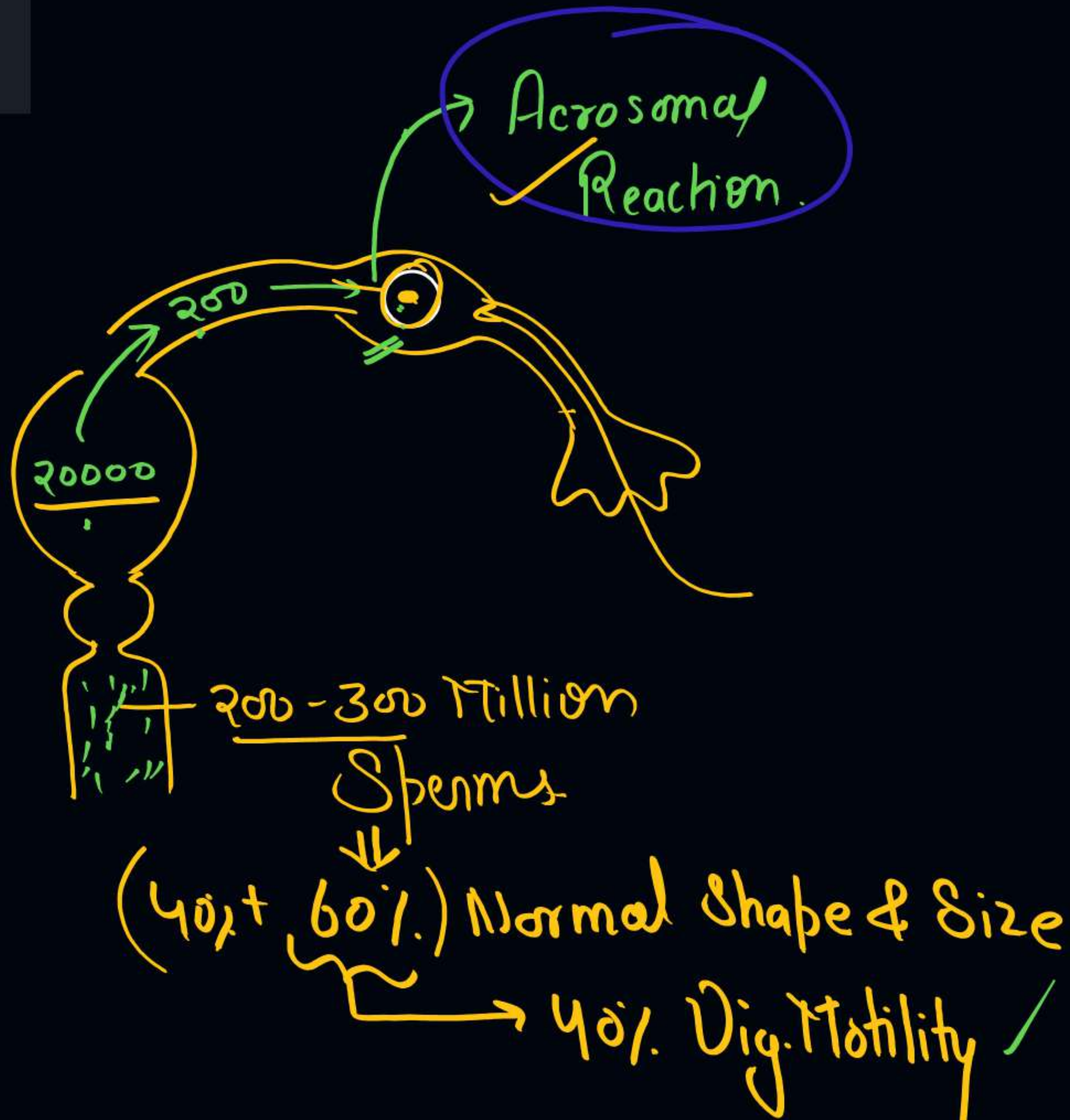


Figure 3.10 Ovum surrounded by few sperms

Fertilization



Hydrolysing Enzymes [Sperm lysin]

1) - Hyluronidase. → Hyluronic acid

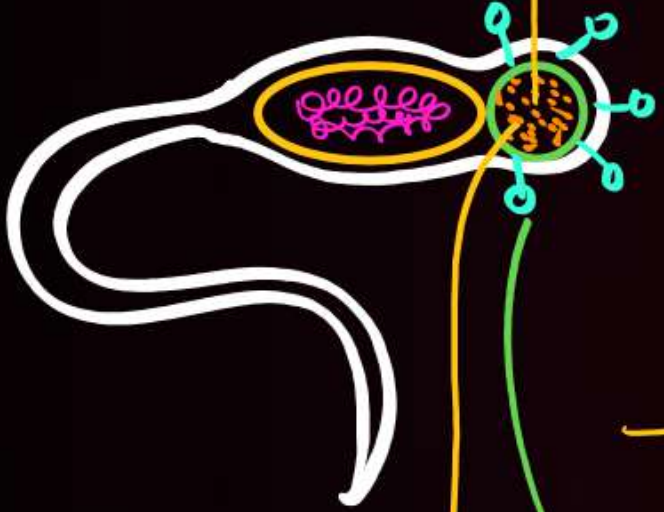
+

2) - Corona penitrating Enzymes → Corona Radiata

+

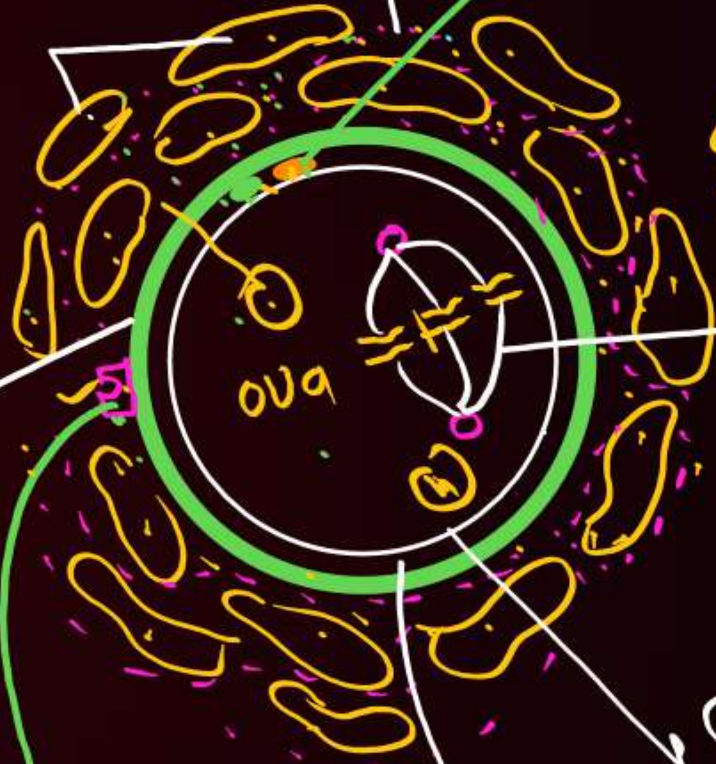
3) - Zonalysin → Zona Pellucida

Or [Acrosin]



Acrosome of Sperm

Antifertilizin ⟷ Fertilizin [Receptor]



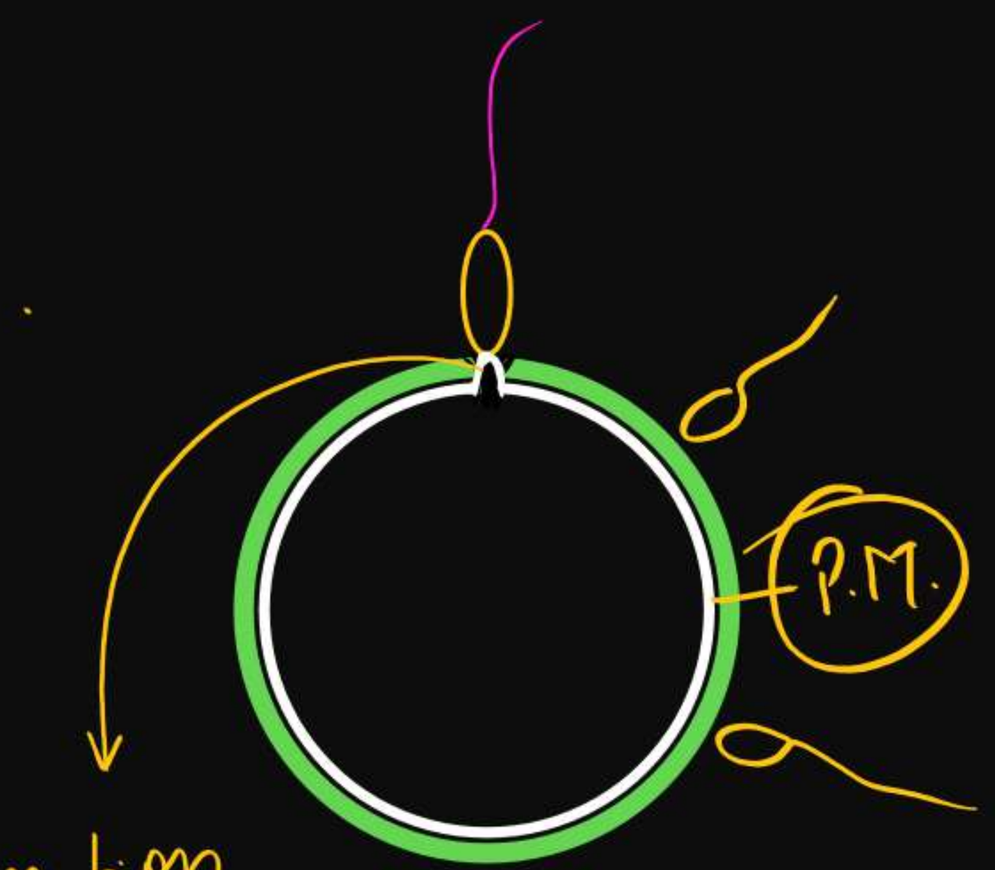
Ist Polar body

Meiosis arrest in Meta-II

P.M

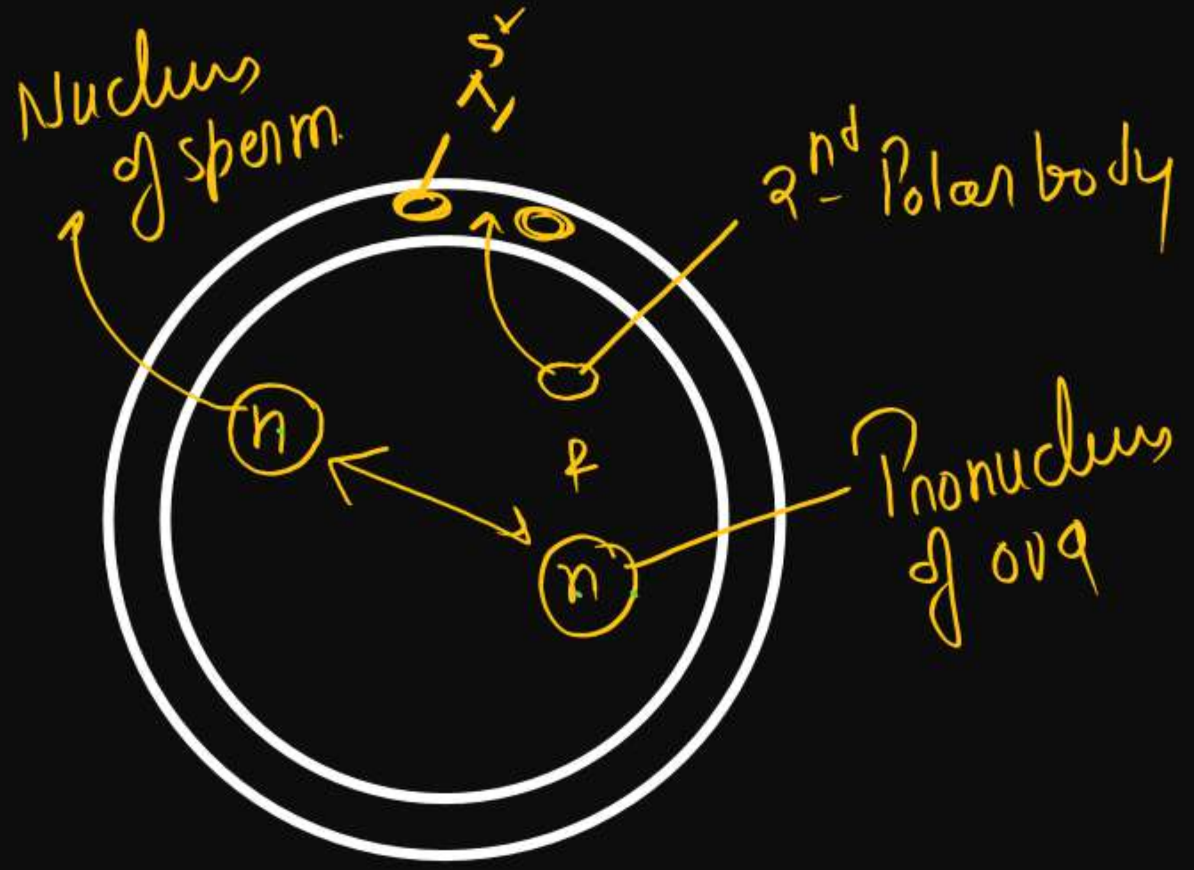
Perivitiline Space

The process of fusion of a sperm with an ovum is called **fertilisation**. During fertilisation, a sperm comes in contact with the *zona pellucida* layer of the ovum (Figure 3.10) and induces changes in the membrane that block the entry of additional sperms. Thus, it ensures that only one sperm can fertilise an ovum. The secretions of the acrosome help the sperm enter into the cytoplasm of the ovum through the zona pellucida and the plasma



Formation of Fertilization Cone → Membrane become depolarised → Block Entery of Additional Sperm.

membrane. This induces the completion of the meiotic division of the secondary oocyte. The second meiotic division is also unequal and results in the formation of a **second polar body** and a haploid ovum (ootid). Soon the haploid nucleus of the sperms and that of the ovum fuse together to form a diploid **zygote**. *How many chromosomes will be there in the zygote?*



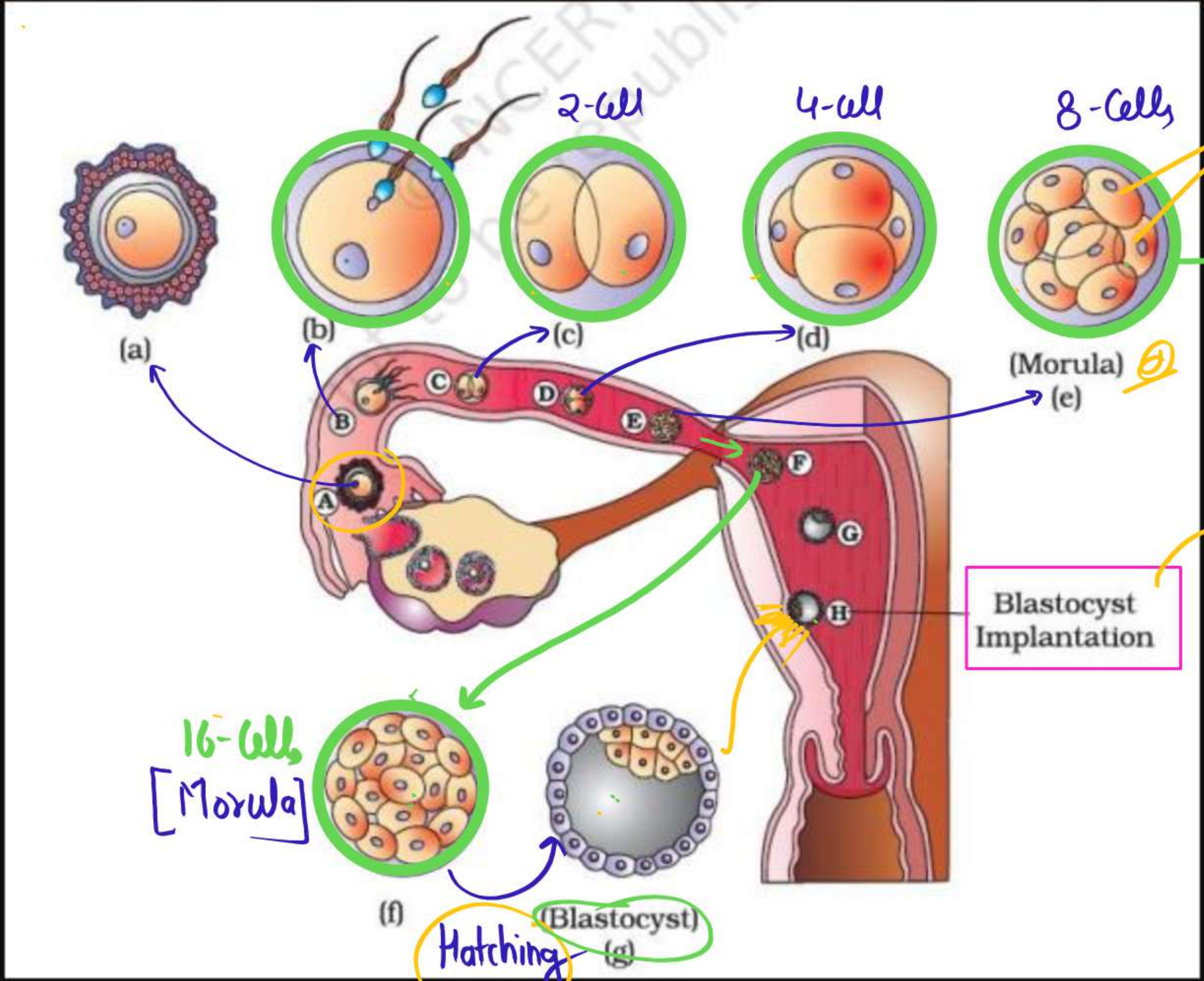
Formation of Zygote

2n

46

Cleavage Start

Rapid Mitotic Division

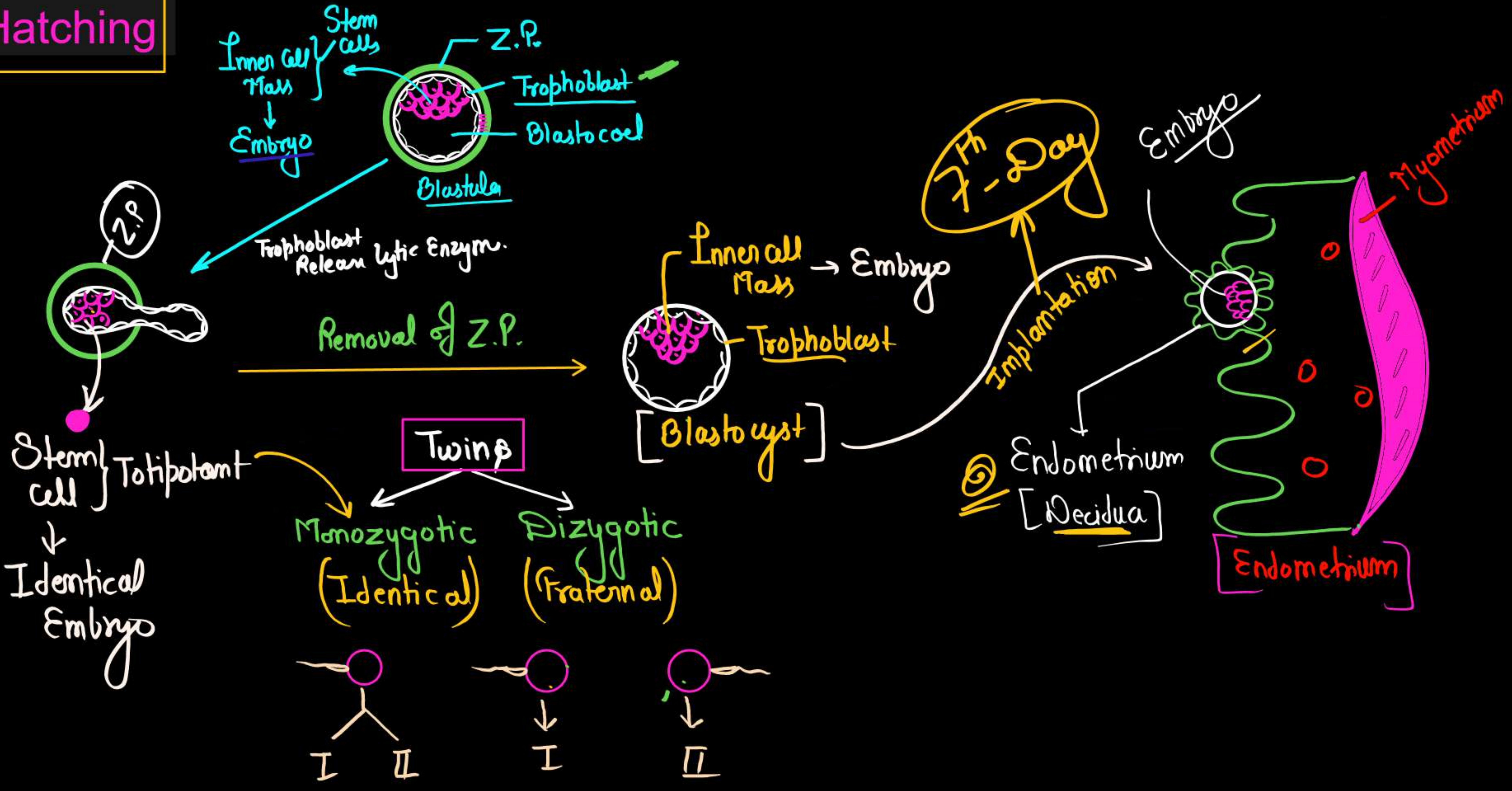


Blastomeres

Z.P

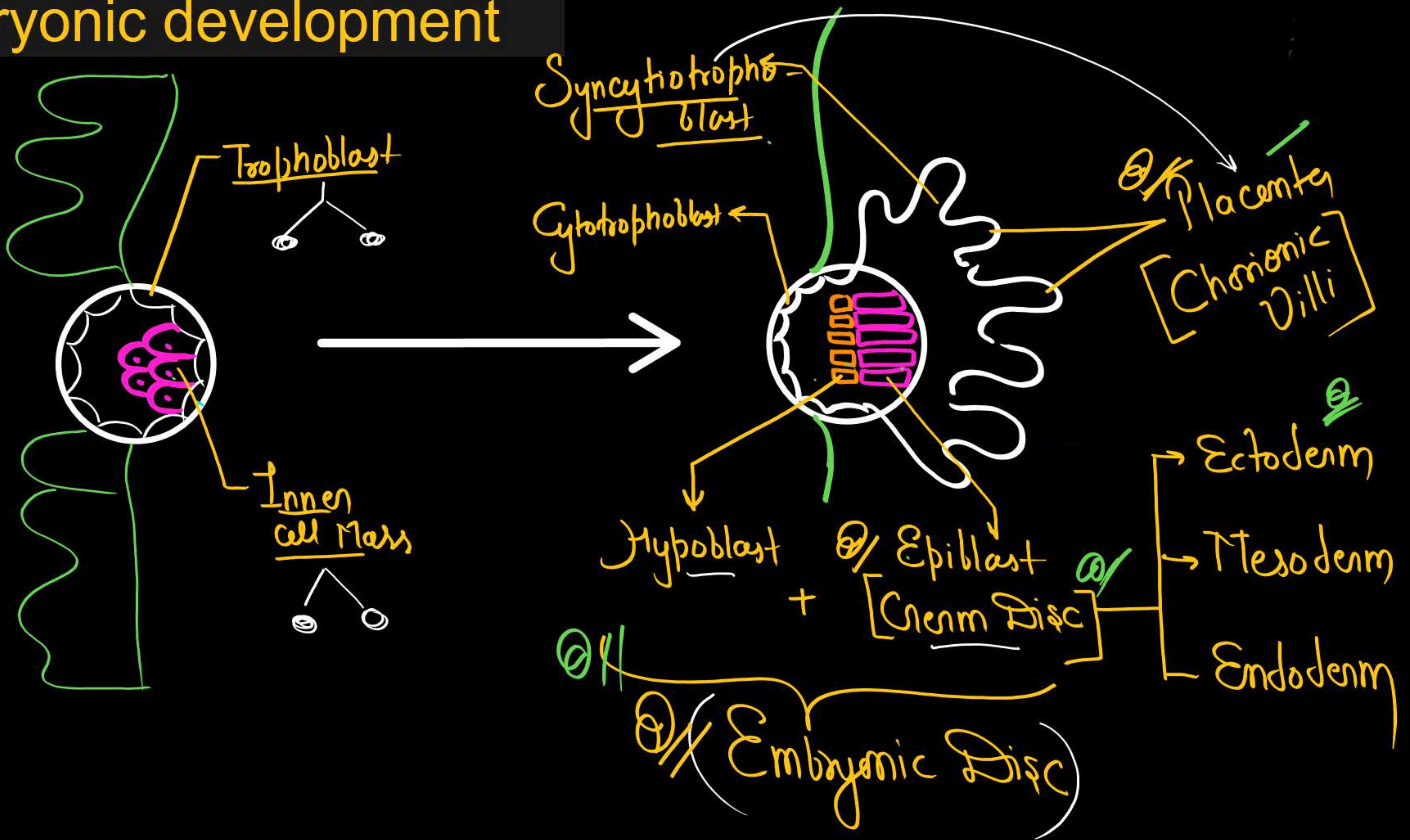
Blastocyst
⊥
Implantation

Hatching



Stem cells
Inner cell Mass
Embryo
Z.P.
Trophoblast
Blastocoel
Blastula
Z.P.
Trophoblast Release lytic Enzyme.
Stem cell
Totipotent
Identical Embryo
Removal of Z.P.
Twins
Monozygotic
(Identical)
Dizygotic
(Fraternal)
I
II
I
II
Inner cell Mass → Embryo
Trophoblast
[Blastocyst]
7th Day
Implantation
Embryo
Myometrium
Endometrium
[Decidua]
Endometrium

Embryonic development



Gastrulation (15th day)

- It is the first major event of the third week of development.
- It occurs about 15 days after fertilization.
- In this process, the bilaminar embryonic disc, consisting of epiblast and hypoblast, transforms into a trilaminar embryonic disc consisting of three primary germ layers:

➢ Ectoderm
➢ Mesoderm
➢ Endoderm

Formation of Embryo.

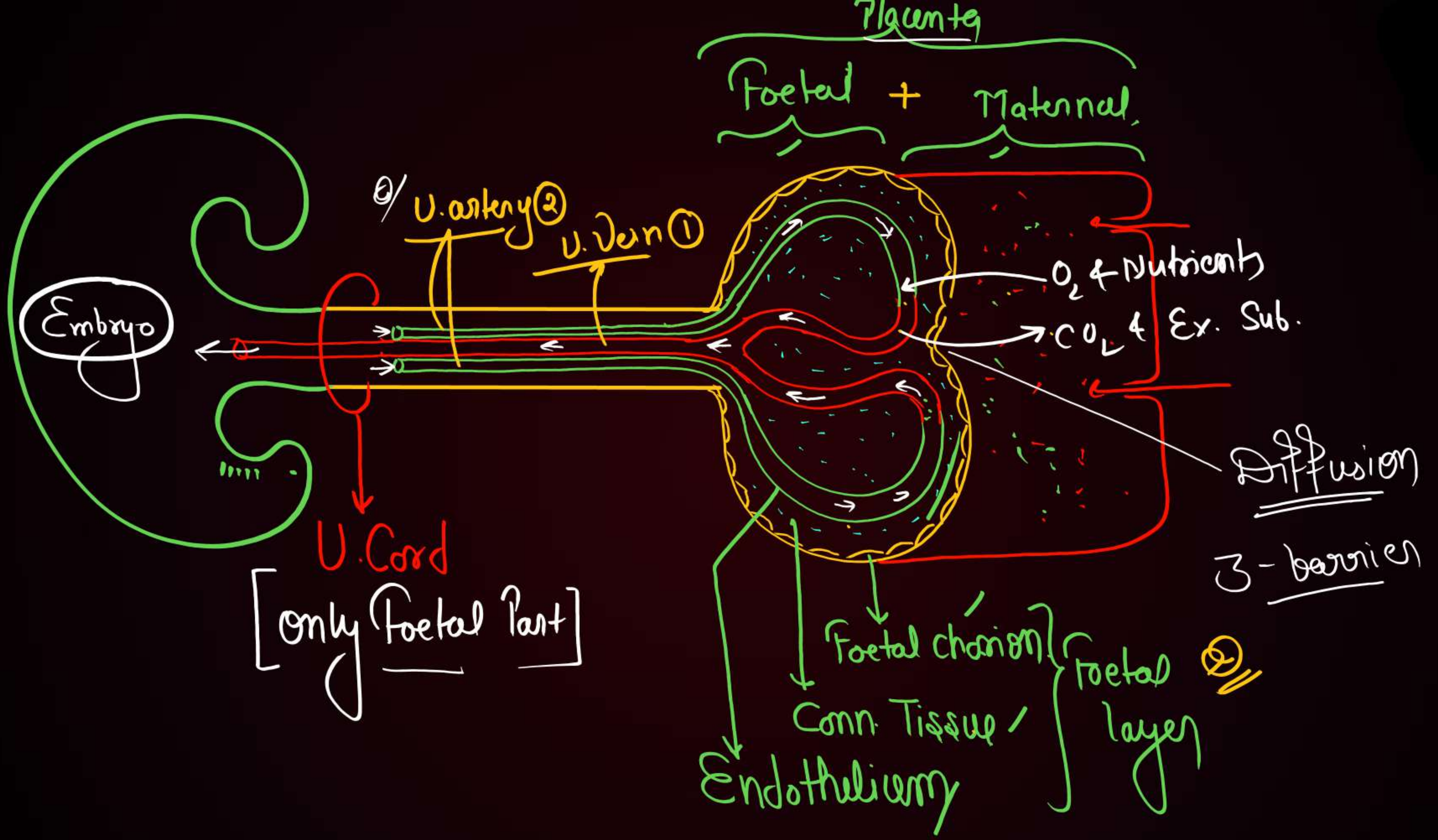
3.6 Pregnancy and Embryonic Development

After implantation, finger-like projections appear on the trophoblast called **chorionic villi** which are surrounded by the uterine tissue and maternal blood. The chorionic villi and uterine tissue become interdigitated with each other and jointly form a structural and functional unit between developing embryo (foetus) and maternal body called **placenta** (Figure 3.12).

= Foetal Membrane
&
= Maternal Blood

Placenta

[$\underbrace{\text{Haemo}}_{\text{Maternal}}\ \underbrace{\text{chorial}}_{\text{Foetal}}$ Placenta]



Placenta
Foetal + Maternal
Embryo
O/ U. artery ②
U. Vein ①
O_2 & Nutrients
CO_2 & Ex. Sub.
Diffusion
3 - barrier
U. Cord
[Only Foetal Part]
Foetal chorion
Conn. Tissue
Endothelium
Foetal layer

Function of Placenta

(1) Placenta act as $\boxed{\text{endocrine gland}}$ and release following hormones-

Maintain Corpus luteum after fertilization.

- Estrogen ✓
- Progesterone ✓
- Human chorionic gonadotrophin (hCG)
- Chorionic thyrotrophin ✓
- Chorionic corticotrophin ✓
- Human placental lactogen (hPL) hormone or chorionic somatomammotrophin ✓
- Relaxing ✓

Release only During Pregnancy

also Release from ovary (C.L)

Gestation Period-

- It is a duration b/w Fertilization to Parturition (Child birth)
- In Human – 9 month/38 week/266 days/280 days from last menstrual
- It divided in to 3 trimester.

Note - NCERT

Gestation Period of –

Dog	–	60-65 days
Cat	–	52-65 days
Elephant	–	607-641 days

Imp

First trimester (3 month)

1 st month heart is formed.
2 nd month develops limbs and digits
3 rd month most of the major organ systems are formed, eg, limbs and external genital organs.
5 th month The first movements of the foetus and appearance of hair on Head
6 th month (end of second trimester), the body is covered with fine hair, eye-lids separate, and eyelashes are formed.
9 months - foetus is fully developed and is ready for delivery.

PARTURITION (Neuro-Endocrine) @

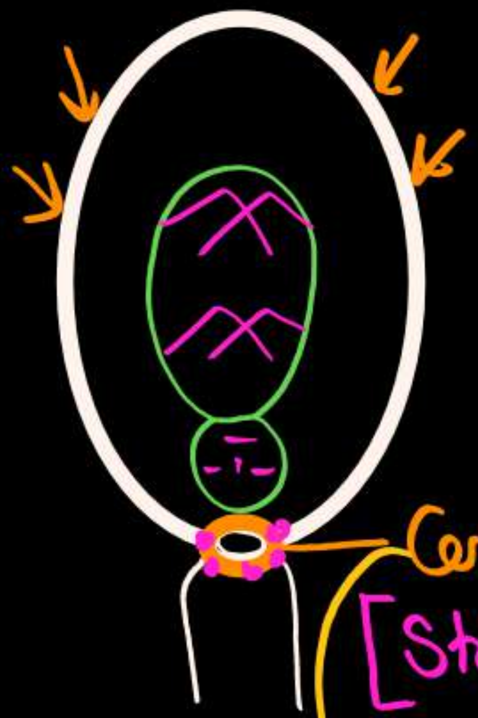
At the end of Gestation = Placenta & Foetus Fully Developed

Placenta Weak (↓ - Progesteron & ↑ - Estrogen)

@ ↑ - Prostoglandin Hor.

Mild Contraction in Uterine Muscles

@ Foetus Ejection Reflex



Cervix

[Stretch Receptor]

Hypothalamus

↓

P. Pituitary

↓

oxytocin ↑↑↑

↑ Ca^{++} ↑ Prostaglandin

↑↑↑ Contraction in Uterus

(10-Cm)

to

Birth

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PDF

Reproductive Healthy

Conditions it may diagnose: This procedure is used to test for the presence of certain genetic disorders such as, **down syndrome, haemoplilia, sickle-cell anemia, muscular dystrophy** etc., and determine the survivability of the foetus.

Misuse of Amniocentesis – @//

It is being used to kill the normal female foetus. @//

RCH

'Saheli'– a new oral contraceptive for the females–was develope by scientists at Central Drug Research Institute (CDRI) in Lucknow, India.

Non Steroidal (Centchroman) = Antiestrogen Effect @//

Stop thickning of Endometrium

Prevent Implantation

@ (Used once in a week)

Contraceptive Methods

1 @ = 100%

vvvvvvImp

Temporary

Natural

1) Rhythm/Periodic abstinence/Safe Period
2) Coitus interruptus or Withdrawal Method
3) Lactational Amenorrhoea (6-Month)
   ↳ PRL → Inhibit GnRH

Artificial

1) Barrier — Condom < ♂ ♀ → Protect from STD/AIDS @
2) IUD vvvImp
3) Oral Contraceptive
4) Injectable
5) Implant

Permanent

1) Vasectomy → ♂
2) Tubectomy → ♀

(1) Natural Methods of Contraception

These work on the principle of avoiding chances of ovum and sperms meeting

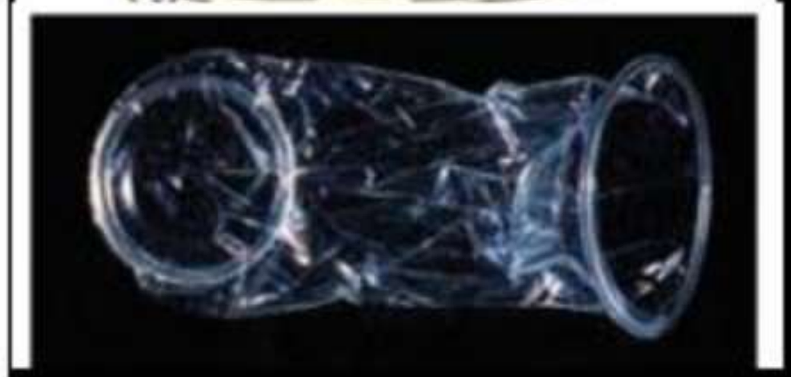
- Rhythm Method or Safe period or Periodic abstinence or Calendar Method
- Withdrawal Method or Coitus Interruptus (High failure Rate)
- Lactational Amenorrhoea

(2) Barrier Methods-

In barrier methods, ovum and sperms are prevented from physically meeting with the help of barriers.

- Condoms — Irreversible
- Fem Shield (Female Condoms)
- Cervical Cap (Reusable)
- Diaphragm(Reusable)
- Vault (Reusable)

Reversible



IUDs

vvvv Imp   1 @ 100%.

- These devices are inserted by doctors or expert nurses in the uterus through vagina.
- IUDs increase phagocytosis of sperms within the uterus and the Cu ions released suppress sperm motility and the fertilizing capacity of sperms.
- IUDs are ideal contraceptives for the females who want to delay pregnancy and/or space children.
- It is one of most widely accepted methods of contraception in India.

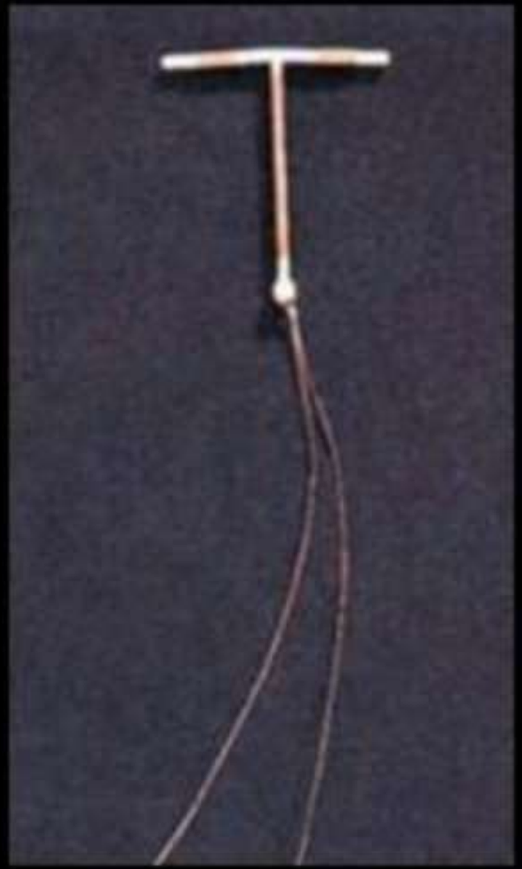
Types of Intra Uterine Devices-

non-medicates

eg - Lippe's loop

Copper Releasing

eg - Cu T-200
Cu T-380
Cu-T-7 etc

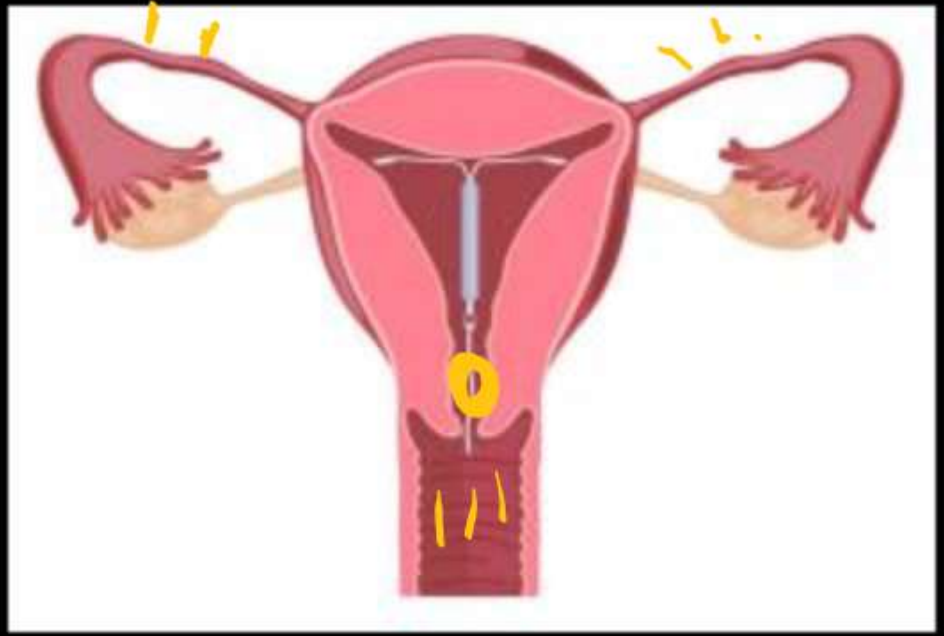


Hormone Releasing

vvvImp eg - Progestasert
- levonorgestrel (LNG 20)

vvvImp

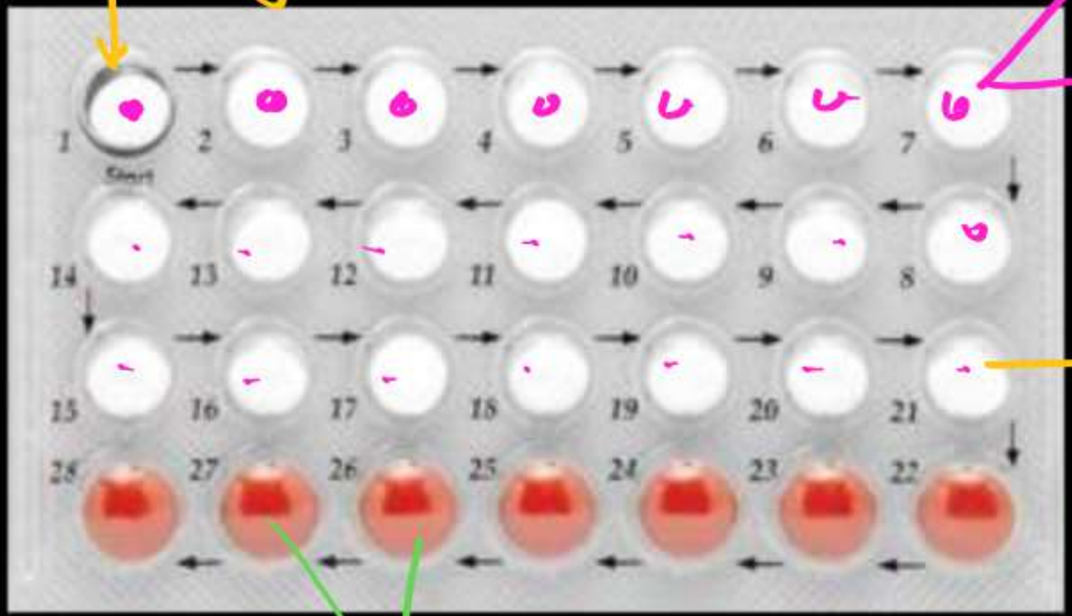
Release Progesteron



(1) Oral Contraceptive Pills(OCPs)

The pills are taken orally for 21 days in a menstrual cycle starting from 5th day and ending on 25th day. However, it is advisable to restart the course after a gap of 7 days.

5th day of M.C



Progesteron Conc = Same.

Estrogen = ↑ - order

25th day of M.C

(Iron Pills) = 7

(2) Injectables (DEPO-PROVERA) -

- They are depot-medroxy progesterone acetate (DEMPA) effective for three month.

(3) Implants - (Subdermal)

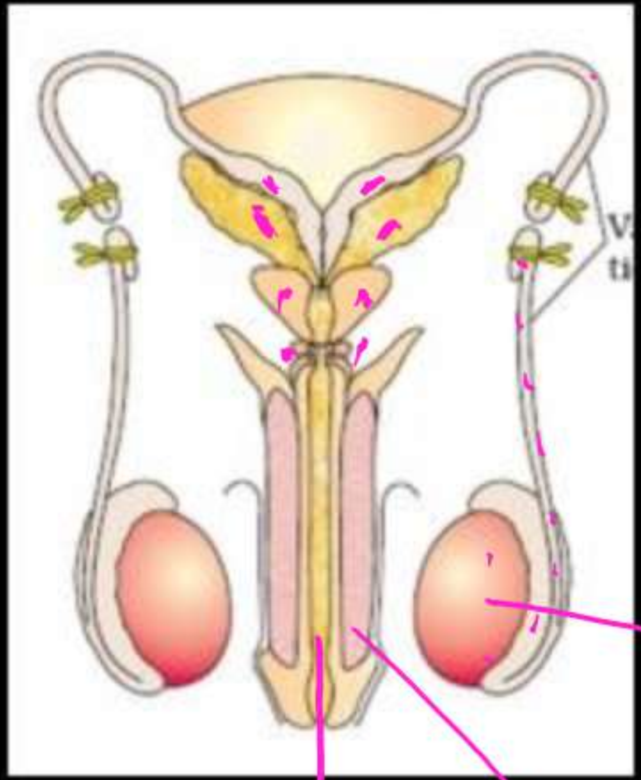
- Norplant is progestin only containing devise has six small silicone (permeable) capsules containing levonorgesterol and these are inserted under skin of left upper arm (5 years).

(4) Emergency Contraception-

- Administration of progestogens or progestogen-estrogen combinations or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives
- Mifepristone (RU486) (antiprogrsterone) is a single pill

(B) Permanent Methods/ Stérilisation

(1) Vasectomy



Spermatogenesis ✓

Erection of Penis ✓

Ejeculation of Semen with Sperm

(2) Tubectomy

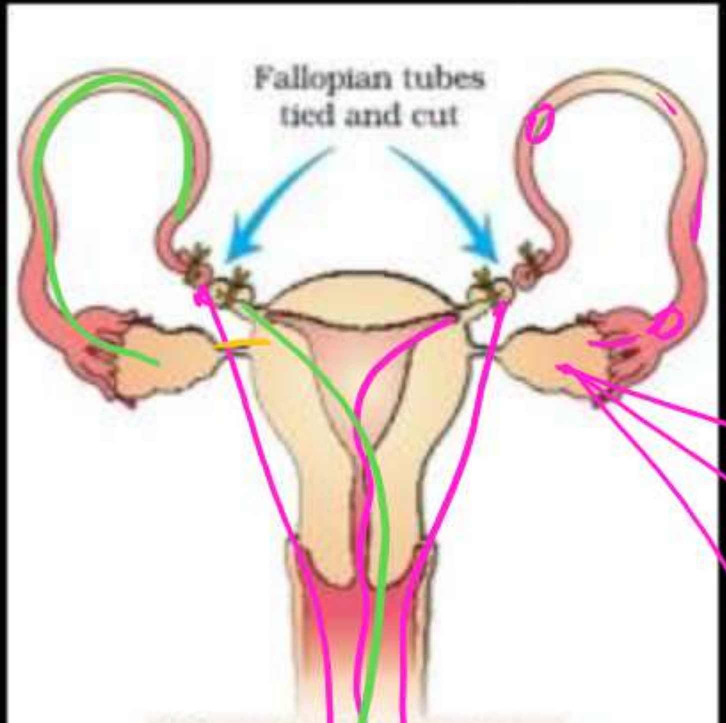


Figure 4.4 (b) Tubectomy

Menstrual ✓

ovulation ✓

Fertilization ✗

MEDICAL TERMINATION OF PREGNANCY (MTP)

- Nearly **45 to 50 million MTPs** are **performed in a year all over the world which accounts to 1/5th of the total number of conceived pregnancies in a year.** 20%
- Government of India **legalised MTP in 1971** with some strict conditions to avoid its misuse.
- MTPs are considered relatively **safe during the first trimester, i.e., upto 12 weeks of pregnancy**. Second trimester abortions are much more riskier.

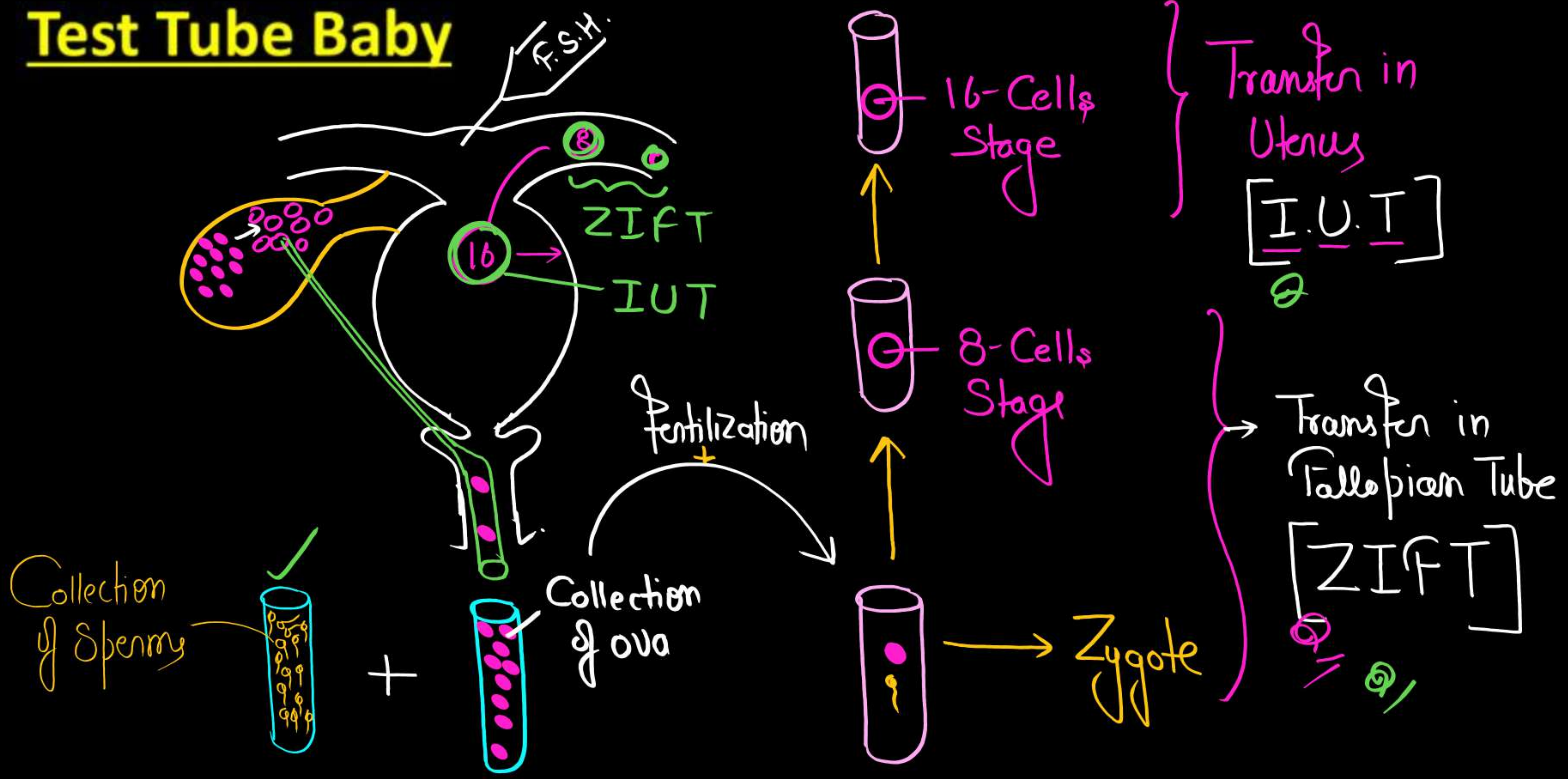
Amendment Act = 2017

- → First = 12 Weak = One Doctor
- → 12 – 24 Weak = Two Doctors

Infertility –

- Inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called infertility.
- A large number of couples all over the world including India are infertile.

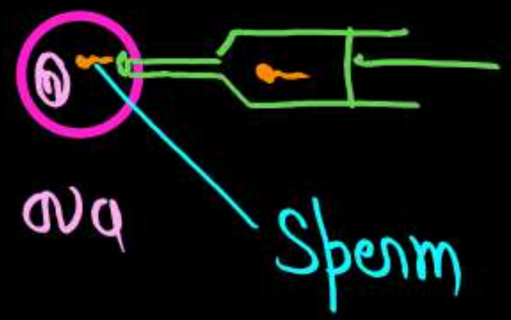
Test Tube Baby



ART

Invitro fertilization (IVF)
[Fertilization Outside the body]
↓
[Embryo transfer] = [E.T.]

- Test Tube baby
  ↓
  - ZIFT = Zygote / 8-cells
  - IUT = 16-cells
- Intracytoplasmic Sperm Injection } ICSI



Invivo fertilization
[Fertilization inside the body]
↓
[Gametes Transfer]

- Male gamets Transfer
  ↓
  - A.I = Vagina
  - IUI = Uterus
- Female gamutes Transfer
  ↓
  - GIFT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Thank You

Human Health and Diseases

Immunity

Innate
Non specific

Acquired
Specific
Memory

Physical barrier → Skin, Mucus Epithe.

Physiological barrier → $\underbrace{\text{Mucus, Saliva, Tears}}_{\text{Lysozymes}}$, $\underbrace{\text{HCl}}_{\text{Stomach}}$

Cellular barrier
- Non Phagocytic cells → Natural Killer cells
  - Cancer cell
  - Virus Infecte cell
- Phagocytic cells
  - Blood = Monocytes, Nutrophils (PMNL)
  - Conn. Tissue = Macrophage

Cytokine barrier → Virus infected cells Release Protein → $\boxed{\text{Interferon}}$

Imp

Interferon protect Non infected cells from Viral infection

ACQUIRED IMMUNITY-

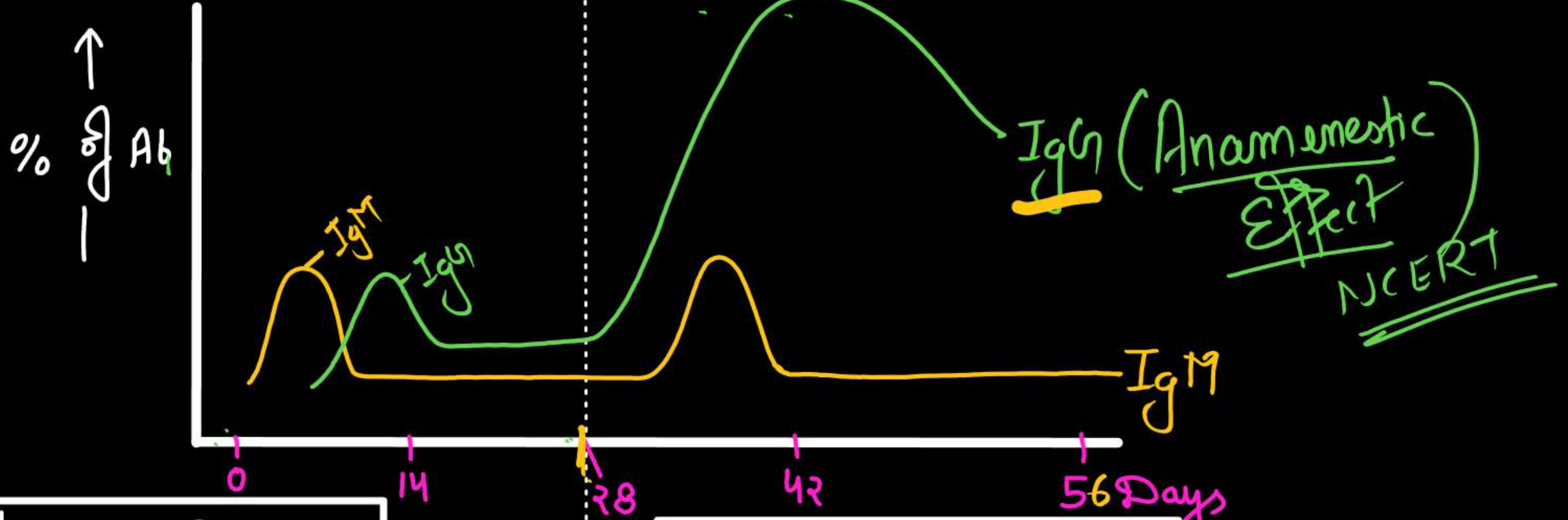
Acquired immunity, on the other hand is pathogen specific. It is characterised by memory. Q/

Features of Acquired Immunity –

I. Specificity
II. Diversity
III. Memory
IV. Discrimination

b/w — Self = No Response. Q/ → Pri. Immun Responses.

Non Self = Responses — Sec. Immun. Responses Q/



Ist Immun Response	IInd Immun Response
• Ist Encounter.	• IInd Encounter - Same antigen
• Involve - B-lym	• Involve Memory cells.
• Formation of Ab = Low Intensity	• High Intensity.
• IgM > IgG	• IgG >>>> IgM

Structure of Antibody (Ab)- (Immunoglobulin) = Glycoprotein

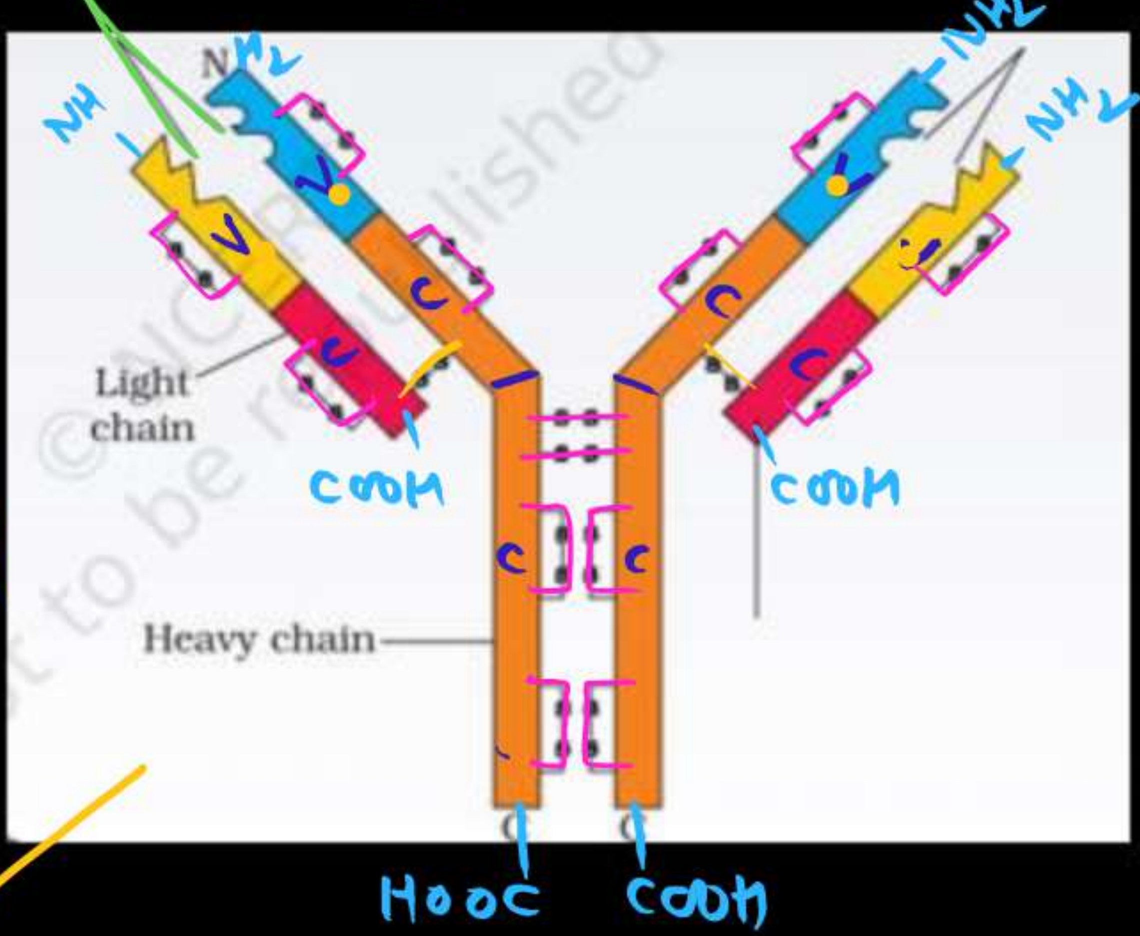
Q/ Antigen Binding Sites } N. Terminus

Total Disulphide bond = 16

b/w Heavy chain = ?

b/w H + L chain = ?

vvImp



E}V

C

4 - Polypeptides (H_2 L_2)

Two Heavy chain

Two light chain

γ - IgG (Max^m)

μ - IgM

α - IgA

δ - IgD

ε - IgE (Min^m)

Types of Acquired Immunity

(B-lymphocytes based)
or
Antibody Mediated Imm. (AMI)
or
Humoral Immunity
(Body fluid)

- Active B-lym.
  [Plasma cells]
  (Formation of Antibody)
- Inactive B-lym.
  [Memory cells]

(T-lymphocytes based)
or
Cell Mediated Immunity (CMI)

Responsible for Tissue grafting or Organ Transplantation

- T_H ⟶ Control all Immun cells.
- T_K ⟶ Kill — [Cancer cells, Virus Infected cells]
- T_S
- T_{Memory}

Tissue graft/ Organ transplantation (Cell Mediated Immunity)

- Matching → Blood Group
  → Human leucocyte Antigen (HLA) → MHC-I, MHC-II
- Immuno suppresive Drugs → Cyclosporin, Cortisol

Active and Passive Immunity

Antibody Produced by Host

A

Natural: All Infection

Artificial: All Vaccination

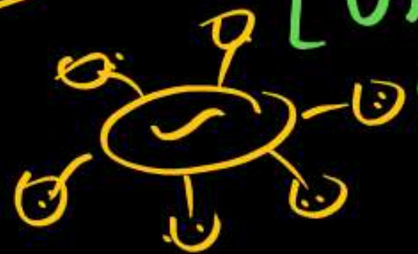
Prepaired antibody Provided to Host

P

Natural:
IgA = Colostrum
IgG = Placenta

Artificial:
A.T.S.
Antivenum

Vaccination

- Ist-generation [Whole Organism]
  
  - Killed
  - Live attenuated (weak)
  - eg- OPV = Live atte, BCG, Small pox etc.
- IInd-generation [Sub-unit]
  - Recombinant → Hepatitis-B (NCERT)
  - Toxoids [Inactive toxin] → D.P.T.
- IIIrd-generation [DNA] Vaccines → Hepatitis-B

Allergy

- Antibody Responsible for allergy = IgE
- Chemicals " " " " = Mast Cells — Histamine, Serotonin
- Anti allergic Drugs = Antihistamin, Adrenalin & Steroids.

Autoimmunity

= Due to Genetic — Unknown Reason
- eg = Rheumatoid arthritis etc.

Types of lymphoid organs

- Pri. Lym. Organ {Formation & Maturation} Red bone marrow & Thymus
- Sec. lym. Organ {Interaction of lympho.} Spleen, Lymph Nodes, MALT etc.
  - MALT — 50%

Types of diseases

100% 10

Congenital diseases

(Genetical Diseases)

Acquired diseases

Communicable (Infectious)

Spread through —
- Bacteria
- Virus
- Fungus
- Helminthes
- Protozoa etc.

eg - AIDS

Noncommunicable (Non infectious)

eg - Diabetes
- Cancer
- Deficiency Disorders
- Drugs & Alcohol Abuse

= etc.

Bacterial Diseases	Pathogen	Symptoms & Pathogenicity
Typhoid	*Salmonella typhi*	➢ Enters small intestine through **contaminated food and water** and migrate to other organs through blood. ➢ Sustained high **fever (39-40°C).** ➢ Weakness, constipation, stomach pain, Headache, loss of appetite. ➢ In severe cases **intestinal perforation, (death).** ➢ Confirmed by **widal test**. ➢ Mary Mallon, **nicknamed-Typhoid Mary** (Carrier of typhoid)
Pneumonia	*Streptococcus pneumoniae* *Haemophilus influenzae*	➢ By **droplet or aerosol infection** ➢ Infects alveoli of the lungs. ➢ Alveoli get filled with fluid leading to severe problem in respiration. ➢ Fever with chills, cough & headache. ➢ In **severe cases lips and nails turns gray to bluish in colour**.

Diphtheria	Corynebacterium diphtheriae	➢ High grade fever, affects throat. ➢ Causes suffocation DPT
Plague (Black death)	Bacteria Yersinia pestis (Parasite of - Rat flea/ Xenopsylla cheopis)	➢ High fever, headache. ➢ Enlargement of axillary lymph nodes ➢ Red patches appear on skin which turns black - Death

Viral Diseases	Pathogen	Symptoms & Pathogenicity
Common cold	Rhino viruses (Group of viruses)	➢ One of the **most infectious** human ailments. Ø// ➢ Transmits through **droplet** resulting from cough, sneeze etc. ➢ Infect nose and respiratory passage **but not the lungs**. Ø// ➢ Nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness. ➢ Usually last **for 3-7 days.** Ø//
Hepatitis-B	HBV (ds DNA)	➢ Severe liver damage, jaundice. ➢ Recombination DNA vaccine ➢ Transmits-through parenteral and sexual route (Can cross placenta)
Polio	Polio virus (ss RNA)	➢ Inflammation of nervous system, stiffness of the neck, weakness of particular skeletal muscles. ➢ High fever, pain in body, paralysis of limb muscles.

DENGUE	CHIKUNGUNYA
1) Causative agent = Dengue Virus – RNA [Flaviviridae - family]	Causative agent = Chikungunya Virus – RNA [Togaviridae - family]
2) = Vector = Female Aedes aegypti Mosquito	2)- Vector = Female Aedes aegypti Mosquito

Helminthic Diseases	Pathogen	Symptoms & Pathogenicity
Ascariasis	Ascaris Lumbricoides (Common Round worm)	➢ Intestinal parasite ➢ Internal bleeding, muscular pain, fever, anaemia, blockage of the intestinal passage. ➢ Eggs of the parasite excreted along with the faeces of infected person which contaminate soil, water, plants etc. ➢ Infection occurs by taking contaminated water, fruits, vegetables etc.
Elephantiasis (Filariasis)	Wuchereria (W. bancrofti) (Filarial worm)	➢ Slowly developing chronic inflammation of the organs in which they live for many years. ➢ Usually the lymphatic vessels of the lower limbs get blocked. ➢ Genital organs are also often affected, (swelling in tests & scrotum). ➢ Transmitted to a healthy person through the bite by the female mosquito vector (Mainly Culex).

Fungal Diseases	Pathogen	Symptoms & Pathogenicity
Ringworms	**Microsporum, Trichophyton & Epidermophyton**	➢ One of the most common infectious disease. ➢ Appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp. ➢ Generally acquired from soil or by using towels, clothes or even the comb of infected individuals.

Figure 8.3 Diagram showing ringworm affected area of the skin

Protozoan Diseases

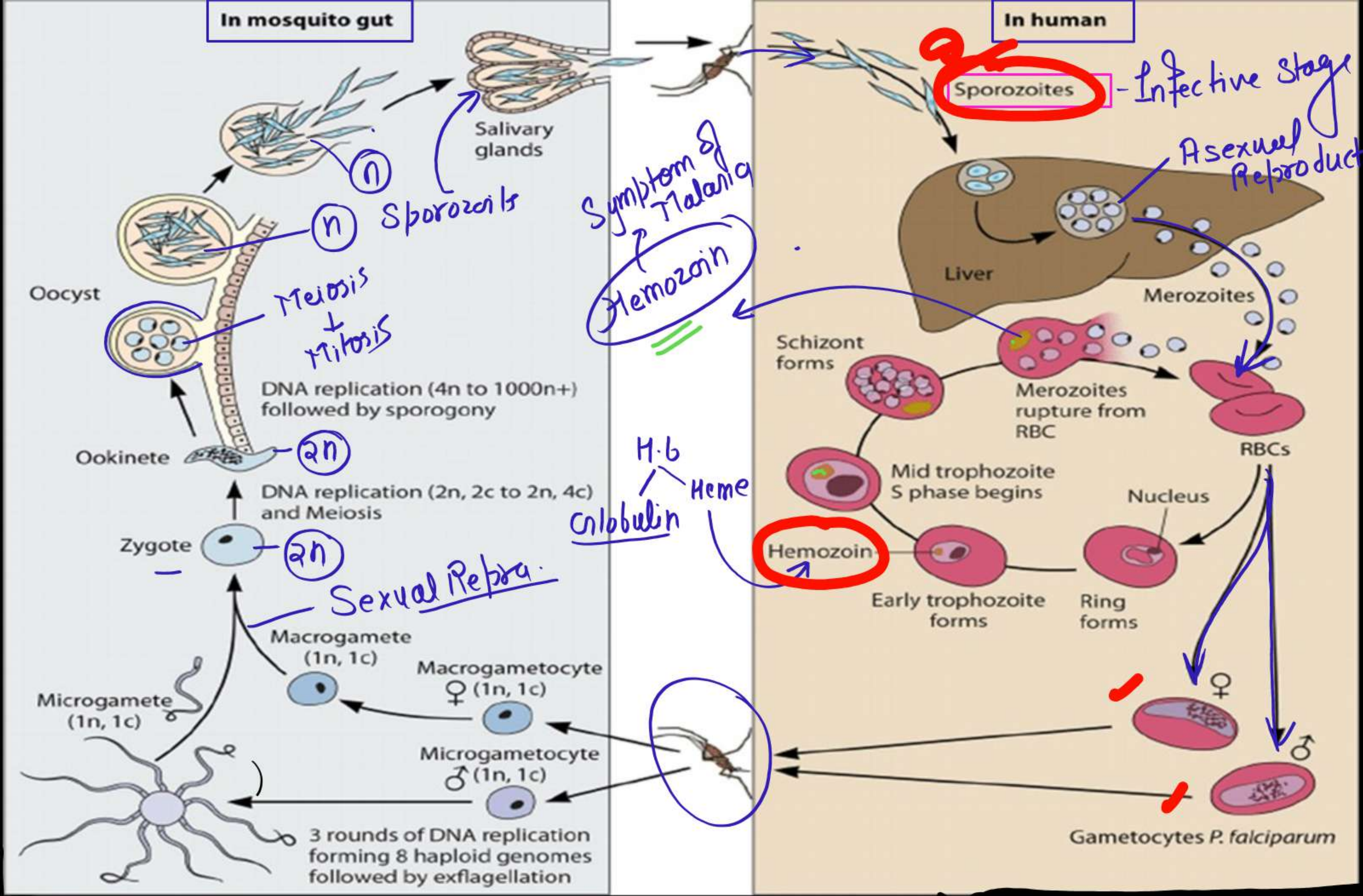
1 **Malaria** → Protozoa (Plasmodium) @

Different species of $\boxed{\text{Plasmodium}}$ which infect humans are-

- P. Vivax [60-65%] → $\boxed{\text{Benign}}$ Tertian Malaria
- P. Malaria [1%] → Quartan Malaria.
- P. falciparum [30-40%] → $\boxed{\text{Malignant Tertian Malaria}}$ NCERT vvvImp @

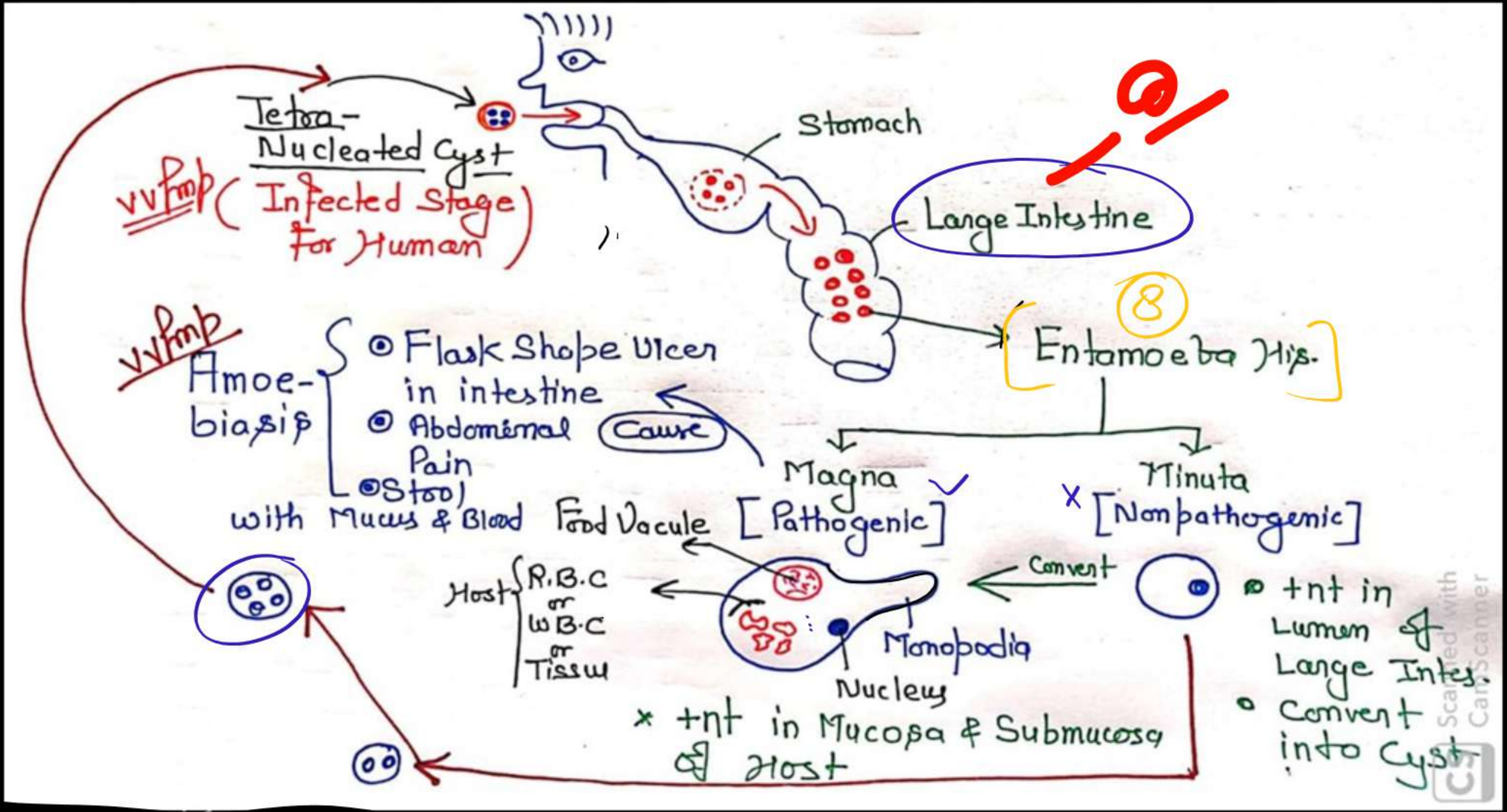
Lifecycle → Digenetic – Pri. Host = Mosquito (Female Anaphilis) = Stomach

Sec. Host = Man – Liver, R.B.C


In mosquito gut
In human
Salivary glands
Sporozoits
n
n
Oocyst
Meiosis + Mitosis
DNA replication (4n to 1000n+) followed by sporogony
Ookinete
2n
DNA replication (2n, 2c to 2n, 4c) and Meiosis
Zygote
2n
Sexual Repro.
Macrogamete (1n, 1c)
Microgamete (1n, 1c)
Macrogametocyte ♀ (1n, 1c)
Microgametocyte ♂ (1n, 1c)
3 rounds of DNA replication forming 8 haploid genomes followed by exflagellation
Symptom of Malaria
Hemozoin
H.b
Globulin
Heme
Sporozoites
- Infective stage
Asexual Reproduction
Liver
Merozoites
Schizont forms
Merozoites rupture from RBC
RBCs
Mid trophozoite S phase begins
Nucleus
Hemozoin
Early trophozoite forms
Ring forms
♀
♂
Gametocytes P. falciparum

(2) Amoebiasis (amoebic dysentery)- Entamoeba histolytica

(parasite in the large intestine of human) - Q *

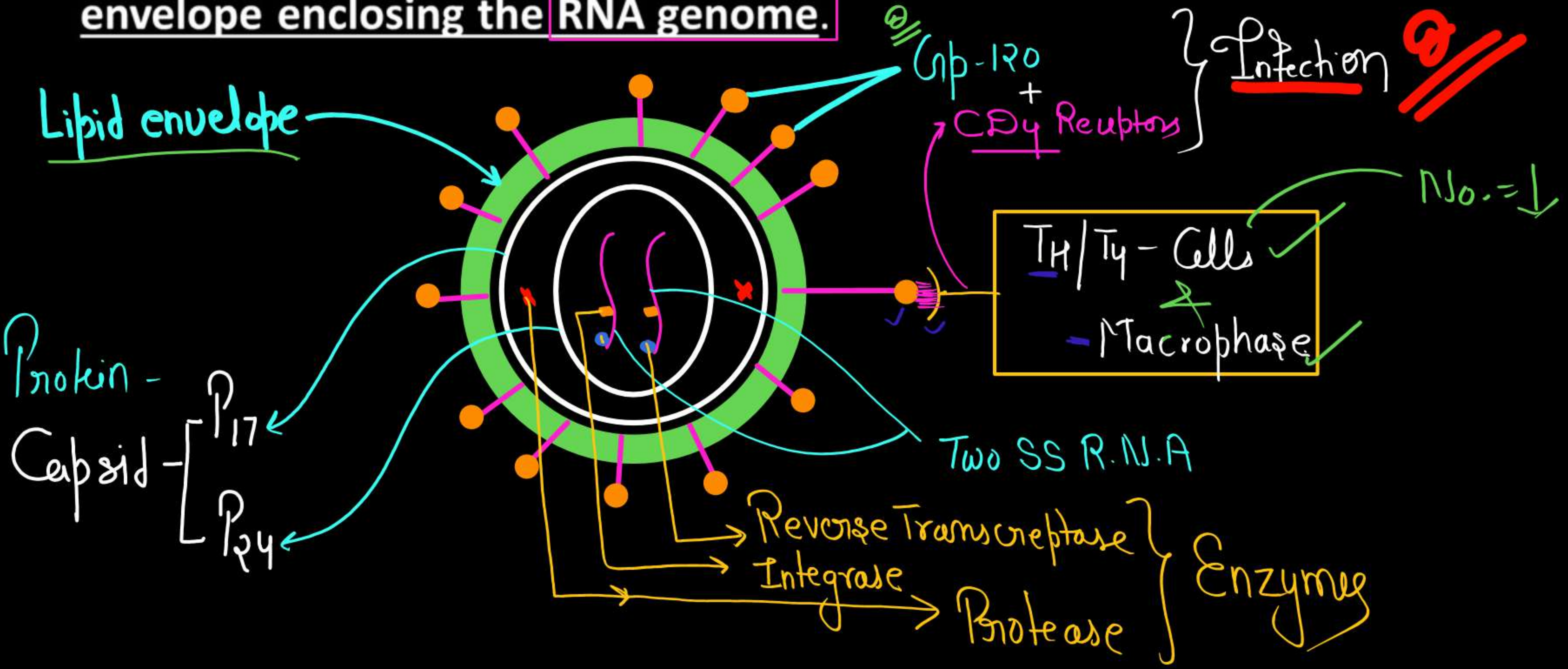


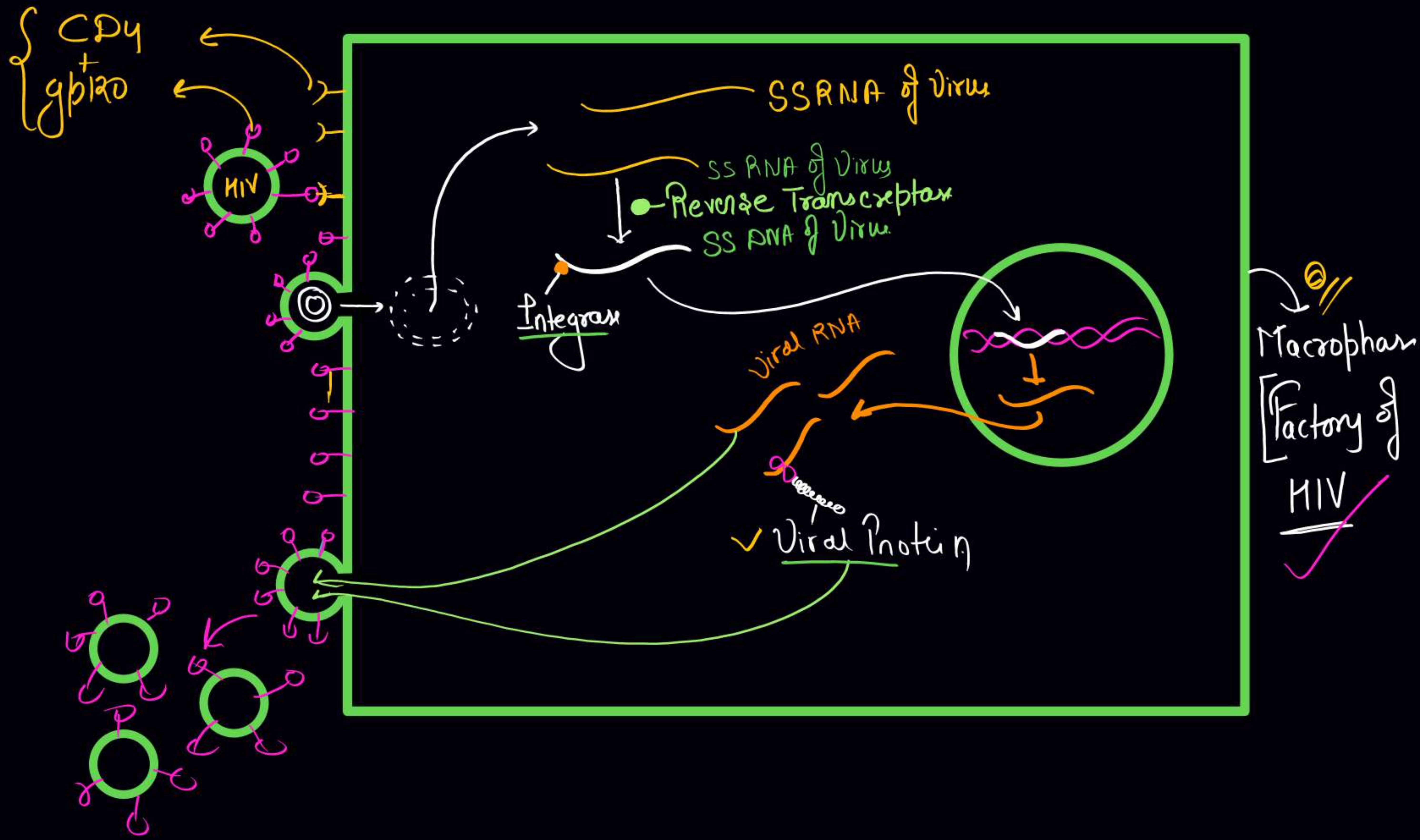
AIDS

- The word AIDS stands for **Acquired Immuno Deficiency Syndrome.** This means deficiency of immune system, acquired during the lifetime of an individual indicating that it is not a congenital disease.
- 'Syndrome' means a group of symptoms.
- AIDS was first reported in **1981** and in the last twenty-five years or so, it has spread all over the world killing more than **25 million persons**

HIV

- AIDS is caused by the Human Immuno deficiency Virus (HIV), a member of a group of viruses called **retrovirus,** which have an **envelope enclosing the RNA genome**.




CD4
+
gp120
HIV
SSRNA of Virus
SS RNA of Virus
Reverse Transcriptase
SS DNA of Virus
Integrase
Viral RNA
Viral Protein
Macrophage
[Factory of
HIV

Diagnostic test and Treatment

- A widely used diagnostic test for AIDS is - Enzyme Linked Immuno-Sorbent Assay $\boxed{\text{(ELISA).}}$ Confirmation test – $\boxed{\text{Western blotting}}$

- Treatment of AIDS with anti-retroviral drugs is only partially effective.
- They can only prolong the life of the patient but cannot prevent death, which is inevitable.

[NACO]

Drugs →

1)- Reverse transcreptase inhibitor → A.Z.T

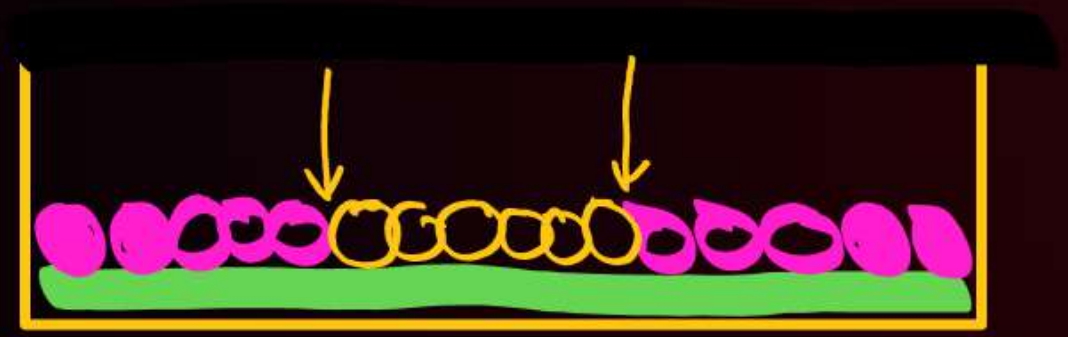
2) → Protease inhibitor → Ritonavire

Cancer

- Cancer is one of the most dreaded diseases of human beings and is a major cause of death all over the globe.
- More than a million Indians suffer from cancer and a large number of them die from it annually

Difference between Normal Cell and Cancer Cell

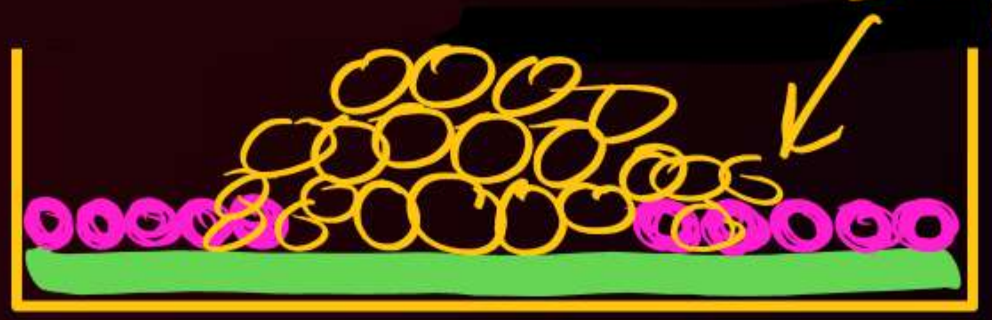
* [Control Cell Division]
* Contact inhibition +nt



* Invegunness = −nt
* Mortality = +nt

· [Uncontrol Cell Division]

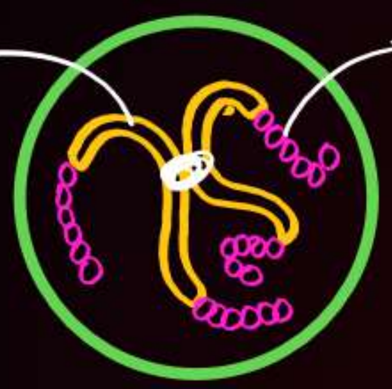
Invegination +nt



+nt

−nt Immortal.

• Telomerase Enzyme ⇒ Inactive



Telomerase Inactive → 50 generation

After 50 generation

Teromer Seq. - (-nt)

→ Program Cell Death [Apoptosis]

✓ • Telomerase = Active

After 50 generation

Telomer Seq -

Maintain by -

Telomerase Enzyme.

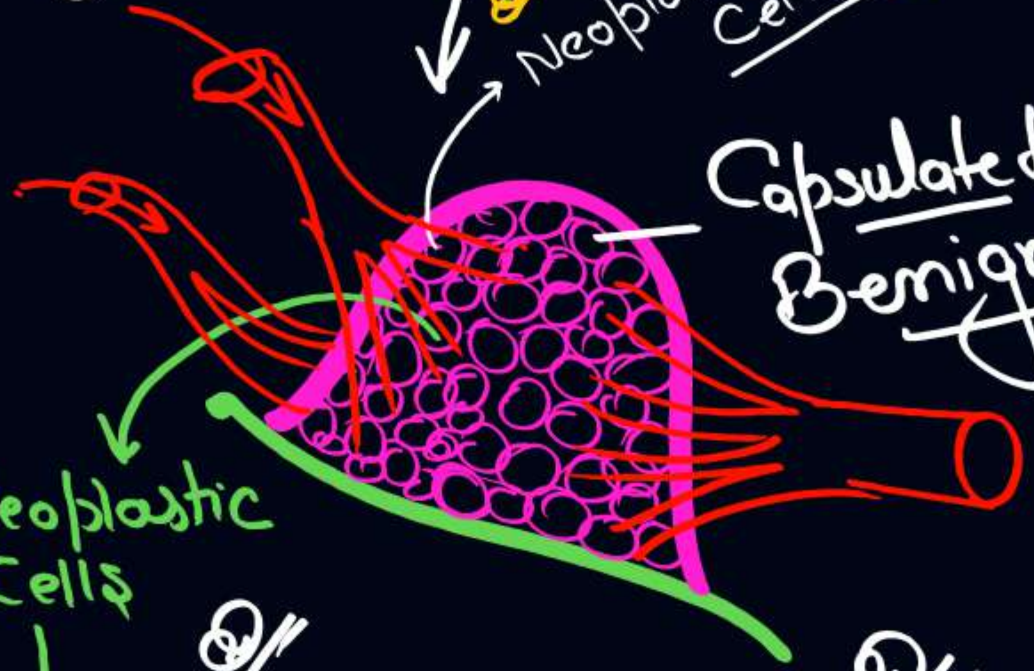
× Apoptosis (-nt)

Chemical Carcinogen
or
Physical Carcino-gen
Biological Carcinogen

Normal cell

Onchogenic Transformation

ⓐ Neoplastic Cells } vvfmp



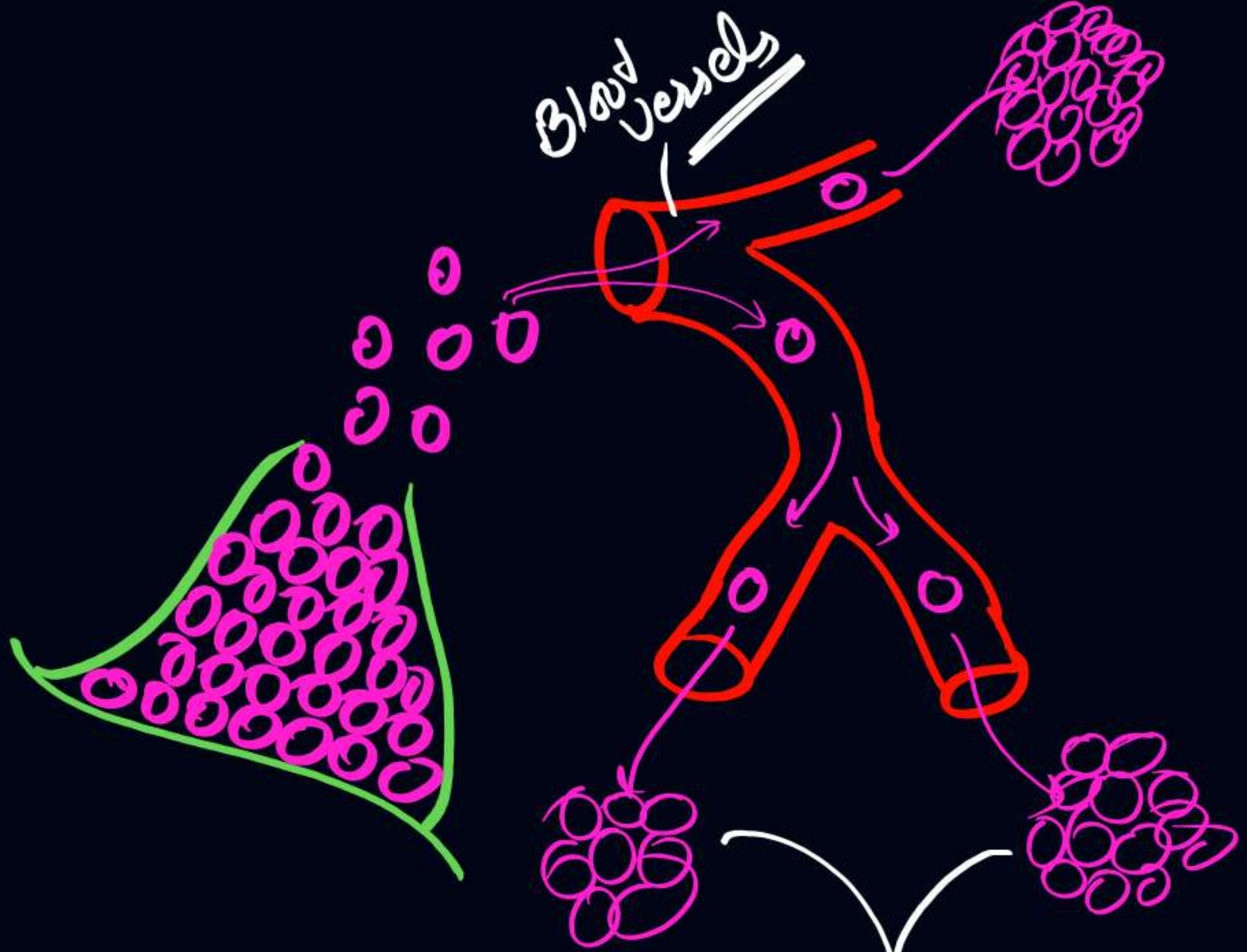
ⓐ// Capsulated Benign Tumor } Neoplasm vfmp

↓

Group of Neoplastic Cells

{ Neoplastic Cells

Release → ⓑ// Angiogenasis Factors. } → ⓑ// ↑ No. of Blood Vessels around Tumor



Malignant Tumor
or
Cancer

Spreading of Tumor Cells

imp

Metastasis

Causes of cancer :

Physical, chemical or biological agents. These agents are called carcinogens.

Physical carcinogens –

Ionising radiations - X-rays and gamma rays

Non-ionizing radiations - UV cause DNA damage

Chemical carcinogens-

Tobacco smoke have been identified as a major cause of lung cancer etc.

Biological agents –

Cancer causing viruses called oncogenic viruses like – HPV, HBV etc.

Cancer detection and diagnosis :

- Biopsy and histopathological studies
- Radiography (use of X-rays),
- CT (computed tomography)
- MRI (magnetic resonance imaging) – Safest technique
- Antibodies against cancer -specific antigens are also used for detection of certain cancers.
- Techniques of molecular biology can be applied to detect genes in individuals with inherited susceptibility to certain cancers.

PetCT

Treatment of cancer :

The common approaches for treatment of cancer are

- Surgery
- Radiation therapy
- Chemotherapy – Vincristine and vinblastine
- Immunotherapy - α-interferon

Drugs

Psychotropic

(Mood altering)

eg-
- × Tranquillizer
- × Opiods
- × Stimulants

Psychedelic

(Hallucinogenic)

eg-
- Cannabinoids
- Datura
- Atropa belladona
- L.S.D (Synthitic)

vvfm

① Tranquillizer

→ Anesthetic
→ Depressent
→ Sleeping drugs
→ ↑ - Tranquility

eg → Benzodiazepine (Valum) → used in
- Anxiety
- ↓ Insomnia
- Schizophrinia

+
Barbiturate

→ Mechanism
↓
↑ GABA in Brain

Opioids →

vvImp

- → Narcotics ✓ नशीले
- → Sedative
- → Depressents
- → Analgesic (Pain Killer) vvvImp
- → Euphoriant

- → Isolate from → Papaver somniferum Q// (opium poppy) → (Latex)
- → Contain ⇒ 20 - Alkaloids
- → Derivatives of Opium poppy →

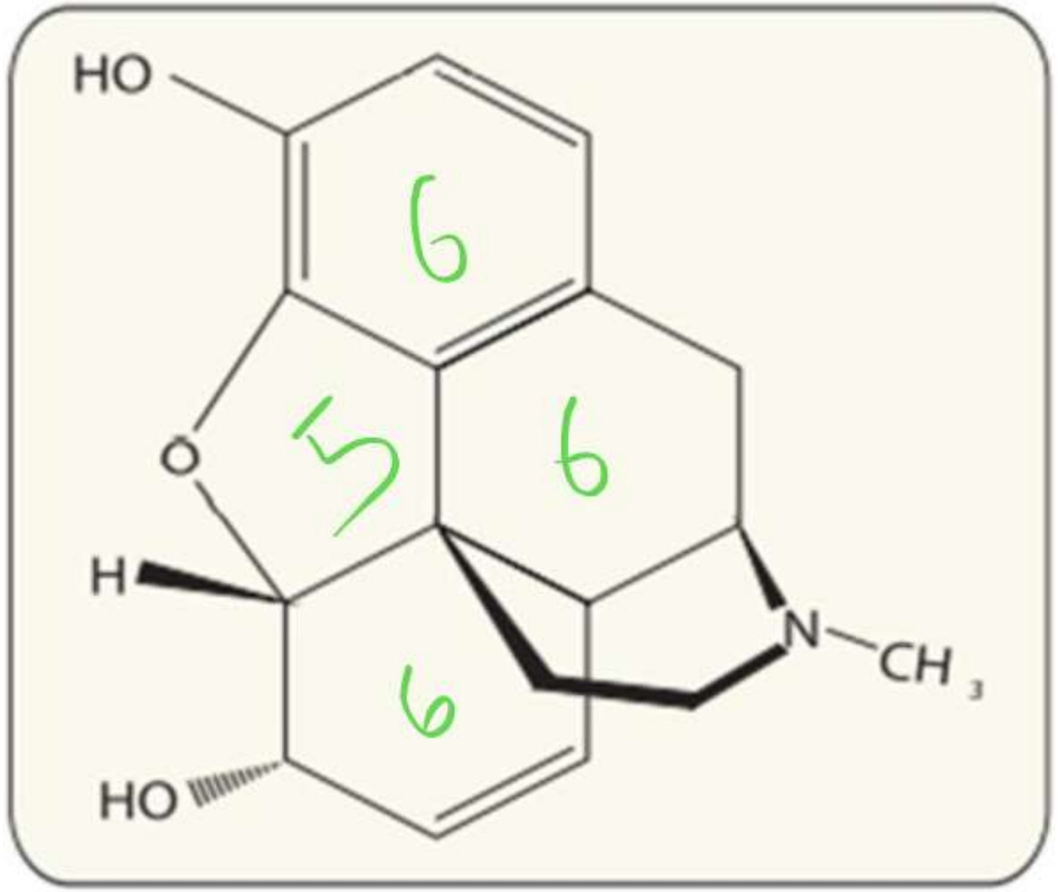


Figure 8.7 Chemical structure of Morphine

Figure 8.8 Opium poppy

* Morphine → • Analgesic (Pain killer)
  • Receptor tnt on — [Brain / Gut
* Codein → • Methy-morphine
  • used in Cough Syrup
* Heroine → • Diacetylation of morphine
* Smack → • Produced from Heroine
  • Brown sugar
  • Taken by ⟶ Snorting & Injuction

Smack ↓
· White Crystals
· Bitter & odourless

Methylation
Acetylation

vvImp

Stimulants

- Cocaine ✓
- Tobaco ✓
- Amphitamines ✗

1) Cocaine or Coke/Crack

Imp

- → Isolate fram = Coca plant (Erythroxylum coca) [Native to South America]
- → Taken by = Snorting
- → Euphoriant, ↑ - Energy
- → Excessive dosage ⇒ Hallucination

Tobaco

Tobaco →
- Nicotine alkaloid +nt
- Taken by = Smoked, Chewed, Snuff
- Stimulate = Adrenal glands

↑ Ardenalin & Nor Adrenalin.

↓

↑ - Glucose level

- Associated with ⇒ ↑ Mouth Cancer
- Smoking –
  - ↑ - CO in blood
  - Lung Cancer

CANNABINOIDS (Hallucinogenic)

vvvImp

* Receptor ⁿt in ⇒ Brain @/
* Isolate from plant ⇒ Cannabis Sativa @/

eg → Marijuana → [Active ingradient (Delta-9-T.H.C.)
[Dried - Flower @/
Bhang → Fresh leaves @/
Ganja → Female inflorescence @/
Charas or Hashish → Resinous Extract @//

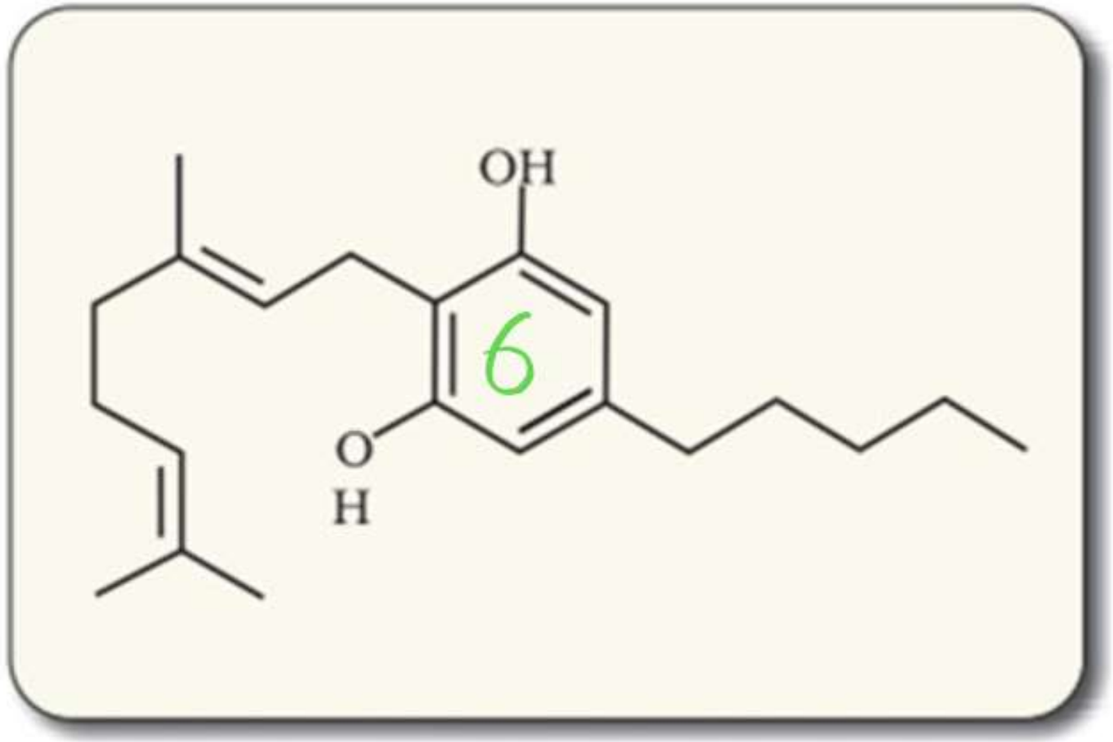


Figure 8.9 Skeletal structure of cannabinoid molecule

Figure 8.10 Leaves of *Cannabis sativa*

Lysergic acid diethylamide (L.S.D)

- Synthetic Hallucinogen
- obtain from = Ergot of fruiting body of fungus ⇒ Claviceps vvImp purpurea

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